Post-Selection Inference for Partially Identified Parameters*

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Abstract

Partial identification of parameters such as average treatment effects, average partial effects and welfare is particularly common when using observational data and experimental data with imperfect compliance due to the endogeneity of individuals' treatment uptake. In this setting, the researcher is typically interested in a treatment or policy that is either selected from the estimated set of best-performers or arises from a data-dependent selection rule. In this paper, we develop new inference tools for partially-identified parameters chosen via these forms of selection. We develop three types of confidence intervals for data-dependent and partially-identified parameters, discuss how they apply to several examples of interest and prove their uniform asymptotic validity under weak assumptions.

Keywords: Partial Identification, Post-Selection Inference, Selective Inference, Conditional Inference, Uniform Validity, Treatment Choice.

1 Introduction

Policymakers are often interested in choosing policies or treatments based on empirical evidence and understanding the welfare consequences of implementing their selections. An empirically-grounded policy is commonly chosen to optimize an estimated criterion (e.g., empirical welfare itself or regret),

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leading to a statistical inference problem for a data-dependent parameter. It is now well-known that selecting an object of interest from data invalidates conventional inference tools (e.g., Andrews et al., 2024). In addition, optimal treatments and policies are often partially identified for various reasons, such as treatment endogeneity in observational data or experimental data with imperfect compliance (Stoye, 2012; Kallus and Zhou, 2021; Pu and Zhang, 2021; D'Adamo, 2021; Yata, 2021), imperfect external validity of experimental evidence (Ishihara and Kitagawa, 2021; Adjaho and Christensen, 2022; Montiel Olea et al., 2026), nonlinearity of welfare (Cui and Han, 2026), and lack of overlap in algorithmic decisions (Ben-Michael et al., 2021). The failure of conventional inference tools after data-dependent selection is only compounded by the presence of such partial identification. At the same time, policymakers may hope to avoid imposing strong point-identifying assumptions on the underlying data-generating processes (e.g., parametric restrictions) lest policy decisions be inadvertently driven by such potentially false assumptions.

This paper aims to address the inference challenges simultaneously posed by partial identification and the data-driven selection of parameters. We focus on a general class of data-dependent parameter selection, called *polyhedral selection*, and develop new inference tools for parameters selected according to this class. We show polyhedral selection captures a wide range of attitudes of a policymaker towards a policy choice, which includes two broad categories: (a) a policymaker, as a statistical decision maker under ambiguity (Manski, 2000), choosing a single policy (e.g., via maximin, minimax) and conducting inference on welfare evaluated at that policy, and (b) one conducting inference on welfare for a policy in the estimated identified set for optimal policy. The confidence intervals (CIs) we propose here provide such a policymaker with the ability to accurately infer how well their selected treatments/policies can be expected to perform, for instance to gauge whether implementation of them is worthwhile from a welfare standpoint.

The current literature has not yet developed valid post-selection inference tools in partially-identified contexts, an important deficiency that the methods proposed in this paper aim to correct. The methods we propose here build upon the ideas of conditional and hybrid inference employed in various point-identified contexts by, e.g., Lee et al. (2016), Fithian et al. (2017), Tibshirani et al. (2018), Andrews et al. (2024) and McCloskey (2024) to produce CIs for partially-identified

parameters such as welfare or ATEs chosen from an estimated set or via a data-dependent selection rule. Our framework formally nests the point-identified winners setting of Andrews et al. (2024).[1] Although Andrews et al. (2023) also propose conditional and hybrid inference methods in the partial identification context of moment inequality models, they do not allow for data-dependent selection of objects of interest, one of the main focuses of the present paper. Finally, this paper directly relies upon results in the literature on partial identification of welfare, treatment effects and partial outcomes such as Manski (1990), Balke and Pearl (1997, 2011), Manski and Pepper (2000), Mogstad et al. (2018), Han and Yang (2024) and Han (2024). We apply our inference methods to a general class of problems nesting these examples.

We start by motivating the problem of parameter selection and inference for a partially-identified parameter in a leading example of selecting dynamic treatment regimes (Han, 2024) in Section 2. After sketching the ideas behind our inference methods in a simplified version of this example in Section 3, we introduce in Section 4 the general inference framework to which our methods can be applied. We formally introduce a (weak) polyhedral selection event and show that it incorporates selections based on maximin welfare, Hurwicz welfare, and minimax regret, and selection from the estimated identified set of optimal policies. These are all leading cases of the types of selection in the literature on treatment choice under partial identification mentioned above. Our general framework also incorporates treatment choice based on Manski bounds for average counterfactual outcomes or ATEs and bounds on treatment parameters derived from linear programming (e.g., Balke and Pearl, 1997, 2011; Mogstad et al., 2018; Han, 2024; Han and Yang, 2024) as well as parameters chosen via asymptotically optimal treatment choice rules (e.g., Christensen et al., 2023). These latter examples are covered in Section 5 and Appendix A.

Within the general inference framework in Section 6, we develop three types of CIs for data-dependent and partially-identified parameters whose coverage requirements can be broken down into two categories: conditional and unconditional. It is natural for a researcher to require conditional

[1] Under the embedding described in Appendix B, their Assumptions 6–8 imply Assumptions 6.1–6.3 of the present paper. Our conditional CIs then coincide with the Andrews et al. (2024) conditional CIs. The projection and hybrid CIs use the same principles as Andrews et al. (2024) but retain separate one-sided safeguards for the lower and upper endpoints of the identified set in order to cover the set in its entirety; see Appendix B for a step-by-step comparison.

validity when they want reliable inference on a particular policy *only* when that policy is selected or retained for consideration from the data, rather than allowing poor coverage in those cases to be offset by better coverage in others. On the other hand, unconditional validity is a natural requirement for a policymaker wishing to evaluate the overall procedure of selecting and assessing a policy according to a selection rule, requiring reliable coverage *on average* across all repeated samples, no matter which particular policy was selected from the data. As the name suggests, the *conditional* CIs are asymptotically valid conditional on the parameter of interest corresponding to a treatment or policy chosen from an estimated set (i.e., case (b)). The construction of this CI does not require a specific rule for choosing the parameter of interest from the estimated set. Importantly, we show that these CIs are asymptotically valid *uniformly* across a large class of data-generating processes (DGPs). Uniform asymptotic validity is especially important for approximately correct finite-sample coverage in post-selection contexts like those in this paper (see, e.g., Andrews and Guggenberger, 2009).

The second and third types of CIs we develop in this paper are designed for inference on parameters chosen by data-dependent selection rules for which the object of interest is uniquely determined by the selection rule (i.e., case (a)). The *projection* CIs do not require knowledge of the selection rule to be asymptotically valid, whereas the *hybrid* CI construction utilizes the particular form of a selection rule to improve upon the length properties of the projection CI. Since they are valid conditionally, the law of total probability implies that the conditional CIs that condition on the realization of the selection rule are also valid unconditionally. Although the conditional CIs are short for selections that occur with high probability, they can become exceptionally long when this probability is small (see, e.g., Kivaranovic and Leeb, 2021). Conversely, projection CIs are overly conservative when selection probabilities are high. Our hybrid CIs interpolate the length properties of the conditional and projection CIs in order to attain good length properties regardless of the value selection probabilities take. In analogy with the conditional CIs, we formally show that both projection and hybrid CIs are asymptotically valid in a uniform sense.

We analyze in Section 7 the finite-sample coverage and length properties of our proposed CIs in the context of a simple Manski bound example. Since, to our knowledge, these are the first uniformly valid CIs for data-dependent selections of partially-identified parameters, there are no existing CIs

to which we can directly compare. Nevertheless, since our CIs can also be used for inference on a priori chosen partially-identified parameters, we conduct a power comparison with one of the leading methods for inference on a partially-identified parameter in Appendix C. In particular, we compare the power of the test implied by our hybrid CIs to the power of the hybrid test of Andrews et al. (2023), a test that applies to a general class of moment-inequality models that is also based upon a (different) hybrid between conditional and projection-based inference. Encouragingly, the power of the test implied by our hybrid CI is quite competitive even in this environment for which it was not designed. We also find that the finite-sample coverage of all of our CIs is approximately correct in a simple Manski bound example. Finally, we analyze the length tradeoffs between the three different CIs across different DGPs, finding the hybrid CI to perform best overall. The final section, Section 8, contains an empirical application corresponding to the leading example in Section 2, where we apply our procedures to dynamic policies of schooling and post-school training.

In the remaining appendices, Appendix B provides a detailed discussion of precisely how the inference framework and methods of this paper relate to those of Andrews et al. (2024), Appendix D contains additional simulation results corresponding to the dynamic treatment regime example detailed in Section 8, Appendix E contains the mathematical proofs of the main results, while Appendix F describes an extension of our inference methods that permits affine endpoint components with zero asymptotic variance when they are uniformly separated from the corresponding population endpoint.

2 Motivating Example: Dynamic Treatment Regimes

Consider schooling and post-school training as a sequence of treatments. A policymaker (PM) may be interested in understanding the welfare consequences of implementing a policy that comprises a particular (potentially adaptive) sequence of the treatments called a *dynamic treatment regime* (Murphy, 2003; Robins, 2004). To be concrete, similar to Han (2024), consider a dynamic treatment regime $\boldsymbol{\delta}(\cdot) \equiv (\delta_1, \delta_2(\cdot, \delta_1)) \in \mathcal{D}$, where δ_1 indicates receipt of a high school (HS) diploma, $\delta_2(y_1, \delta_1)$ indicates receipt of the job training program given pre-program employment status y_1 and first-stage policy δ_1 and $\mathcal{D}$ denotes the set of treatment regimes under consideration. By having policy δ_2 as a function of y_1, the treatment allocation policy adaptively incorporates information about

Regime #	δ_1	$\delta_2(1,\delta_1)$	$\delta_2(0,\delta_1)$
1	0	0	0
2	1	0	0
3	0	1	0
4	1	1	0
5	0	0	1
6	1	0	1
7	0	1	1
8	1	1	1

Table 1: Possible Adaptive Policies $\boldsymbol{\delta}(\cdot)$ When $T=2$ and δ_1 Does Not Depend on Covariates

unobserved characteristics of the individuals reflected in the response Y_1 to δ_1. By implementing policy $\boldsymbol{\delta}(\cdot)$, the policymaker aims to improve the employment status of individuals in the terminal stage. Let $Y_1(d_1)$ and $Y_2(d_1,d_2)$ denote the counterfactual employment status each respective period for a fixed (i.e., non-adaptive) treatment (sequence) and define the counterfactual employment status given $\boldsymbol{\delta}(\cdot)$ as $Y_2(\boldsymbol{\delta}(\cdot)) \equiv Y_2(\delta_1, \delta_2(\cdot, \delta_1))$. Given this definition, the policymaker may be interested in maximizing the employment share $W(\boldsymbol{\delta}) \equiv \mathbb{E}[Y_2(\boldsymbol{\delta}(\cdot))]$ as a welfare objective over the set of possible policies. All possible adaptive policies in $\mathcal{D}$ are listed in Table 1.

In practice, this policymaker would collect data to conduct inference on the welfare of the policy they select from the data in order to gauge how well the policy is expected to perform. This turns out to be a challenging problem due to modeling and sampling uncertainty. Suppose the policymaker observes (D_1,D_2,Y_1,Y_2), where D_1 indicates whether an individual received a HS diploma, D_2 indicates whether the individual participated in the job training program, Y_1 indicates the employment status after D_1 (but before D_2) is realized, and Y_2 indicates the employment status after D_2 is realized. As (D_1,D_2) can be endogenous, suppose the policymaker is equipped with IVs (Z_1,Z_2) for them, respectively, where Z_1 is an indicator for the density of high schools and Z_2 is an indicator for whether the individual was randomly assigned to the job training program. Note that even with random assignment in the second stage (Z_2), noncompliance is a common source of endogeneity in D_2.

In general, the marginal and joint distributions of $(Y_1(d_1),Y_2(d_1,d_2))$ are partially identified for reasons analogous to those for which average treatment effects are partially identified (Manski, 1990), as will become clear in later sections. This implies that the welfare $W(\boldsymbol{\delta})$ is also partially

identified *for any given* $\boldsymbol{\delta}$, because

$$W(\boldsymbol{\delta}) \equiv \mathbb{P}[Y_2(\boldsymbol{\delta}(\cdot))=1] = \sum_{y_1 \in \{0,1\}} \mathbb{P}[Y_1(\delta_1)=y_1, Y_2(\delta_1, \delta_2(y_1,\delta_1))=1]$$
$$= \mathbb{P}[Y_1(\delta_1)=1, Y_2(\delta_1,\delta_2(1,\delta_1))=1] + \mathbb{P}[Y_1(\delta_1)=0, Y_2(\delta_1,\delta_2(0,\delta_1))=1]. \quad (2.1)$$

Among the candidate parameters $\{W(\boldsymbol{\delta}) : \boldsymbol{\delta} \in \mathcal{D}\}$, for which parameter should the policymaker conduct inference (e.g., construct CIs)? Ideally, she may want to select $W(\boldsymbol{\delta}^*)$ as a parameter of interest, where $\boldsymbol{\delta}^* \in \mathcal{D}$ is the optimal policy that maximizes $W(\boldsymbol{\delta})$. This choice of parameter is infeasible for at least two reasons. First, $\boldsymbol{\delta}^*$ is partially identified in general, as $W(\boldsymbol{\delta})$ is partially identified even with IVs. Second, even if it were point-identified, say, from imposing strong parametric assumptions, the value of $\boldsymbol{\delta}^* \in \mathcal{D}$ is unknown to the PM in practice because it is a population object. On the other hand, the PM would naturally be interested in inference on a value of $W(\boldsymbol{\delta})$ for a value of $\boldsymbol{\delta}$ that she has selected to perform well from the data because this is the welfare objective for a policy the PM may implement in practice.

Given that the PM is naturally interested in inference on $W(\boldsymbol{\delta})$ for some data-dependent value of $\boldsymbol{\delta}$, we consider the following two scenarios:

Case (a). When the PM is willing to choose the parameter from among $\{W(\boldsymbol{\delta}) : \boldsymbol{\delta} \in \mathcal{D}\}$, we assume that the PM is a statistical decision maker under ambiguity (Manski, 2000): She would map the data to a policy in the form of $r : \{S_i\} \mapsto \widehat{\boldsymbol{\delta}}$, where $S \equiv (D_1, D_2, Y_1, Y_2, Z_1, Z_2)$ is the vector of observables and $\{S_i\}$ denotes the sample of observables. We consider a class of selection rules for r that we call *polyhedral selection rules*, for reasons that will become clear later. Examples are a selection rule that maximizes worst-case welfare and one that minimizes worst-case regret. Given the selection $\widehat{\boldsymbol{\delta}}$, $W(\widehat{\boldsymbol{\delta}})$ becomes the parameter for which the PM will form CIs.

Case (b). Alternatively, the PM may be interested in conducting inference on $W(\boldsymbol{\delta})$ for a value of $\boldsymbol{\delta}$ in the estimated identified set for $\boldsymbol{\delta}^*$ (or equivalently the set of undominated policies). That is, the PM is interested in inference on $W(\boldsymbol{\delta})$ conditional on the *polyhedral selection event* that $\boldsymbol{\delta} \in \widehat{\mathcal{D}}$. Since $\boldsymbol{\delta}^*$ is the ideal infeasible evaluation point for welfare, it is natural for the PM to be

interested in conducting inference that is conditional on this feasible event.

When the parameter is selected in a data-dependent manner (case (a)) or inference is conditioned on a data-dependent event (case (b)), additional econometric challenges emerge; see the next section. We propose a computationally simple inference procedure that overcomes these challenges and yields CIs that are asymptotically valid uniformly across the DGPs that could have generated S.

For our application to the adaptive policies described above, we construct CIs for $W(\boldsymbol{\delta})$ evaluated at $\boldsymbol{\delta}=\widehat{\boldsymbol{\delta}}$ obtained from a maximin rule, which can be expressed as $\widehat{\boldsymbol{\delta}}=\operatorname{argmax}_{\boldsymbol{\delta}\in\mathcal{D}}\widehat{L}(\boldsymbol{\delta})$, where $\widehat{L}(\boldsymbol{\delta})$ is the estimated lower bound on $W(\boldsymbol{\delta})$. When we apply our inference method to the dataset constructed for the application (details of this application are contained in Section 8), we obtain the following 95% CIs for the welfare (i.e., the probability of being employed in the terminal stage):[2]

Conventional	(a) Hybrid	(b) Conditional
(0.32, 0.78)	(0.28, 0.73)	(0.28, ∞)

In this table, the conventional CI for $W(\boldsymbol{\delta})$ is invalid since $\boldsymbol{\delta}$ was selected from the data and $W(\boldsymbol{\delta})$ is partially identified. On the other hand, the hybrid CI is valid unconditionally while the conditional CI is valid conditional on the event that the regime $\boldsymbol{\delta}=\operatorname{argmax}_{\boldsymbol{\delta}\in\mathcal{D}}\widehat{L}(\boldsymbol{\delta})$ is realized in the data.[3] The following are the coverage probabilities for the three CIs at the 0.95 nominal level using simulated data based on the same setting as the above application (details are contained in Appendix D):

	Conventional	(a) Hybrid	(b) Conditional
DGP 1	0.66	0.94	0.94
DGP 2	0.88	0.95	0.94

3 Theoretical Challenges and How We Address Them: An Illustration

We first illustrate our proposed methods using a simplified version of the motivating example from the previous section. Consider a binary outcome of interest Y, a binary treatment indicator D

[2] In the actual application in Section 8, the welfare is the probability that 30-month earnings are above the median; the CIs below come from that application.

[3] See Section 8 for an in-depth discussion of when conditional vs unconditional validity would be desirable in the context of this application.

and a binary treatment assignment Z. Furthermore, let $Y(1)$ and $Y(0)$ denote counterfactual outcomes under treatment ($D=1$) and no treatment ($D=0$). Assuming $\mathbb{E}[Y(d)|Z]=\mathbb{E}[Y(d)]$, Manski (1990) shows that we can bound the average counterfactual outcomes $W(d)=\mathbb{E}[Y(d)]$ in the absence and presence of treatment as follows: for $d=0,1$,

$$L(d)\equiv\max\{p^{1d0},p^{1d1}\}\leq W(d)\leq\min\{1-p^{0d0},1-p^{0d1}\}\equiv U(d), \tag{3.1}$$

where $p^{ydz}\equiv Pr(Y=y,D=d|Z=z)$. Given (3.1) it is natural to define the set of best-performing options $\mathcal{D}^*\subset\mathcal{D}\equiv\{0,1\}$, as a subset of the two options of treatment and no treatment, to be those that are *undominated* options; from an observed dataset of outcomes, treatments and treatment assignments $\{(Y_i,D_i,Z_i)\}_{i=1}^n$, such a set can be estimated as:

$$\widehat{\mathcal{D}}=\left\{d\in\{0,1\}:\nexists d'\in\{0,1\}\text{ s.th. }\widehat{U}(d)<\widehat{L}(d')\right\}=\left\{d\in\{0,1\}:\widehat{U}(d)\geq\widehat{L}(d')\ \forall d'\in\{0,1\}\text{ s.th. }d'\neq d\right\},$$

where $\widehat{L}(d)\equiv\max\{\widehat{p}^{1d0},\widehat{p}^{1d1}\}$ and $\widehat{U}(d)\equiv\min\{1-\widehat{p}^{0d0},1-\widehat{p}^{0d1}\}$ with $\widehat{p}^{ydz}$ being an empirical estimate of the fitted probability p^{ydz}.

We are interested in inference on the identified interval $[L(d),U(d)]$ for the average counterfactual outcome $W(d)$ of option $d\in\{0,1\}$, which will immediately provide inference on $W(d)$ itself, after the researcher selects this option from $\widehat{\mathcal{D}}$ (in the sense of either (a) or (b) in the previous section). In other words, we would like to provide statistically precise statements about the true average counterfactual outcome of an option selected from the data to give the researcher an idea of how well this selected option should be expected to perform in the population. We first provide some intuition for why standard inference techniques based upon asymptotic normality fail and then sketch our proposals for valid inference in this context.

3.1 Why Does Standard Inference Fail?

To fix ideas, let us focus for now on inference for the lower bound $L(d)$ of a selected option rather than the entire identified set $[L(d),U(d)]$. More specifically, since $L(d)$ is a lower bound for the average counterfactual outcome $W(d)$, we would like to obtain a lower confidence bound for $L(d)$.

Under standard conditions, a central limit theorem implies $\widehat{p} = (\widehat{p}^{100},\widehat{p}^{010},\widehat{p}^{110},\widehat{p}^{101},\widehat{p}^{011},\widehat{p}^{111})'$ is normally distributed in large samples. So why can't a researcher form a conventional CI using $\widehat{L}(d)$ and quantiles from a normal distribution as the basis for inference? There are two reasons such an approach is (asymptotically) invalid:

1. Even in the absence of data-dependent selection, $\widehat{L}(d) = \max\{\widehat{p}^{1d0},\widehat{p}^{1d1}\}$ is not normally distributed in large samples.

2. Data-dependent selection of d further complicates the distribution of $\widehat{L}(d)$.

Reason 1. is easy to see since $\widehat{L}(d)$ is the maximum of two normally distributed random variables in large samples when d is chosen a priori. To better understand reason 2., note that the distribution of $\widehat{L}(d)$ given $d \in \widehat{\mathcal{D}}$ is the conditional distribution of the maximum of two normally distributed random variables given that the minimum of two other normally distributed random variables, $\widehat{U}(d) \equiv \min\{1-\widehat{p}^{0d0},1-\widehat{p}^{0d1}\}$, exceeds the maximum of yet another set of two normally distributed random variables, $\widehat{L}(d') = \max\{\widehat{p}^{1d'0},\widehat{p}^{1d'1}\}$ for $d' \neq d$. Unconditionally, $\widehat{L}(\widehat{d})$ for any data-dependent choice of $\widehat{d}$ is distributed as a mixture of the distributions of $\widehat{L}(0)$ and $\widehat{L}(1)$, neither of which are themselves normally distributed. Even this simple example, in which the set of candidate parameters contains only two elements (i.e., $\{W(0),W(1)\}$), poses interesting theoretical challenges: complex non-normal distributions arise due to the non-smoothness of the lower bound in the underlying parameters and data-dependent selection.

3.2 Case (b): Conditional Confidence Intervals

Even though Case (a) is the leading case, we consider Case (b) first, because the former relies on the same conditional inference framework. Suppose that a researcher's interest in inference on $L(d)$ only arises when d is estimated to be in the set of best-performing options, viz., $d \in \widehat{\mathcal{D}}$. In such a case, we are interested in a lower confidence bound for $L(d)$ that is approximately valid across repeated samples for which $d \in \widehat{\mathcal{D}}$ because, as alluded to in the motivating example above, a PM would only be interested in inference on $W(d)$ precisely because $d \in \widehat{\mathcal{D}}$. Formally, we would

like to form a *conditionally* valid lower confidence bound $\widehat{L}(d)_\alpha^C$ such that

$$\mathbb{P}\Big(L(d) \geq \widehat{L}(d)_\alpha^C \Big| d \in \widehat{\mathcal{D}} \Big) \geq 1-\alpha \tag{3.2}$$

for some $\alpha \in (0,1)$ in large samples. To do so, we characterize the conditional distribution of $\widehat{L}(d)$. Specifically, let $\widehat{j}_L(d) \equiv \text{argmax}_{j\in\{0,1\}} \widehat{p}^{1dj}$ be the value of Z at which the maximum between the two estimated probabilities is achieved. Then $\widehat{L}(d) = \widehat{p}^{1d\widehat{j}_L(d)}$. The conditioning event $\{d \in \widehat{\mathcal{D}}, \widehat{j}_L(d) = j_L^*\}$, where j_L^* is a potential value of Z, can be written as a polyhedron in $\widehat{p}$ and $\widehat{L}(d)$ is equal to a (random) element of $\widehat{p}$ so that Lemma 5.1 of Lee et al. (2016) implies

$$\widehat{L}(d) \Big| \Big\{ d \in \widehat{\mathcal{D}}, \widehat{j}_L(d) = j_L^* \Big\} \sim \widehat{p}^{1dj_L^*} \Big| \Big\{ \widehat{\mathcal{L}}_L(\mathcal{Z}_L) \leq \widehat{p}^{1dj_L^*} \leq \widehat{\mathcal{U}}_L(\mathcal{Z}_L) \Big\} \tag{3.3}$$

for some known functions $\widehat{\mathcal{L}}_L(\cdot)$ and $\widehat{\mathcal{U}}_L(\cdot)$, where $\widehat{p}^{1dj_L^*} \sim \mathcal{N}(p^{1dj_L^*}, \text{Var}(\widehat{p}^{1dj_L^*}))$ in large samples and $\mathcal{Z}_L$ is a sufficient statistic for the nuisance parameter $p = (p^{100}, p^{010}, p^{110}, p^{101}, p^{011}, p^{111})'$ that is asymptotically independent of $\widehat{p}^{1dj_L^*}$. The characterization in (3.3) permits the straightforward computation of a conditionally quantile-unbiased estimator for $p^{1d\widehat{j}_L(d)} = p^{1dj_L^*}$, since after further conditioning on the realized value of $\mathcal{Z}_L$ in sample, the latter is equal to the mean of the underlying normally distributed random variable $\widehat{p}^{1dj_L^*} \sim \mathcal{N}(p^{1dj_L^*}, \text{Var}(\widehat{p}^{1dj_L^*}))$ that is subject to truncation.[4] Denoting this $1-\alpha$ quantile-unbiased estimator as $\widehat{L}(d)_\alpha^C$, we have

$$\mathbb{P}\Big(p^{1d\widehat{j}_L(d)} \geq \widehat{L}(d)_\alpha^C \Big| d \in \widehat{\mathcal{D}} \Big) = 1-\alpha \tag{3.4}$$

in large samples. However, noting that $L(d) \geq p^{1d\widehat{j}_L(d)}$ with probability one, we can see that (3.2) holds for this choice of $\widehat{L}(d)_\alpha^C$.

Although (3.2) does not hold with exact equality, we note that the left-hand side cannot be much larger than the right-hand side. In other words, although $\widehat{L}(d)_\alpha^C$ is a conservative lower confidence bound for $L(d)$, it is not very conservative. This can be seen heuristically by working through the two possible values that $L(d)$ can take:

[4] See, e.g., Pfanzagl (1994) for quantile-unbiased estimation of the underlying mean of a truncated normal.

1. If $L(d)=p^{1d\widehat{j}_L(d)}$ (i.e., the correct probability is selected), then (3.2) holds with equality in large samples by (3.4).

2. If $L(d)\neq p^{1d\widehat{j}_L(d)}$ (i.e., an incorrect probability is selected), then $L(d)\approx p^{1d\widehat{j}_L(d)}$ (i.e., the correct and incorrect probabilities cannot differ by much) since $\widehat{L}(d)=\widehat{p}^{1d\widehat{j}_L(d)}$ so that the left-hand side of (3.2) cannot be much larger than the right-hand side.

Finally, a construction analogous to that described above for producing a lower confidence bound for $L(d)$ produces a conditionally valid upper confidence bound $\widehat{U}(d)^C_{1-\alpha}$ for $U(d)$ that satisfies

$$\mathbb{P}\Big(U(d)\leq\widehat{U}(d)^C_{1-\alpha}\Big|d\in\widehat{\mathcal{D}}\Big)\geq 1-\alpha \tag{3.5}$$

for some $\alpha\in(0,1)$ in large samples. The lower and upper confidence bounds can then be combined to form a CI, $[\widehat{L}(d)^C_{\alpha/2},\widehat{U}(d)^C_{1-\alpha/2}]$, that is conditionally valid for $[L(d),U(d)]$ in large samples since

$$\begin{aligned}&\mathbb{P}\Big(L(d)\geq\widehat{L}(d)^C_{\alpha/2},U(d)\leq\widehat{U}(d)^C_{1-\alpha/2}\Big|d\in\widehat{\mathcal{D}}\Big) \\ \geq 1-\mathbb{P}\Big(L(d)<\widehat{L}(d)^C_{\alpha/2}\Big|d\in\widehat{\mathcal{D}}\Big)-\mathbb{P}\Big(U(d)>\widehat{U}(d)^C_{1-\alpha/2}\Big|d\in\widehat{\mathcal{D}}\Big)\geq 1-\alpha.\end{aligned} \tag{3.6}$$

3.3 Case (a): Unconditional Confidence Intervals

Given the ingredients from Case (b), now we are ready to illustrate Case (a). Suppose now that the researcher uses a data-dependent rule to select a *unique* option of inferential interest. For example, suppose the researcher is interested in choosing the option with the highest counterfactual outcome in the worst case across its identified set (i.e., maximin welfare) so that she chooses $\widehat{d}=\text{argmax}_{d\in\{0,1\}}\widehat{L}(d)$. In such a case, it is natural to form a lower confidence bound for $L(\widehat{d})$ that is valid across *all* repeated samples when the PM is interested in the average performance of $W(\widehat{d})$ overall, not just at the $\widehat{d}$ value realized in the data. Formally, we would like to form an *unconditionally* valid lower confidence bound $\widehat{L}(\widehat{d})^U_\alpha$ such that

$$\mathbb{P}\Big(L(\widehat{d})\geq\widehat{L}(\widehat{d})^U_\alpha\Big)\geq 1-\alpha \tag{3.7}$$

for some $\alpha \in (0,1)$ in large samples. Given its conditional validity (3.2), the conditional lower confidence bound $\widehat{L}(d)^C_\alpha$ also satisfies (3.7) upon changing the definition of $\widehat{\mathcal{D}}$ to $\widehat{\mathcal{D}} = \{\widehat{d}\}$ in its construction. However, it is well known in the literature on selective inference that conditionally-valid confidence bounds can be very uninformative (i.e., far below the true value) when the probability of the conditioning event is small (e.g., Kivaranovic and Leeb, 2021, Andrews et al., 2024 and McCloskey, 2024). See Section 6.3 for a detailed explanation. Here, we propose two additional forms of confidence bounds that are only unconditionally valid but do not suffer from this drawback.

First, we can form a lower confidence bound for $L(\widehat{d})$ by projecting a one-sided rectangular simultaneous confidence lower bound for all possible values $L(\widehat{d})$ can take: $\widehat{L}(\widehat{d})^P_\alpha \equiv \widehat{L}(\widehat{d}) - \widehat{c}_{1-\alpha,L}\sqrt{\widehat{\Sigma}_{L,2\widehat{d}+1+\widehat{j}_L(\widehat{d})}}$, where $\widehat{c}_{1-\alpha,L}$ is the $1-\alpha$ quantile of $\max_i \widehat{\zeta}_i/\sqrt{\widehat{\Sigma}_{L,i}}$ for $\widehat{\zeta} \sim \mathcal{N}(0,\widehat{\Sigma}_L)$, $\widehat{\Sigma}_L$ is a consistent estimator of $\Sigma_L \equiv \text{Var}(\widehat{p}^{100},\widehat{p}^{101},\widehat{p}^{110},\widehat{p}^{111})$ and, for any square matrix A, A_i denotes the ith element of the main diagonal of A. Here, the maximum is taken to guarantee simultaneous coverage of all possible values of $L(\widehat{d})$. Since $p^{1\widehat{d}\widehat{j}_L(\widehat{d})} \in \{\widehat{p}^{100},\widehat{p}^{101},\widehat{p}^{110},\widehat{p}^{111}\}$ with probability one,

$$\mathbb{P}\left(p^{1\widehat{d}\widehat{j}_L(\widehat{d})} \geq \widehat{L}(\widehat{d})^P_\alpha\right) = \mathbb{P}\left(p^{1\widehat{d}\widehat{j}_L(\widehat{d})} \geq \widehat{p}^{1\widehat{d}\widehat{j}_L(\widehat{d})} - \widehat{c}_{1-\alpha,L}\sqrt{\widehat{\Sigma}_{L,2\widehat{d}+1+\widehat{j}_L(\widehat{d})}}\right)$$
$$\geq \mathbb{P}\left((p^{100},p^{101},p^{110},p^{111}) \geq (\widehat{p}^{100},\widehat{p}^{101},\widehat{p}^{110},\widehat{p}^{111}) - \widehat{c}_{1-\alpha,L}\sqrt{\text{Diag}(\widehat{\Sigma}_L)}\right) = 1-\alpha$$

in large samples and (3.7) holds for $\widehat{L}(\widehat{d})^U_\alpha = \widehat{L}(\widehat{d})^P_\alpha$ because $L(\widehat{d}) \geq p^{1\widehat{d}\widehat{j}_L(\widehat{d})}$. However, $\widehat{L}(\widehat{d})^P_\alpha$ suffers from a converse drawback to that of $\widehat{L}(\widehat{d})^C_\alpha$: it is unnecessarily conservative when $\widehat{d} = d$ is chosen with high probability (see e.g., Andrews et al., 2024 and McCloskey, 2024).

We propose a favorable second lower confidence bound for $L(\widehat{d})$ that combines the complementary strengths of $\widehat{L}(d)^C_\alpha$ and $\widehat{L}(\widehat{d})^P_\alpha$. Construction of this hybrid lower bound $\widehat{L}(\widehat{d})^H_\alpha$ proceeds analogously to the construction of $\widehat{L}(d)^C_\alpha$ after adding the additional condition $\{p^{1\widehat{d}\widehat{j}_L(\widehat{d})} \geq \widehat{L}(\widehat{d})^P_\beta\}$ for $\beta < \alpha$ to the conditioning event and instead computing a conditionally quantile-unbiased estimator for $p^{1\widehat{d}\widehat{j}_L(\widehat{d})}$, denoted as $\widehat{L}(\widehat{d})^H_\alpha$, satisfying

$$\mathbb{P}\left(p^{1\widehat{d}\widehat{j}_L(\widehat{d})} \geq \widehat{L}(\widehat{d})^H_\alpha \middle| \widehat{d} = d^*, p^{1\widehat{d}\widehat{j}_L(\widehat{d})} \geq \widehat{L}(\widehat{d})^P_\beta\right) = \frac{1-\alpha}{1-\beta}$$

in large samples, where d^* is any realized value of the random variable $\widehat{d}$. Imposing this additional condition in the formation of the hybrid bound ensures that $\widehat{L}(\widehat{d})^H_\alpha$ is always greater than $\widehat{L}(\widehat{d})^P_\beta$, limiting its worst-case performance relative to $\widehat{L}(\widehat{d})^P_\beta$ when $\mathbb{P}(\widehat{d}=d^*)$ is small. On the other hand, when $\mathbb{P}(\widehat{d}=d^*)$ is large, the additional condition $\{p^{1\widehat{d}j_L(\widehat{d})} \geq \widehat{L}(\widehat{d})^P_\beta\}$ is far from binding with high probability so that $\widehat{L}(\widehat{d})^H_\alpha$ becomes very close to $\widehat{L}(d)^C_{(\alpha-\beta)/(1-\beta)}$. In this case, $\widehat{L}(d)^C_{(\alpha-\beta)/(1-\beta)}$ is close to the conventional lower confidence bound based upon the normal distribution $\widehat{L}(\widehat{d}) - z_{(1-\alpha)/(1-\beta)}\sqrt{\text{Var}(\widehat{p}^{1d j^*_L})}$ because the truncation bounds in (3.3) are very wide (Proposition 3 in Andrews et al., 2024).

To see how (3.7) holds for $\widehat{L}(\widehat{d})^U_\alpha = \widehat{L}(\widehat{d})^H_\alpha$, note first that

$$\mathbb{P}\Big(L(\widehat{d}) \geq \widehat{L}(\widehat{d})^H_\alpha \Big| \widehat{d}=d^*, p^{1\widehat{d}j_L(\widehat{d})} \geq \widehat{L}(\widehat{d})^P_\beta\Big) \geq \mathbb{P}\Big(p^{1\widehat{d}j_L(\widehat{d})} \geq \widehat{L}(\widehat{d})^H_\alpha \Big| \widehat{d}=d^*, p^{1\widehat{d}j_L(\widehat{d})} \geq \widehat{L}(\widehat{d})^P_\beta\Big) = \frac{1-\alpha}{1-\beta}$$

for all $d^* \in \{0,1\}$. Then, note that

$$\begin{aligned}\mathbb{P}\Big(L(\widehat{d}) \geq \widehat{L}(\widehat{d})^H_\alpha\Big) &\geq \mathbb{P}\Big(L(\widehat{d}) \geq \widehat{L}(\widehat{d})^H_\alpha \Big| p^{1\widehat{d}j_L(\widehat{d})} \geq \widehat{L}(\widehat{d})^P_\beta\Big) \cdot \mathbb{P}\Big(p^{1\widehat{d}j_L(\widehat{d})} \geq \widehat{L}(\widehat{d})^P_\beta\Big) \\ &\geq \frac{1-\alpha}{1-\beta}(1-\beta) = 1-\alpha\end{aligned}$$

by the law of total probability.

By similar reasoning to that used for the conditional CIs in Section 3.2 above, $\widehat{L}(\widehat{d})^H_\alpha$ is not very conservative as a lower confidence bound for $L(\widehat{d})$. The researcher's choice of $\beta \in (0,\alpha)$ trades off the performance of $\widehat{L}(\widehat{d})^H_\alpha$ across scenarios for which $\mathbb{P}(\widehat{d}=d^*)$ is large and small with a small β corresponding to better performance when $\mathbb{P}(\widehat{d}=d^*)$ is large. See McCloskey (2024) for an in-depth discussion of these tradeoffs. We recommend $\beta = \alpha/10$.

Finally, analogous constructions to those above produce unconditional projection and hybrid confidence upper bounds $\widehat{U}(\widehat{d})^P_{1-\alpha}$ and $\widehat{U}(\widehat{d})^H_{1-\alpha}$ that can then be combined with the lower bounds to form CIs $[\widehat{L}(d)^P_{\alpha/2}, \widehat{U}(d)^P_{1-\alpha/2}]$ and $[\widehat{L}(d)^H_{\alpha/2}, \widehat{U}(d)^H_{1-\alpha/2}]$ for $[L(d),U(d)]$ that are unconditionally valid in large samples by the same arguments as those used in (3.2) above.

4 General Inference Framework

We now introduce the general inference framework that we propose, nesting the examples in the previous sections as special cases. After introducing the general framework, we describe several additional example applications that fall within this framework.

In the setting of this paper, a researcher's interest in $W(d)$ arises when the policymaker selects $d \in \mathcal{D}$ in a data-driven manner. Suppose the policymaker selects a rule given data summarized in the empirical distribution $\widehat{p}$. We first define the type of policymaker's selection rules we focus on.

Definition 4.1 (Polyhedral Selection)**.** *Let a random set $\mathcal{D}(\widehat{p}) \subseteq \mathcal{D} \equiv \{d^0,...,d^K\}$ be a correspondence of $\widehat{p}$, an estimator of a finite-dimensional parameter p. For all $d \in \mathcal{D}$, the event $\{d \in \mathcal{D}(\widehat{p})\}$ is a weak polyhedral selection in $\widehat{p}$ if*

$$\{d \in \mathcal{D}(\widehat{p})\} = \bigcup_{j=1}^{J} \{A_j \widehat{p} \leq c_j\} \tag{4.1}$$

almost surely (a.s.) for some matrix A_j and vector c_j for $j \in \{1,...,J\}$; the event is a polyhedral selection in $\widehat{p}$ if (4.1) holds with $J = 1$.

Weak polyhedral selection events are common in the selective inference literature. See, e.g., Lee et al. (2016) for an example in the context of LASSO-based model selection and Andrews et al. (2021) for an example in the context of inference after estimating a regression threshold or structural break.

The (estimated) policy set $\mathcal{D}(\widehat{p})$ could correspond to an estimated set of optimal treatments or policies or other data-dependent index sets of interest. Continuing with the motivating example in Section 2, Scenario (a) corresponds to a policymaker selecting $d \in \mathcal{D}(\widehat{p}) = \{\widehat{d}\}$ (i.e., a singleton selection $d = \widehat{d}$) where $\widehat{d}$ can be the minimax or maximin rule for $W(d)$. Scenario (b) corresponds to $\mathcal{D}(\widehat{p}) = \widehat{\mathcal{D}}$ where $\widehat{\mathcal{D}}$ is the estimated set for $\operatorname{argmax}_{d \in \mathcal{D}} W(d)$. In these examples, $\mathcal{D}(\widehat{p})$ is typically constructed via the estimated (sharp) lower and upper bounds on welfare $W(d)$:[5]

[5] In the first two examples, we implicitly assume that some tie-breaking procedure is applied in the maximum and minimum (respectively) to produce the final rule.

Example 1 (Maximin Welfare Rule). *Let* $\mathcal{D}(\widehat{p})=\{\widehat{d}\}$ *a.s. with*

$$\widehat{d}=\operatorname*{argmax}_{d\in\mathcal{D}}\widehat{L}(d),$$

where $\widehat{L}(d)$ *is the estimated lower bound on* $W(d)$.

Example 2 (Hurwicz Welfare Rule). *Let* $\mathcal{D}(\widehat{p})=\{\widehat{d}\}$ *a.s. with*

$$\widehat{d}=\operatorname*{argmax}_{d\in\mathcal{D}}\{w_L\widehat{L}(d)+w_U\widehat{U}(d)\},$$

for some fixed known weights $w_L,w_U\geq 0$ *where* $\widehat{L}(d)$ *and* $\widehat{U}(d)$ *are the estimated lower and upper bounds on* $W(d)$. *This includes Example 1* ($w_U=0$) *and the maximax rule* ($w_L=0$) *as special cases.*

Example 3 (Minimax Regret Rule). *Let* $\mathcal{D}(\widehat{p})=\{\widehat{d}\}$ *a.s. with* $\widehat{d}=\operatorname{argmin}_{d\in\mathcal{D}}\widehat{R}(d)$ *where* $\widehat{R}(d)\equiv\max_{d'\in\mathcal{D}}\widehat{U}(d',d)$ *and* $\widehat{U}(d',d)$ *estimates the upper bound on* $W(d')-W(d)$, *with* $\widehat{U}(d,d)=0$. *Suppose, for every* $d'\neq d$, $U(d')$ *and* $L(d)$ *are attainable in the DGP.*[6] *Then* $U(d',d)=U(d')-L(d)$, *and so*

$$\widehat{d}=\operatorname{argmin}_{d\in\mathcal{D}}\max\left\{0,\max_{d'\neq d}\left[\widehat{U}(d')-\widehat{L}(d)\right]\right\}.$$

Example 4 (Set of the Undominated). *Let* $\mathcal{D}(\widehat{p})=\widehat{\mathcal{D}}$ *a.s. with*[7]

$$\begin{aligned}\widehat{\mathcal{D}}&=\left\{d\in\{d^0,\ldots,d^K\}:\nexists d'\in\{d^0,\ldots,d^K\}\ \text{s.th. }\widehat{U}(d)<\widehat{L}(d')\right\}\\&=\left\{d\in\{d^0,\ldots,d^K\}:\widehat{U}(d)\geq\max_{d'\in\{d^0,\ldots,d^K\}}\widehat{L}(d')\right\}.\end{aligned}$$

Example 5 (Predetermined Policy). *Let* $\mathcal{D}(\widehat{p})=\{d^*\}$ *where* $d^*\in\mathcal{D}$ *is a policy chosen a priori without consulting data.*

Examples 1–2 concern maximizing the weighted sum of (estimated) upper and lower bounds on the welfare (Hurwicz, 1951), with maximizing the lower bound as the most conservative special

[6] Formally, for every $d',d\in\mathcal{D}$ with $d'\neq d$, there exists a DGP consistent with p under which $W(d')=U(d')$ and $W(d)=L(d)$.

[7] This estimated set and $\widehat{d}$ in Example 1 are related as follows: $\widehat{\mathcal{D}}=\left\{d\in\mathcal{D}:\widehat{U}(d)\geq\widehat{L}(\widehat{d})\right\}$.

case from the policymaker's perspective. Example 3 considers minimizing worst-case regret, which is considered to be less pessimistic than maximizing the worst-case welfare of Example 1 (Savage, 1951). The estimated set $\widehat{\mathcal{D}}$ in Example 4 can be viewed as the sample counterpart of the identified set $\mathcal{D}^*$ of $d^* \in \operatorname{argmax}_{d\in\mathcal{D}} W(d)$. Examples 1–3 are established rules in the statistical decision theory (Manski, 2004, 2021, 2023) and Example 4 can be viewed as a reasonable set-valued target (Han, 2024). The inherent feature of these examples is that the welfare $W(d)$ is partially identified. This *model ambiguity* in choosing policies can originate from endogeneity (Kallus and Zhou, 2021; Pu and Zhang, 2021; D'Adamo, 2021; Yata, 2021), imperfect internal and external validity (Stoye, 2012; Ishihara and Kitagawa, 2021; Montiel Olea et al., 2026), or the welfare being a function of the joint distribution of potential outcomes (Cui and Han, 2026). Moreover, many decision problems involve a finite menu (Manski, 2004; Manski and Tetenov, 2016; Christensen et al., 2023; Han, 2024) and a finite menu is practically relevant even when a hypothetical infinite menu is available because a policymaker may only find it feasible to choose the policy within a finite set (e.g., finite set of programs, finite types of characteristics).

Under the construction of $\mathcal{D}(\widehat{p})$ reflected in Examples 1–4, the following assumption guarantees that the selection events $d\in\mathcal{D}(\widehat{p})$ in such examples are (weak) polyhedral selections. This assumption posits that our parameter of inferential interest $W(d)$ for some $d\in\mathcal{D}$ belongs to an identified set $[L(d),U(d)]$ taking a particular interval form that is common to many applications of interest.

Assumption 4.1. *For an unknown finite-dimensional parameter p,*

1. $L(d) \equiv \max_{j\in\{1,\ldots,J_L\}}\{\tilde{\ell}_{d,j} + \ell_{d,j}p\} \leq W(d)$ *for some fixed and known* J_L, $\tilde{\ell}_{d,1},\ldots,\tilde{\ell}_{d,J_L}$ *and nonzero row vectors* $\ell_{d,1},\ldots,\ell_{d,J_L}$ *such that* $\ell_{d,j}\neq\ell_{d,j'}$ *for* $j\neq j'$.

2. $U(d) \equiv \min_{j\in\{1,\ldots,J_U\}}\{\tilde{u}_{d,j} + u_{d,j}p\} \geq W(d)$ *for some fixed and known* J_U, $\tilde{u}_{d,1},\ldots,\tilde{u}_{d,J_U}$ *and nonzero row vectors* $u_{d,1},\ldots,u_{d,J_U}$ *such that* $u_{d,j}\neq u_{d,j'}$ *for* $j\neq j'$.

The lower and upper endpoints of identified sets for the welfare, average counterfactual outcome, or ATE typically take the form of $L(d)$ and $U(d)$, especially when (sequences of) outcomes, treatments and instruments are discrete.[8] See Section 5 and Appendix A for such examples.

[8]Our general framework applies to settings with dynamic outcomes, treatments and instruments observed over

Assume the estimated lower and upper bounds on $W(d)$ are defined as the sample counterpart of Assumption 4.1:

$$\widehat{L}(d) \equiv \max_{j \in \{1,\ldots,J_L\}} \{\tilde{\ell}_{d,j} + \ell_{d,j}\widehat{p}\} \tag{4.2}$$

$$\widehat{U}(d) \equiv \min_{j \in \{1,\ldots,J_U\}} \{\tilde{u}_{d,j} + u_{d,j}\widehat{p}\}. \tag{4.3}$$

Proposition 4.1. *Given (4.2)–(4.3), the selection events in Examples 1–3 are weak polyhedral selections and the selection event in Example 4 is a polyhedral selection.*

In developing our inference procedures, it is useful to consider an additional event that a particular affine component is active in (4.2)–(4.3). Define

$$\widehat{j}_L(d) \equiv \operatorname*{argmax}_{j \in \{1,\ldots,J_L\}} \{\tilde{\ell}_{d,j} + \ell_{d,j}\widehat{p}\}, \quad \widehat{j}_U(d) \equiv \operatorname{argmin}_{j \in \{1,\ldots,J_U\}} \{\tilde{u}_{d,j} + u_{d,j}\widehat{p}\}, \tag{4.4}$$

which are the indices at which the estimated lower and upper bounds are realized. Then, the estimated bounds can be expressed as $\widehat{L}(d) = \tilde{\ell}_{d,\widehat{j}_L(d)} + \ell_{d,\widehat{j}_L(d)}\widehat{p}$ and $\widehat{U}(d) = \tilde{u}_{d,\widehat{j}_U(d)} + u_{d,\widehat{j}_U(d)}\widehat{p}$. We work under the high-level assumption that the following joint event can be written as a polyhedron in $\widehat{p}$: (i) an option index d is in the policy set of interest $\mathcal{D}(\widehat{p})$, (ii) the estimated bounds on $W(d)$ are realized at a given value (i.e., $\widehat{j}_L(d) = j_L^*$ and $\widehat{j}_U(d) = j_U^*$) and (iii) (optionally) an additional random vector is realized at any given value (i.e., $\widehat{\gamma}_L(d) = \gamma_L^*$ and $\widehat{\gamma}_U(d) = \gamma_U^*$). Although the event (i) is typically a weak polyhedral selection, the joint event (i)–(iii) is typically a polyhedral selection.[9]

Recall our parameter of interest $W(d)$ for some $d \in \mathcal{D}$ and the policy set $\mathcal{D}(\widehat{p})$ in Definition 4.1.

Assumption 4.2. *1. The joint event $\{d \in \mathcal{D}(\widehat{p}),\ \widehat{j}_L(d) = j_L^*$ and $\widehat{\gamma}_L(d) = \gamma_L^*\}$ is equivalent to $\{A^L\widehat{p} \le c^L\}$ a.s. for some matrix A^L and vector c^L that depend on $d \in \mathcal{D}$, $j_L^* \in \{1,\ldots,J_L\}$, and γ_L^* in the support of $\widehat{\gamma}_L(d)$.*

time. This is what is meant here by "sequences".

[9] In conjunction with Assumption 4.1, the property (ii) is also useful for establishing coverage separately for the true bounds $L(d)$ and $U(d)$ via a dominance argument: by forming a lower (upper) confidence bound for $\tilde{\ell}_{d,j_L^*} + \ell_{d,j_L^*}p$ ($\tilde{u}_{d,j_U^*} + u_{d,j_U^*}p$), we automatically obtain a lower (upper) confidence bound for $L(d) \ge \tilde{\ell}_{d,j_L^*} + \ell_{d,j_L^*}p$ ($U(d) \le \tilde{u}_{d,j_U^*} + u_{d,j_U^*}p$). See, e.g., the argument following (3.4).

2. *The joint event* $\{d \in \mathcal{D}(\widehat{p}),\ \widehat{j}_U(d) = j_U^*$ *and* $\widehat{\gamma}_U(d) = \gamma_U^*\}$ *is equivalent to* $\{A^U \widehat{p} \leq c^U\}$ *a.s. for some matrix* A^U *and vector* c^U *that depend on* $d \in \mathcal{D}$, $j_U^* \in \{1,...,J_U\}$, *and* γ_U^* *in the support of* $\widehat{\gamma}_U(d)$.

Although not immediately obvious, this assumption holds in a variety of settings; see the examples below. In particular, the joint event can be shown to be polyhedral in $\widehat{p}$ even if the marginal event $\{d \in \mathcal{D}(\widehat{p})\}$ is only weakly polyhedral; see Proposition 4.2 below. The auxiliary vectors $\widehat{\gamma}_L(d)$ and $\widehat{\gamma}_U(d)$ are a mechanical bookkeeping device. Conditional on $\widehat{j}_M(d) = j_M^*$ $(M = L,U)$, they record any other random variables needed to express the event $\{d \in \mathcal{D}(\widehat{p})\}$ as a single polyhedron in $\widehat{p}$ after their realizations are conditioned upon.[10]

For example, selection rules based on estimated intersection bounds are piecewise affine (Examples 1–3): before the relevant active indices $\widehat{j}_M(d)$ and auxiliary vectors $\widehat{\gamma}_M(d)$ for $M \in \{L,U\}$ are fixed, the selection event can be a union of polyhedra; after they are fixed, it is described by linear inequalities. When $\{d \in \mathcal{D}(\widehat{p}), \widehat{j}_M(d) = j_M^*\}$ is already polyhedral, no additional label is needed and $\widehat{\gamma}_M(d)$ can be vacuously set to a constant.[11] A leading example of these cases is the selection of empirically non-dominated options (Example 4) for which $\mathcal{D}(\widehat{p}) = \widehat{\mathcal{D}}$. See the proof of Proposition 4.2 below for details. We also note that Assumption 4.2 is compatible with Example 5, where $\mathcal{D}(\widehat{p}) = \{d^*\}$: absent data-dependent selection, the researcher is interested in forming a CI for an identified interval $[L(d^*),U(d^*)]$ chosen by the researcher a priori. In this case, $\widehat{\gamma}_M(d^*)$ can be vacuously set to a fixed constant and A^M and c^M no longer depend on γ. Indeed, we examine an example of this special case when conducting a finite-sample power comparison in Section C below.

The conditional CIs below do not require $\mathcal{D}(\widehat{p})$ to be a singleton. Under Assumptions 4.1–4.2 (and other regularity conditions introduced later), they provide correct (asymptotic) coverage for $W(d)$ for each d conditional on the event $d \in \mathcal{D}(\widehat{p})$. By contrast, unconditional inference on a

[10] Conditioning on the realizations of these random variables produces computational tractability by allowing us to express the asymptotic conditional distributions of $\widehat{L}(d)$ and $\widehat{U}(d)$ as normal distributions truncated to a single interval. This computational tractability may be at the expense of some power loss relative to not conditioning on $\widehat{j}_M(d)$ and/or $\widehat{\gamma}_M(d)$, a common choice in the literature. See, e.g., Lee et al. (2016), who analogously condition on the additional random variables that characterize the signs of estimated regression coefficients after selecting a model via LASSO.

[11] Because additional conditioning may reduce power, $\widehat{\gamma}_M(d)$ should contain only the indices required for the polyhedral representation.

data-dependent parameter requires the final target itself to be a well-defined random variable. We therefore impose the next assumption only for the projection and hybrid procedures that target $W(\widehat{d})$ unconditionally. The assumption concerns the final sample recommendation and places no restriction on whether the population identified set of best-performing policies $\mathcal{D}^*$ is a singleton.

Assumption 4.3. *For the unconditional inference problem, the researcher specifies a selection rule whose final output is a singleton $\mathcal{D}(\widehat{p})=\{\widehat{d}\}$ a.s., where $\widehat{d}$ takes values in $\mathcal{D}\equiv\{d^0,...,d^K\}$.*

If an initial procedure produces an estimated set $\widehat{\mathcal{D}}$ and a policymaker subsequently chooses one element using a rule r, the relevant final recommendation is $\widehat{d}=r(\widehat{\mathcal{D}},\widehat{p},\eta)$, where η may be an independent randomization device.[12] Assumption 4.3 applies to this final rule, not to the intermediate set.

Under Assumptions 4.1–4.3 (and other regularity conditions), the unconditional CIs provide correct (asymptotic) coverage for $W(\widehat{d})$. Recall Examples 1–3 for the type of selection rules considered in Assumption 4.3. On the other hand, only Assumptions 4.1 and 4.2 (and other regularity conditions) are required for the conditional CIs to provide correct coverage for $W(d)$ conditional on $d\in\mathcal{D}(\widehat{p})$, such as $\mathcal{D}(\widehat{p})=\widehat{\mathcal{D}}$ in Example 4.

Taken together, Assumptions 4.2 and 4.3 hold naturally when $\widehat{d}$ is selected by uniquely maximizing a linear combination of $(L(d),U(d))$ (Examples 1–2) or minimizing the regret (Example 3) and the additional conditioning vectors $\widehat{\gamma}_L(d)$ and $\widehat{\gamma}_U(d)$ are defined appropriately. Assumption 4.2 holds when the policy is selected from the estimated set $\widehat{\mathcal{D}}$ (Example 4). The following proposition formalizes these statements.

Proposition 4.2. *Suppose (4.2)–(4.3) hold. When $\mathcal{D}(\widehat{p})=\{\widehat{d}\}$ a.s. (i.e., Assumption 4.3), Assumption 4.2 holds in*

1. *Example 1 for $\widehat{\gamma}_L(d)$ constant and $\widehat{\gamma}_U(d)=\widehat{j}_L(d)$;*
2. *Example 2 with $w_L=0$ for $\widehat{\gamma}_L(d)=\widehat{\gamma}_U(d)=(\widehat{j}_U(d^0),...,\widehat{j}_U(d^K))'$;*
3. *Example 2 with $w_L,w_U>0$ for $\widehat{\gamma}_L(d)=\widehat{\gamma}_U(d)=(\widehat{j}_L(d^0),...,\widehat{j}_L(d^K),\widehat{j}_U(d^0),...,\widehat{j}_U(d^K))'$;*

[12] A deterministic tie-breaking convention is enough when ties otherwise have probability zero. A randomized policy rule can also be represented by augmenting the data with the realized randomization draw, provided that the resulting selection event satisfies the polyhedral structure needed for Assumption 4.2.

4. Example 3 for $\widehat{\gamma}_L(d)=\widehat{\gamma}_U(d)=\big(\widehat{j}_L(a),\widehat{j}_U(a),\widehat{b}(a)\big)_{a\in\mathcal{D}}$*, where*

$$\widehat{b}(a)\equiv\underset{b\in\mathcal{D}}{\operatorname{argmax}}\,\widehat{U}(b,a),\qquad \widehat{U}(b,a)\equiv\begin{cases}0, & b=a,\\ \widehat{U}(b)-\widehat{L}(a), & b\neq a.\end{cases}$$

When $\mathcal{D}(\widehat{p})=\widehat{\mathcal{D}}$ *(Example 4) or* $\mathcal{D}(\widehat{p})=\{d^*\}$ *(Example 5), Assumption 4.2 holds with* $\widehat{\gamma}_L(d)$ *and* $\widehat{\gamma}_U(d)$ *constant.*

This proposition establishes that the joint event stated in Assumption 4.2 is polyhedral in all the leading examples. Expressions for A^M and c^M $(M=L,U)$, which are needed to construct the CIs we introduce below, in the settings of Proposition 4.2 are available for reference in its proof in Appendix E. As this proposition makes clear, the additional conditioning vectors needed for Assumption 4.2 to hold depend upon the particular form of selection rule used by the researcher. For example, if $\widehat{d}$ maximizes $\widehat{L}(d)$, then $\{\widehat{d}=d,\widehat{j}_L(d)=j\}$ is polyhedral and $\widehat{\gamma}_L(d)$ can be constant. To conduct upper-endpoint inference for the same selected policy, however, conditioning on $\widehat{j}_U(d)$ does not identify the lower-bound component that generated selection, so we additionally set $\widehat{\gamma}_U(d)=\widehat{j}_L(d)$. If $\widehat{d}$ instead maximizes $\widehat{U}(d)$, comparisons across policies require the active upper-bound index for each competitor, so the vector $(\widehat{j}_U(d^0),...,\widehat{j}_U(d^K))'$ is included in the auxiliary conditioning information. Weighted combinations of lower and upper bounds generally require the corresponding vectors of both types of active indices.

5 Examples

In this section, we show that Assumptions 4.1 and 4.2 are satisfied in the motivating example in Section 2. We then briefly mention other examples, while deferring the verification of the assumptions for them to Appendix A.

5.1 Dynamic Treatment Regime (Example in Section 2)

Continuing with the motivating example in Section 2, consider constructing a CI for the welfare $W(\boldsymbol{\delta})\equiv\mathbb{P}[Y_2(\boldsymbol{\delta}(\cdot))=1]$ defined in (2.1) that is evaluated at a policy chosen from $\mathcal{D}$, which is a

finite set (Table 1). For the selection rule, we focus on implementing the maximin welfare rule (Example 1) or the Hurwicz welfare rule (Example 2). In both cases, we need to derive bounds on $W(\boldsymbol{\delta})$ based on the distribution of data, which is expressed as the vector p of the form

$$p \equiv \{\mathbb{P}[D_1=d_1,Y_1=y_1,D_2=d_2,Y_2=y_2|Z_1=z_1,Z_2=z_2]\}_{(d_1,y_1,d_2,y_2,z_1,z_2)}.$$

First, consider bounds on $W_y(d) \equiv \mathbb{P}[Y(d)=y]$ for $d \equiv (d_1,d_2)$, which are $L_y(d) \equiv \max_z L_y(d;z)$ and $U_y(d) \equiv \min_z U_y(d;z)$ where

$$\begin{aligned}
L_y(d;z) &\equiv \mathbb{P}[Y=y,D=d|Z=z], && (5.1)\\
U_y(d;z) &\equiv \mathbb{P}[Y=y,D=d|Z=z]+\mathbb{P}[Y_1=y_1,D_1=d_1,D_2=1-d_2|Z=z]\\
&\quad +\mathbb{P}[D_1=1-d_1,D_2=d_2|Z=z]+\mathbb{P}[D_1=1-d_1,D_2=1-d_2|Z=z]\\
&= \mathbb{P}[Y=y,D=d|Z=z]+\mathbb{P}[Y_1=y_1,D_1=d_1,D_2=1-d_2|Z=z]+\mathbb{P}[D_1=1-d_1|Z=z]. && (5.2)
\end{aligned}$$

Using these bounds, we can calculate bounds on $W(\boldsymbol{\delta})$, which are

$$\begin{aligned}
L(\boldsymbol{\delta}) &\equiv \max_z L_{(1,1)}(\delta_1,\delta_2(1,\delta_1);z)+\max_z L_{(0,1)}(\delta_1,\delta_2(0,\delta_1);z), && (5.3)\\
U(\boldsymbol{\delta}) &\equiv \min_z U_{(1,1)}(\delta_1,\delta_2(1,\delta_1);z)+\min_z U_{(0,1)}(\delta_1,\delta_2(0,\delta_1);z). && (5.4)
\end{aligned}$$

For example, for the fourth regime in Table 1,

$$\begin{aligned}
\mathbb{P}[Y_2(\boldsymbol{\delta}_{(4)}(\cdot))=1] &= \sum_{y_1\in\{0,1\}} \mathbb{P}[Y_1(\delta_1)=y_1,Y_2(\delta_1,\delta_2(y_1,\delta_1))=1]\\
&= \mathbb{P}[Y_1(1)=1,Y_2(1,\delta_2(1,1))=1]+\mathbb{P}[Y_1(1)=0,Y_2(1,\delta_2(0,1))=1]\\
&= \mathbb{P}[Y_1(1)=1,Y_2(1,1)=1]+\mathbb{P}[Y_1(1)=0,Y_2(1,0)=1]
\end{aligned}$$

is bounded by

$$L(\boldsymbol{\delta}_{(4)}) \equiv \max_z L_{(1,1)}(1,1;z)+\max_z L_{(0,1)}(1,0;z),$$

$$U(\boldsymbol{\delta}_{(4)}) \equiv \min_z U_{(1,1)}(1,1;z) + \min_z U_{(0,1)}(1,0;z).$$

Since $\max_z L(z) + \max_z \tilde{L}(z) = \max_{z,\tilde{z}} \left\{ L(z) + \tilde{L}(\tilde{z}) \right\}$ for any functions L and $\tilde{L}$, we can express (5.3) as

$$\begin{aligned} L(\boldsymbol{\delta}) &= \max_{z,\tilde{z}} \left\{ L_{(1,1)}(\delta_1, \delta_2(1,\delta_1); z) + L_{(0,1)}(\delta_1, \delta_2(0,\delta_1); \tilde{z}) \right\} \\ &\equiv \max_{z,\tilde{z}} L(\boldsymbol{\delta}; z, \tilde{z}), \end{aligned}$$

and, analogously, (5.4) as $U(\boldsymbol{\delta}) \equiv \min_{z,\tilde{z}} U(\boldsymbol{\delta}; z, \tilde{z})$. Therefore, given (5.1)–(5.2), the bounds satisfy Assumption 4.1. Now, consider choosing a single optimal regime that maximizes $L(\boldsymbol{\delta})$. Then, for example, we can show that the following event can be characterized as a polyhedron in the space of p

$$\{\boldsymbol{\delta}_{(4)} = \operatorname*{argmax}_{\boldsymbol{\delta}} L(\boldsymbol{\delta}) \text{ and } L(\boldsymbol{\delta}_{(4)}; z^*, \tilde{z}^*) \geq L(\boldsymbol{\delta}_{(4)}; z, \tilde{z}) \text{ for all } z, \tilde{z}\},$$

where $(z^*, \tilde{z}^*) = \arg\max_{z,\tilde{z}} L(\boldsymbol{\delta}_{(4)}; z, \tilde{z})$. Note that this event is equivalent to $\{L(\boldsymbol{\delta}_{(4)}; z^*, \tilde{z}^*) \geq L(\boldsymbol{\delta}; z, \tilde{z})$ for all $\boldsymbol{\delta}$ and $z, \tilde{z}\}$ or equivalently,

$$\{L(\boldsymbol{\delta}; z, \tilde{z}) - L(\boldsymbol{\delta}_{(4)}; z^*, \tilde{z}^*) \leq 0 \text{ for all } \boldsymbol{\delta} \text{ and } z, \tilde{z}\}. \tag{5.5}$$

Note that each $L(\boldsymbol{\delta}; z, \tilde{z})$ is a linear combination of the elements in p as shown in (5.1) and (5.2). More formally, we can show that (5.5) is equivalent to $A_L p \leq 0$ for some A_L. Note that

$$\begin{aligned} L(\boldsymbol{\delta}; z, \tilde{z}) &= L_{(1,1)}(\delta_1, \delta_2(1,\delta_1); z) + L_{(0,1)}(\delta_1, \delta_2(0,\delta_1); \tilde{z}) \\ &= A^1(\boldsymbol{\delta}; z) p + A^0(\boldsymbol{\delta}; \tilde{z}) p \end{aligned}$$

for some row vectors A^1 and A^0 because, for $\boldsymbol{\delta} = \boldsymbol{\delta}_{(4)}$,

$$L_{(1,1)}(\delta_1, \delta_2(1,\delta_1); z) = L_{(1,1)}(1,1;z) = \mathbb{P}[Y = (1,1), D = (1,1) | Z = z]$$

$$= \mathbb{P}[D_1 = 1, Y_1 = 1, D_2 = 1, Y_2 = 1 | Z = z],$$

$$L_{(0,1)}(\delta_1, \delta_2(0,\delta_1); \tilde{z}) = L_{(0,1)}(1,0;\tilde{z}) = \mathbb{P}[Y = (0,1), D = (1,0) | Z = \tilde{z}]$$

$$= \mathbb{P}[D_1 = 1, Y_1 = 0, D_2 = 0, Y_2 = 1 | Z = \tilde{z}].$$

An analogous argument can be applied to $U(\boldsymbol{\delta}; z, \tilde{z})$. The sample counterpart of this characterization implies that Assumption 4.2 holds in this setting. Also, this characterization facilitates the CI calculations in the following section.

5.2 Other examples

In Appendix A, we list other examples that fit into our inference framework. These examples can be broadly categorized as (i) Manski bounds with a continuous outcome; (ii) bounds for dynamic treatment effects; (iii) bounds derived from linear programming; (iv) optimal treatment assignment with partially identified ATE; and (v) empirical welfare maximization with observational data. In these examples, parameters of interest are chosen in data-driven manners that correspond to weak polyhedral selection. In particular, we show in the appendix how Assumptions 4.1 and 4.2 hold in these examples.

6 Confidence Interval Construction and Validity

We now generalize the CI construction heuristically described in Section 3 to apply in the general framework of Section 4, covering all example applications discussed above and in Appendix A. After introducing a few additional mild assumptions on the DGP, we construct conditional CIs and then move to unconditional CIs.

6.1 Additional Assumptions on the DGP

We suppose that the sample of data with n observations is drawn from some unknown distribution $\mathbb{P} \in \mathcal{P}_n$. As an estimator for p, we assume that $\widehat{p}$ is uniformly asymptotically normal under $\mathbb{P} \in \mathcal{P}_n$.

Assumption 6.1. *For the class of Lipschitz functions that are bounded in absolute value by one and have Lipschitz constant bounded by one, BL_1, there exist functions $p(\mathbb{P})$ and $\Sigma(\mathbb{P})$ such that*

for $\xi_{\mathbb{P}} \sim \mathcal{N}(0, \Sigma(\mathbb{P}))$ *with*

$$\lim_{n\to\infty} \sup_{\mathbb{P}\in\mathcal{P}_n} \sup_{f\in BL_1} \left| E_{\mathbb{P}}\left[f\left(\sqrt{n}(\widehat{p} - p(\mathbb{P}))\right)\right] - E_{\mathbb{P}}[f(\xi_{\mathbb{P}})] \right| = 0.$$

The notation of this assumption makes explicit that the parameter p and the asymptotic variance Σ depend upon the unknown distribution of the data $\mathbb{P}$. It holds naturally for standard estimators $\widehat{p}$ under random sampling or weak dependence in the presence of bounds on the moments and dependence of the underlying data. This assumption holds in all the examples considered in the previous section and Appendix A: with appropriate random samples, $\widehat{p}$ are sample means. For example, Assumption 6.1 is satisfied in the example of dynamic treatment regime when the data $\{Y_i, D_i, Z_i\}_{i=1}^n$ form a random sample, since the entries of $\widehat{p}$ are sample means.

Next, we assume that the asymptotic variance of $\widehat{p}$ can be uniformly consistently estimated by an estimator $\widehat{\Sigma}$.

Assumption 6.2. *For all* $\varepsilon > 0$*, the estimator* $\widehat{\Sigma}$ *satisfies*

$$\lim_{n\to\infty} \sup_{\mathbb{P}\in\mathcal{P}_n} \mathbb{P}\left(\left\| \widehat{\Sigma} - \Sigma(\mathbb{P}) \right\| > \varepsilon \right) = 0.$$

This assumption is again naturally satisfied when using a standard sample analog estimator of Σ under random sampling or weak dependence in the presence of moment and dependence bounds.

When a raw covariance estimator is not positive semidefinite, we replace it by its projection onto the cone of positive-semidefinite matrices and continue to denote the resulting estimator by $\widehat{\Sigma}$. This replacement does not affect Assumption 6.2: projection onto the positive-semidefinite cone is nonexpansive under the Frobenius norm, and all matrix norms are equivalent in the fixed-dimensional setting considered here.

We make a final assumption imposing directional nondegeneracy on the covariance matrix Σ uniformly across $\mathcal{P}_n$.

Assumption 6.3. *For some finite* $\bar{\lambda} > 0$ *and every* $\mathbb{P} \in \mathcal{P}_n$*:*

1. $\Sigma(\mathbb{P})$ *is positive semidefinite and* $\lambda_{\max}\{\Sigma(\mathbb{P})\} \leq \bar{\lambda}$.

2. *For every* $d \in \{d^0,...,d^K\}$, $\ell_{d,j}\Sigma(\mathbb{P})\ell'_{d,j} \geq \|\ell_{d,j}\|^2/\bar{\lambda}$ *for* $j=1,...,J_L$ *and* $u_{d,j}\Sigma(\mathbb{P})u'_{d,j} \geq \|u_{d,j}\|^2/\bar{\lambda}$ *for* $j=1,...,J_U$.

3. *For* $M=L,U$, *every* $d \in \{d^0,...,d^K\}$, *every* $j \in \{1,...,J_M\}$, *every* γ *in the support of* $\widehat{\gamma}_M(d)$, *and every nonzero row* v *of* $A^M(d,j,\gamma)$, $v\Sigma(\mathbb{P})v' \geq \|v\|^2/\bar{\lambda}$.

Part 2 ensures that every scalar affine statistic used to estimate a lower or upper endpoint has variance uniformly bounded away from zero. Part 3 rules out degeneracy of comparisons that determine the binding component or a potentially active polyhedral selection inequality. Identically zero or redundant rows should be deleted from $A^M(d,j,\gamma)$ before applying the assumption. No lower bound is imposed on the eigenvalues of $\Sigma(\mathbb{P})$ in directions orthogonal to these collections. The assumption therefore permits the full covariance matrix to be singular, including when some reduced-form probabilities are zero or when the vector $\widehat{p}$ contains exact linear redundancies, provided that the singular directions do not determine an endpoint or a potentially active selection boundary. Moreover, this assumption immediately holds if the eigenvalues of $\Sigma(\mathbb{P})$ are uniformly bounded above and away from zero. For projection inference, only Parts 1 and 2 are imposed. The conditional and hybrid procedures additionally use Part 3 to control the behavior of the truncation limits.

Part 2 of Assumption 6.3 can be relaxed when an affine component has zero asymptotic variance but is uniformly separated from the corresponding population endpoint. Intuitively, such a component is asymptotically irrelevant for the maximum or minimum defining the endpoint because the probability that it is selected as binding converges uniformly to zero. Appendix F formalizes this relaxation and modifies the CI constructions only by omitting components whose estimated variances fall below a vanishing threshold from the studentized maxima used for projection inference and, if such a component is nevertheless selected, setting the corresponding conditional, projection, and hybrid bound equal to its affine estimate. This situation can arise when reduced-form probabilities are exactly zero or one, making some empirical probabilities and affine endpoint components degenerate, as in the Balke–Pearl calibration used in Section 7.

6.2 Conditional Confidence Intervals

We first generalize the conditional CI construction described in Section 3.2. As in Section 3.2, we are interested in forming lower and upper confidence bounds $\widehat{L}(\widehat{d})^C_\alpha$ and $\widehat{U}(\widehat{d})^C_{1-\alpha}$ that satisfy (3.2) and (3.5) for all $d \in \mathcal{D} \equiv \{d^0,...,d^K\}$ as endpoints in the formation of a conditionally valid CI. This is because the researcher's interest in inference on option d only arises when it is a member of the policy set $\mathcal{D}(\widehat{p})$.

To begin, we characterize the conditional distributions of $\widehat{L}(d)$ given the event $\{d \in \mathcal{D}(\widehat{p}), \widehat{j}_L(d) = j^*_L$ and $\widehat{\gamma}_L(d) = \gamma^*_L\}$ characterized by Assumption 4.2. These conditional distributions depend upon the nuisance parameter p. As a first step, we form sufficient statistics for p that are asymptotically independent of $\widehat{L}(d)$ given $\widehat{j}_L(d) = j^*_L$ and $\widehat{U}(d)$ given $\widehat{j}_U(d) = j^*_U$. Since $\widehat{L}(d) = \tilde{\ell}_{d,j^*_L} + \ell_{d,j^*_L}\widehat{p}$ given $\widehat{j}_L(d) = j^*_L$ by Assumption 4.1 and (4.4) (and similarly for $\widehat{U}(d)$), such sufficient statistics can be constructed as

$$\widehat{\mathcal{Z}}_L(d,j^*_L) \equiv \sqrt{n}\widehat{p} - \widehat{b}_L(d,j^*_L)\sqrt{n}(\tilde{\ell}_{d,j^*_L} + \ell_{d,j^*_L}\widehat{p})$$

and

$$\widehat{\mathcal{Z}}_U(d,j^*_U) \equiv \sqrt{n}\widehat{p} - \widehat{b}_U(d,j^*_U)\sqrt{n}(\tilde{u}_{d,j^*_U} + u_{d,j^*_U}\widehat{p})$$

with

$$\widehat{b}_L(d,j^*_L) \equiv \widehat{\Sigma}\ell'_{d,j^*_L}\left(\ell_{d,j^*_L}\widehat{\Sigma}\ell'_{d,j^*_L}\right)^{-1} \quad \text{and} \quad \widehat{b}_U(d,j^*_U) \equiv \widehat{\Sigma}u'_{d,j^*_U}\left(u_{d,j^*_U}\widehat{\Sigma}u'_{d,j^*_U}\right)^{-1},$$

by the asymptotic normality of $\widehat{p}$ and asymptotic absence of correlation between $\sqrt{n}\widehat{p}$ and $\widehat{\mathcal{Z}}_L(d,j^*_L)$ and $\widehat{\mathcal{Z}}_U(d,j^*_U)$. Next, Assumption 4.2 characterizes the conditioning events $\{d \in \mathcal{D}(\widehat{p}), \widehat{j}_L(d) = j^*_L, \widehat{\gamma}_L(d) = \gamma^*_L\}$ and $\{d \in \mathcal{D}(\widehat{p}), \widehat{j}_U(d) = j^*_U, \widehat{\gamma}_U(d) = \gamma^*_U\}$ as polyhedra in $\widehat{p}$, which can in turn be expressed as intervals for $\tilde{\ell}_{d,j^*_L} + \ell_{d,j^*_L}\widehat{p}$ and $\tilde{u}_{d,j^*_U} + u_{d,j^*_U}\widehat{p}$. Then, again since $\widehat{L}(d) = \tilde{\ell}_{d,j^*_L} + \ell_{d,j^*_L}\widehat{p}$ given $\widehat{j}_L(d) = j^*_L$ by Assumption 4.1, Lemma 1 of McCloskey (2024) implies

$$\sqrt{n}\widehat{L}(d) \Big| \left\{ d \in \mathcal{D}(\widehat{p}), \widehat{j}_L(d) = j^*_L, \widehat{\gamma}_L(d) = \gamma^*_L \right\}$$

$$\sim \sqrt{n}(\tilde{\ell}_{d,j_L^*}+\ell_{d,j_L^*}\widehat{p})\Big|\Big\{\widehat{\mathcal{V}}_L^-\Big(\widehat{\mathcal{Z}}_L(d,j_L^*),d,j_L^*,\gamma_L^*\Big)\le\sqrt{n}(\tilde{\ell}_{d,j_L^*}+\ell_{d,j_L^*}\widehat{p})\le\widehat{\mathcal{V}}_L^+\Big(\widehat{\mathcal{Z}}_L(d,j_L^*),d,j_L^*,\gamma_L^*\Big),$$
$$\widehat{\mathcal{V}}_L^0\Big(\widehat{\mathcal{Z}}_L(d,j_L^*),d,j_L^*,\gamma_L^*\Big)\ge 0\Big\} \tag{6.1}$$

and

$$\sqrt{n}\widehat{U}(d)\Big|\Big\{d\in\mathcal{D}(\widehat{p}),\widehat{j}_U(d)=j_U^*,\widehat{\gamma}_U(d)=\gamma_U^*\Big\}$$
$$\sim \sqrt{n}(\tilde{u}_{d,j_U^*}+u_{d,j_U^*}\widehat{p})\Big|\Big\{\widehat{\mathcal{V}}_U^-\Big(\widehat{\mathcal{Z}}_U(d,j_U^*),d,j_U^*,\gamma_U^*\Big)\le\sqrt{n}(\tilde{u}_{d,j_U^*}+u_{d,j_U^*}\widehat{p})\le\widehat{\mathcal{V}}_U^+\Big(\widehat{\mathcal{Z}}_U(d,j_U^*),d,j_U^*,\gamma_U^*\Big),$$
$$\widehat{\mathcal{V}}_U^0\Big(\widehat{\mathcal{Z}}_U(d,j_U^*),d,j_U^*,\gamma_U^*\Big)\ge 0\Big\}$$

with

$$\widehat{\mathcal{V}}_M^-(z,d,j,\gamma)\equiv\max_{k:(A^M(d,j,\gamma)\widehat{b}_M(d,j))_k<0}\frac{\sqrt{n}(c^M(d,j,\gamma))_k-(A^M(d,j,\gamma)z)_k}{(A^M(d,j,\gamma)\widehat{b}_M(d,j))_k},$$
$$\widehat{\mathcal{V}}_M^+(z,d,j,\gamma)\equiv\min_{k:(A^M(d,j,\gamma)\widehat{b}_M(d,j))_k>0}\frac{\sqrt{n}(c^M(d,j,\gamma))_k-(A^M(d,j,\gamma)z)_k}{(A^M(d,j,\gamma)\widehat{b}_M(d,j))_k},$$
$$\widehat{\mathcal{V}}_M^0(z,d,j,\gamma)\equiv\min_{k:(A^M(d,j,\gamma)\widehat{b}_M(d,j))_k=0}\sqrt{n}(c^M(d,j,\gamma))_k-(A^M(d,j,\gamma)z)_k$$

for $M=L,U$.

Now, under Assumptions 6.1 and 6.2, the distribution of $\sqrt{n}(\tilde{\ell}_{d,j_L^*}+\ell_{d,j_L^*}\widehat{p})$ can be approximated by a $\mathcal{N}(\sqrt{n}(\tilde{\ell}_{d,j_L^*}+\ell_{d,j_L^*}p),\ell_{d,j_L^*}\widehat{\Sigma}\ell'_{d,j_L^*})$-distributed random variable that is asymptotically independent of $\widehat{\mathcal{Z}}_L(d,j_L^*)$. Using the distributional characterization in (6.1), we can therefore use the corresponding truncated normal cumulative distribution function to produce quantile-unbiased estimators of the underlying mean $\sqrt{n}(\tilde{\ell}_{d,j_L^*}+\ell_{d,j_L^*}p)$. Let $F_{TN}(\cdot;\mu,\sigma^2|\mathcal{V}^-,\mathcal{V}^+)$ denote the truncated normal cumulative distribution function for an underlying normally-distributed random variable with mean μ and variance σ^2 that is truncated to lie between $\mathcal{V}^-$ and $\mathcal{V}^+$. For $\alpha\in(0,1)$, define $\widehat{L}(d)_\alpha^C$ to solve

$$F_{TN}\Big(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}\widehat{p});\mu,\ell_{d,\widehat{j}_L(d)}\widehat{\Sigma}\ell'_{d,\widehat{j}_L(d)}\Big|\widehat{\mathcal{V}}_L^-\Big(\widehat{\mathcal{Z}}_L(d,\widehat{j}_L(d)),d,\widehat{j}_L(d),\widehat{\gamma}_L(d)\Big),$$
$$\widehat{\mathcal{V}}_L^+\Big(\widehat{\mathcal{Z}}_L(d,\widehat{j}_L(d)),d,\widehat{j}_L(d),\widehat{\gamma}_L(d)\Big)\Big)=1-\alpha$$

in μ. Similarly, define $\widehat{U}(d)^C_\alpha$ to solve

$$F_{TN}\Big(\sqrt{n}(\tilde{u}_{d,\widehat{j}_U(d)}+u_{d,\widehat{j}_U(d)}\widehat{p});\mu,u_{d,\widehat{j}_U(d)}\widehat{\Sigma}u'_{d,\widehat{j}_U(d)}\Big|\widehat{\mathcal{V}}^-_U\Big(\widehat{\mathcal{Z}}_U(d,\widehat{j}_U(d)),d,\widehat{j}_U(d),\widehat{\gamma}_U(d)\Big),$$
$$\widehat{\mathcal{V}}^+_U\Big(\widehat{\mathcal{Z}}_U(d,\widehat{j}_U(d)),d,\widehat{j}_U(d),\widehat{\gamma}_U(d)\Big)\Big)=1-\alpha$$

in μ. Then, results in Pfanzagl (1994) imply that $\widehat{L}(d)^C_\alpha$ and $\widehat{U}(d)^C_\alpha$ are optimal α quantile-unbiased estimators of $\sqrt{n}(\tilde{\ell}_{d,j^*_L}+\ell_{d,j^*_L}p)$ and $\sqrt{n}(\tilde{u}_{d,j^*_U}+u_{d,j^*_U}p)$ asymptotically.

Finally, combine these quantile-unbiased estimators to form a conditional CI for the identified interval $[L(d),U(d)]$:

$$(n^{-1/2}\widehat{L}(d)^C_{\alpha_1},n^{-1/2}\widehat{U}(d)^C_{1-\alpha_2}). \tag{6.2}$$

We establish the conditional uniform asymptotic validity of this CI.

Theorem 6.1. *Suppose Assumptions 4.1, 4.2 and 6.1–6.3 hold. Then, for any $d\in\{d^0,...,d^K\}$ and $0<\alpha_1,\alpha_2<1/2$,*

$$\liminf_{n\to\infty}\inf_{\mathbb{P}\in\mathcal{P}_n}\left\{\left[\mathbb{P}\Big([L(d),U(d)]\subseteq\Big(n^{-1/2}\widehat{L}(d)^C_{\alpha_1},n^{-1/2}\widehat{U}(d)^C_{1-\alpha_2}\Big)\Big|d\in\mathcal{D}(\widehat{p})\Big)-(1-\alpha_1-\alpha_2)\right]\cdot\mathbb{P}(d\in\mathcal{D}(\widehat{p}))\right\}\geq 0.$$

6.3 Unconditional Confidence Intervals

In parallel with the previous subsection, we now generalize the unconditional CI constructions described in Section 3.3 to the general framework of Section 4. While conditional inference is well-defined for any fixed d after conditioning on $d\in\mathcal{D}(\widehat{p})$ even when $\mathcal{D}(\widehat{p})$ contains multiple policies, unconditional inference instead averages over the random identity of the selected object. It therefore requires the researcher to specify which policy is ultimately selected. Assumption 4.3 formalizes this requirement by defining a unique final recommendation $\mathcal{D}(\widehat{p})=\{\widehat{d}\}$.

As mentioned above, if only unconditional coverage of $[L(\widehat{d}),U(\widehat{d})]$ is desired, the conditional CI (6.2) with $d=\widehat{d}$ can be unnecessarily long. This length problem can be understood directly from the truncated-normal distribution. Conditional on the sufficient statistic and selection labels, the end-point estimates are normal truncated to an interval $[a,b]$. The truncated-normal distribution function

divides the probability mass between a and the observation by the probability of the full truncation event. When selection is rare, the observed statistic often lies in a far tail and only slightly inside a truncation boundary. Conditional on selection, crossing that boundary is expected and is therefore no longer evidence of a large underlying mean. For example, if $T \sim N(\mu,\sigma^2)$ is observed conditional on $T \geq a$ and $T = a+\delta$, inversion of the conditional distribution function implies approximately $\mu = a+\sigma^2 \log(1-c)/\delta+\delta/2$ at conditional quantile c. This value diverges to $-\infty$ as $\delta \downarrow 0$. Hence, a winner that barely exceeds the runner-up, or a policy that barely satisfies its selection inequalities, can generate a very long conditional CI. To avoid this potential extreme length problem, we instead describe two different methods—projection and hybrid methods—to form the unconditional confidence bounds that constitute the endpoints of these unconditional CIs in this general framework.

The general formation of the confidence bounds based upon projecting simultaneous confidence bounds for all possible values of $\sqrt{n}L(\widehat{d})$ and $\sqrt{n}U(\widehat{d})$ proceeds by computing $\widehat{c}_{1-\alpha,M}$, the $1-\alpha$ quantile of $\max_{i \in \{1,\ldots,(K+1)J_M\}} \widehat{\zeta}_{M,i}/\sqrt{\widehat{\Sigma}_{M,i}}$, where $\widehat{\zeta}_M \sim \mathcal{N}(0,\widehat{\Sigma}_M)$ for $M = L,U$ with $\widehat{\Sigma}_L = \ell^{mat} \widehat{\Sigma} \ell^{mat\prime}$, $\widehat{\Sigma}_U = u^{mat} \widehat{\Sigma} u^{mat\prime}$, $\ell^{mat} = (\ell'_{0,1},\ldots,\ell'_{0,J_L},\ldots,\ell'_{K,1},\ldots,\ell'_{K,J_L})'$ and $u^{mat} = (u'_{0,1},\ldots,u'_{0,J_U},\ldots,u'_{K,1},\ldots,u'_{K,J_U})'$, recalling that Υ_i denotes the ith element of the main diagonal of any square matrix Υ. Here, the maximum is taken to guarantee simultaneous coverage. The lower level $1-\alpha$ projection confidence bound for $\sqrt{n}L(\widehat{d})$ is $\widehat{L}(\widehat{d})^P_\alpha = \sqrt{n}\widehat{L}(\widehat{d}) - \widehat{c}_{1-\alpha,L}\sqrt{\widehat{\Sigma}_{L,\widehat{d}J_L+\widehat{j}_L(\widehat{d})}}$ and the upper level $1-\alpha$ projection confidence bound for $\sqrt{n}U(\widehat{d})$ is $\widehat{U}^P_{1-\alpha}(\widehat{d}) = \sqrt{n}\widehat{U}(\widehat{d}) + \widehat{c}_{1-\alpha,U}\sqrt{\widehat{\Sigma}_{U,\widehat{d}J_U+\widehat{j}_U(\widehat{d})}}$, because e.g., $\sqrt{n}L(\widehat{d})$ can take value equal to any entry of the vector $\sqrt{n}(\tilde{\ell}_{0,1},\ldots,\tilde{\ell}_{0,J_L},\ldots,\tilde{\ell}_{K,1},\ldots,\tilde{\ell}_{K,J_L})' + \sqrt{n}\ell^{mat}p$, $\sqrt{n}(\ell^{mat}\widehat{p} - \ell^{mat}p)$ is asymptotically distributed $\mathcal{N}(0,\Sigma_L)$ for $\Sigma_L = \ell^{mat}\Sigma\ell^{mat\prime}$ by Assumption 6.1 and $\widehat{\Sigma}_L$ is consistent for Σ_L by Assumption 6.2.

Combining these two confidence bounds at appropriate levels yields an unconditional CI for the identified interval $[L(\widehat{d}),U(\widehat{d})]$ of the selected $\widehat{d}$,

$$(n^{-1/2}\widehat{L}(\widehat{d})^P_{\alpha_1}, n^{-1/2}\widehat{U}(\widehat{d})^P_{1-\alpha_2}), \tag{6.3}$$

with uniformly correct asymptotic coverage, regardless of how $\widehat{d}$ is selected from the data.

Theorem 6.2. *Suppose Assumptions 4.1, 6.1, 6.2 and Parts 1–2 of Assumption 6.3 hold. Then,*

for any (random) $\widehat{d} \in \{d^0,...,d^K\}$ *and* $0<\alpha_1,\alpha_2<1/2$,

$$\liminf_{n\to\infty} \inf_{\mathbb{P}\in\mathcal{P}_n} \mathbb{P}\Big([L(\widehat{d}),U(\widehat{d})] \subseteq \Big(n^{-1/2}\widehat{L}(\widehat{d})^P_{\alpha_1}, n^{-1/2}\widehat{U}(\widehat{d})^P_{1-\alpha_2}\Big)\Big) \geq 1-\alpha_1-\alpha_2.$$

Note that the projection CI (6.3) has the benefit of correct coverage regardless of how $\widehat{d}$ is chosen from the data. In this sense, it is more robust than the other CIs we propose in this paper. On the other hand, by using the common selection structure of Assumption 4.2, we are able to produce a hybrid CI that combines the strengths of the conditional CI (6.2) and the projection CI (6.3) which, as described in Section 3.3, are shorter under complementary scenarios.

In analogy with the construction of the conditional CIs, to construct the hybrid CIs we begin by characterizing the conditional distributions of $\widehat{L}(\widehat{d})$ and $\widehat{U}(\widehat{d})$ but now adding an additional component to the conditioning events. More specifically, under Assumptions 4.2 and 4.3, by intersecting the events

$$\begin{aligned}
&\Big\{\widehat{d}=d^*,\widehat{j}_L(\widehat{d})=j_L^*,\widehat{\gamma}_L(\widehat{d})=\gamma_L^*\Big\} = \Big\{d^*\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d^*)=j_L^*,\widehat{\gamma}_L(d^*)=\gamma_L^*\Big\} \\
&= \Big\{\widehat{\mathcal{V}}_L^-\Big(\widehat{\mathcal{Z}}_L(d^*,j_L^*),d^*,j_L^*,\gamma_L^*\Big) \leq \sqrt{n}(\tilde{\ell}_{d^*,j_L^*}+\ell_{d^*,j_L^*}\widehat{p}) \leq \widehat{\mathcal{V}}_L^+\Big(\widehat{\mathcal{Z}}_L(d^*,j_L^*),d^*,j_L^*,\gamma_L^*\Big), \\
&\qquad \widehat{\mathcal{V}}_L^0\Big(\widehat{\mathcal{Z}}_L(d^*,j_L^*),d^*,j_L^*,\gamma_L^*\Big)\geq 0\Big\}
\end{aligned}$$

and

$$\begin{aligned}
&\Big\{\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p) \geq \widehat{L}^P_\beta(\widehat{d})\Big\} \\
&= \Big\{\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}\widehat{p}) \leq \sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p)+\widehat{c}_{1-\beta,L}\sqrt{\widehat{\Sigma}_{L,\widehat{d}J_L+\widehat{j}_L(\widehat{d})}}\Big\}
\end{aligned}$$

for some $0<\beta<\alpha<1$, we have

$$\begin{aligned}
&\sqrt{n}\widehat{L}(\widehat{d})\Big|\Big\{\widehat{d}=d^*,\widehat{j}_L(\widehat{d})=j_L^*,\widehat{\gamma}_L(\widehat{d})=\gamma_L^*,\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p)\geq\widehat{L}^P_\beta(\widehat{d})\Big\} \\
&\sim \sqrt{n}(\tilde{\ell}_{d^*,j_L^*}+\ell_{d^*,j_L^*}\widehat{p})\Big|\Big\{\widehat{\mathcal{V}}_L^-\Big(\widehat{\mathcal{Z}}_L(d^*,j_L^*),d^*,j_L^*,\gamma_L^*\Big)\leq\sqrt{n}(\tilde{\ell}_{d^*,j_L^*}+\ell_{d^*,j_L^*}\widehat{p}) \\
&\leq \widehat{\mathcal{V}}_L^{+,H}\Big(\widehat{\mathcal{Z}}_L(d^*,j_L^*),d^*,j_L^*,\gamma_L^*,\sqrt{n}(\tilde{\ell}_{d^*,j_L^*}+\ell_{d^*,j_L^*}p)\Big),\widehat{\mathcal{V}}_L^0\Big(\widehat{\mathcal{Z}}_L(d^*,j_L^*),d^*,j_L^*,\gamma_L^*\Big)\geq 0\Big\},
\end{aligned}$$

where

$$\widehat{\mathcal{V}}_L^{+,H}(z,d,j,\gamma,\mu) \equiv \min\left\{\widehat{\mathcal{V}}_L^{+}(z,d,j,\gamma),\mu+\widehat{c}_{1-\beta,L}\sqrt{\widehat{\Sigma}_{L,dJ_L+j}}\right\}.$$

Similarly,

$$\begin{aligned}
&\sqrt{n}\widehat{U}(\widehat{d})\Big|\left\{\widehat{d}=d^*,\widehat{j}_U(\widehat{d})=j_U^*,\widehat{\gamma}_U(\widehat{d})=\gamma_U^*,\sqrt{n}(\tilde{u}_{\widehat{d},\widehat{j}_U(\widehat{d})}+u_{\widehat{d},\widehat{j}_U(\widehat{d})}p)\le\widehat{U}_{1-\beta}^{P}(\widehat{d})\right\}\\
&\sim\sqrt{n}(\tilde{u}_{d^*,j_U^*}+u_{d^*,j_U^*}\widehat{p})\Big|\left\{\widehat{\mathcal{V}}_U^{-,H}\left(\widehat{\mathcal{Z}}_U(d^*,j_U^*),d^*,j_U^*,\gamma_U^*,\sqrt{n}(\tilde{u}_{d^*,j_U^*}+u_{d^*,j_U^*}p)\right)\right.\\
&\left.\le\sqrt{n}(\tilde{u}_{d^*,j_U^*}+u_{d^*,j_U^*}\widehat{p})\le\widehat{\mathcal{V}}_U^{+}\left(\widehat{\mathcal{Z}}_U(d^*,j_U^*),d^*,j_U^*,\gamma_U^*\right),\widehat{\mathcal{V}}_U^{0}\left(\widehat{\mathcal{Z}}_U(d^*,j_U^*),d^*,j_U^*,\gamma_U^*\right)\ge 0\right\},
\end{aligned}$$

where

$$\widehat{\mathcal{V}}_U^{-,H}(z,d,j,\gamma,\mu) \equiv \max\left\{\widehat{\mathcal{V}}_U^{-}(z,d,j,\gamma),\mu-\widehat{c}_{1-\beta,U}\sqrt{\widehat{\Sigma}_{U,dJ_U+j}}\right\}.$$

Since the distribution of $\sqrt{n}(\tilde{\ell}_{d^*,j_L^*}+\ell_{d^*,j_L^*}\widehat{p})$ can be approximated by $\mathcal{N}(\sqrt{n}(\tilde{\ell}_{d^*,j_L^*}+\ell_{d^*,j_L^*}p),\ell_{d^*,j_L^*}\Sigma\ell'_{d^*,j_L^*})$ asymptotically and $\widehat{\mathcal{Z}}_L(d^*,j_L^*)$ is asymptotically independent, we again work with the truncated normal distribution to compute a hybrid lower confidence bound for $\sqrt{n}L(\widehat{d})$: for $0<\beta<\alpha<1$, define $\widehat{L}(d)_\alpha^H$ to solve

$$\begin{aligned}
&F_{TN}\left(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}\widehat{p});\mu,\ell_{d,\widehat{j}_L(d)}\widehat{\Sigma}\ell'_{d,\widehat{j}_L(d)}\Big|\widehat{\mathcal{V}}_L^{-}\left(\widehat{\mathcal{Z}}_L(d,\widehat{j}_L(d)),d,\widehat{j}_L(d),\widehat{\gamma}_L(d)\right),\right.\\
&\left.\widehat{\mathcal{V}}_L^{+,H}\left(\widehat{\mathcal{Z}}_L(d,\widehat{j}_L(d)),d,\widehat{j}_L(d),\widehat{\gamma}_L(d),\mu\right)\right)=\frac{1-\alpha}{1-\beta}
\end{aligned}$$

in μ. Similarly, define $\widehat{U}(d)_\alpha^H$ to solve

$$\begin{aligned}
&F_{TN}\left(\sqrt{n}(\tilde{u}_{d,\widehat{j}_U(d)}+u_{d,\widehat{j}_U(d)}\widehat{p});\mu,u_{d,\widehat{j}_U(d)}\widehat{\Sigma}u'_{d,\widehat{j}_U(d)}\Big|\widehat{\mathcal{V}}_U^{-,H}\left(\widehat{\mathcal{Z}}_U(d,\widehat{j}_U(d)),d,\widehat{j}_U(d),\widehat{\gamma}_U(d),\mu\right),\right.\\
&\left.\widehat{\mathcal{V}}_U^{+}\left(\widehat{\mathcal{Z}}_U(d,\widehat{j}_U(d)),d,\widehat{j}_U(d),\widehat{\gamma}_U(d)\right)\right)=\frac{1-\alpha}{1-\beta}
\end{aligned}$$

in μ.

Here, $\widehat{L}(\widehat{d})_\alpha^H$ is an unconditionally valid lower confidence bound for $L(\widehat{d})$ and $\widehat{U}(\widehat{d})_{1-\alpha}^H$ is an unconditionally valid upper confidence bound for $U(\widehat{d})$. Combining these two confidence bounds at

appropriate levels yields an unconditional CI for the identified interval $[L(\widehat{d}),U(\widehat{d})]$ of the selected $\widehat{d}$,

$$(n^{-1/2}\widehat{L}(\widehat{d})^H_{\alpha_1},n^{-1/2}\widehat{U}(\widehat{d})^H_{1-\alpha_2}),$$

with uniformly correct asymptotic coverage when $\widehat{d}$ is selected from the data according to Assumptions 4.2 and 4.3.

Theorem 6.3. *Suppose Assumptions 4.1–4.3 and 6.1–6.3 hold. Then, for any* $0<\alpha_1,\alpha_2<1/2$,

$$\liminf_{n\to\infty}\inf_{\mathbb{P}\in\mathcal{P}_n}\mathbb{P}\Big([L(\widehat{d}),U(\widehat{d})]\subseteq\Big(n^{-1/2}\widehat{L}(\widehat{d})^H_{\alpha_1},n^{-1/2}\widehat{U}(\widehat{d})^H_{1-\alpha_2}\Big)\Big)\geq 1-\alpha_1-\alpha_2.$$

7 Finite Sample Performance of Confidence Intervals

Moving now to a context in which the object of interest is selected from the data, we compare the finite sample performance of our conditional, projection and hybrid CIs again in the setting of the Manski bounds example of Section 3. In this case, we are interested in inference on the average counterfactual outcome $W(\widehat{d})$ for $W(d)=\mathbb{E}[Y(d)]$, where interest arises either in the average counterfactual outcome for treatment $(\widehat{d}=1)$ or control $(\widehat{d}=0)$ depending upon which has the largest estimated lower bound: $\widehat{d}=\text{argmax}_{d\in\{0,1\}}\widehat{L}(d)$. This form of $\widehat{d}$ corresponds to Example 1, treated in case 1. of Proposition 4.2, which we use to specify $\widehat{\gamma}_L(\widehat{d})$ and $\widehat{\gamma}_U(\widehat{d})$ in the construction of the conditional and hybrid CIs. We report analogous simulation results for the dynamic treatment regime example in Appendix D with $\widehat{p}$ generated from a multinomial, rather than normal, distribution. In addition, we conduct a power comparison of the hypothesis tests associated with our CIs to one of the leading methods for inference on a partially-identified parameter in Appendix C in the Example 5 setting of inference on an a priori chosen policy. Encouragingly, the power of the test implied by our hybrid CI is quite competitive in this environment.

We strengthen the mean independence assumption $\mathbb{E}[Y(d)|Z]=\mathbb{E}[Y(d)]$ to full statistical independence $(Y(1),Y(0))\perp Z$ and use the sharp bounds on the ATE $W(1)-W(0)=\mathbb{E}[Y(1)]-\mathbb{E}[Y(0)]$

derived by Balke and Pearl (1997, 2011):

$$L = \max\left\{\begin{array}{c} p^{111}+p^{000}-1 \\ p^{110}+p^{001}-1 \\ p^{110}-p^{111}-p^{101}-p^{010}-p^{100} \\ p^{111}-p^{110}-p^{100}-p^{011}-p^{101} \\ -p^{011}-p^{101} \\ -p^{010}-p^{100} \\ p^{001}-p^{011}-p^{101}-p^{010}-p^{000} \\ p^{000}-p^{010}-p^{100}-p^{011}-p^{001} \end{array}\right\}$$

and

$$U = \min\left\{\begin{array}{c} 1-p^{011}-p^{100} \\ 1-p^{010}-p^{101} \\ -p^{010}+p^{011}+p^{001}+p^{110}+p^{000} \\ -p^{011}+p^{111}+p^{001}+p^{010}+p^{000} \\ p^{111}+p^{001} \\ p^{110}+p^{000} \\ -p^{101}+p^{111}+p^{001}+p^{110}+p^{100} \\ -p^{100}+p^{110}+p^{000}+p^{111}+p^{101} \end{array}\right\}.$$

For a sample size of $n = 100$, we generated $\widehat{p}$ from a $\mathcal{N}(p, \Sigma)$ distribution.[13] We examine three different DGPs within the framework of this example. The DGP corresponding to $p =$ (.081,0,0,.073,.139,.473)′ is calibrated to the probabilities estimated by Balke and Pearl (2011) in the context of a treatment for high-cholesterol (specifically, by the drug cholestyramine). The DGP corresponding to $p=$(.25,.25,.25,.25,.25,.25)′ generates completely uninformative bounds for the ATE. And the DGP corresponding to $p=$(.01,.44,.01,.01,.01,.54)′ generates quite informative bounds.

For the Balke and Pearl calibration, the two zero values in p make the second lower-endpoint component and the second upper-endpoint component used in this selected-welfare exercise have

[13]Note that in this problem, the value of Σ is implied by the value of p.

zero variance. A mechanical implementation of the formulas in Section 6 would therefore involve division by zero when studentizing these components for projection inference or when constructing the sufficient statistic and truncation limits after a degenerate component is selected. We instead use the degenerate-endpoint-robust versions of the conditional, projection, and hybrid procedures described in Appendix F: components classified as degenerate are omitted from the studentized maxima used to compute the projection critical values and, if a degenerate component is selected, the corresponding bound is set equal to its affine estimate. At this calibration, the degenerate components are uniformly separated from the corresponding population endpoints, so Theorem F.1 implies that the modified procedures retain the asymptotic coverage guarantees established in Section 6.

We compute the unconditional coverage frequencies of the conditional, projection and hybrid CIs. For comparison, we also report the coverage of the conventional CI that applies normal critical values to the estimated endpoints while treating the selected policy $\widehat{d}$ and the estimated binding indices $\widehat{j}_L(\widehat{d})$ and $\widehat{j}_U(\widehat{d})$ as if they were fixed in advance. Specifically, its nominal $(1-\alpha)$ form is

$$\left[\widehat{L}(\widehat{d})-\frac{z_{1-\alpha/2}}{\sqrt{n}}\sqrt{\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}\widehat{\Sigma}\ell'_{\widehat{d},\widehat{j}_L(\widehat{d})}},\widehat{U}(\widehat{d})+\frac{z_{1-\alpha/2}}{\sqrt{n}}\sqrt{u_{\widehat{d},\widehat{j}_U(\widehat{d})}\widehat{\Sigma}u'_{\widehat{d},\widehat{j}_U(\widehat{d})}}\right], \tag{7.1}$$

where z_q is the q quantile of the standard normal distribution.[14] These coverage frequencies are reported in Table 2. Consistent with the asymptotic results of Theorems 6.1, 6.2 and 6.3, the conditional, projection and hybrid CIs all have correct coverage for all DGPs and the modest sample size of $n=100$. Also consistent with Theorem 6.2, we note that the projection CI tends to be conservative with true coverage above the nominal level of 95%. Finally, we note that the conventional CI can substantially under-cover, consistent with the discussion in Section 3.1. Such under-coverage by the conventional CI is even more pronounced in our simulations for dynamic treatment regimes in Appendix D.

Next, we compare the length quantiles of the CIs with correct coverage for these same DGPs. Figure 1 plots the ratios of the 5th, 25th, 50th, 75th and 95th quantiles of the length of the conditional, projection and hybrid CIs relative to those same length quantiles of the projection CI. As can

[14] When a selected affine component is classified as degenerate, we set the corresponding endpoint of the conventional CI equal to its affine estimate without a standard-error adjustment, following the convention in Appendix F.

Table 2: Unconditional Coverage Frequencies

	Confidence Interval			
Data-Generating Process	Conv	Cond	Proj	Hyb
$p=(.081,0,0,.073,.139,.473)'$	0.95	0.95	0.98	0.95
$p=(.25,.25,.25,.25,.25,.25)'$	0.86	0.95	0.97	0.95
$p=(.01,.44,.01,.01,.01,.54)'$	0.95	0.95	0.99	0.96

This table reports unconditional coverage frequencies for the counterfactual outcome selected by maximizing the estimated lower bound on the counterfactual outcomes of either treatment or control, all evaluated at the nominal coverage level of 95%. Coverage frequencies are reported for conventional ("Conv"), conditional ("Cond"), projection ("Proj") and hybrid ("Hyb") CIs for a sample size of $n=100$.

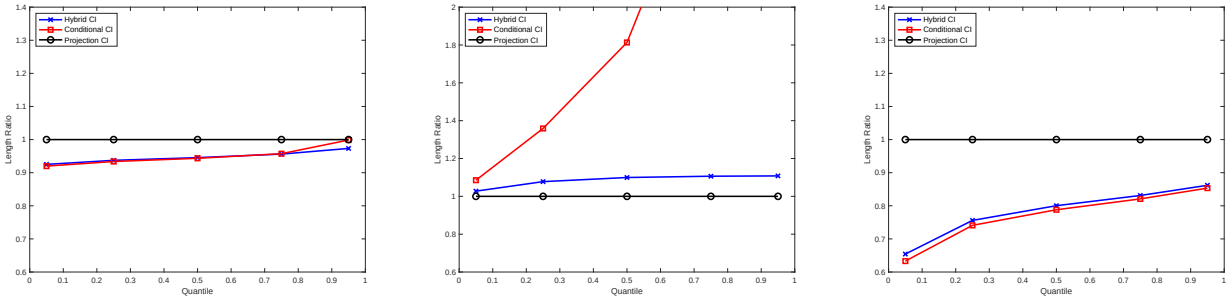


Figure 1: Ratios of CI length quantiles relative to those of the projection CI for the conditional CI (red), projection CI (black) and hybrid CI (blue) of the average counterfactual outcome selected by maximizing the estimated lower bound when $p=(.081,0,0,.073,.139,.473)'$ (left), $p=(.25,.25,.25,.25,.25,.25)'$ (middle) and $p=(.01,.44,.01,.01,.01,.54)'$ (right) and $n=100$.

be seen from the figure, the conditional CI has the tendency to become very long, especially at high quantile levels for certain DGPs, whereas the hybrid CI tends to perform the best overall by limiting the worst-case length performance of the conditional CI relative to the projection CI. Relative to projection, the hybrid CI enjoys length reductions of 20-30% for favorable DGPs while only showing length increases of 5-10% for unfavorable DGPs.

8 Empirical Application: Dynamic Treatment Regime

Continuing the motivating example in Sections 2 and 5, we revisit Han's (2024) application in which schooling and post-school training are viewed as a sequence of treatments. Han (2024) considers the Job Training Partnership Act (JTPA) program for post-school training and a high school diploma (HS) (or its equivalents) for schooling. He considers high school diplomas rather than college degrees because the former are more relevant for the disadvantaged population of Title II of the JTPA

program. In this paper, we are interested in conducting inference on welfare—earnings—evaluated at the regime chosen in a data-driven manner. The dataset is constructed by combining the JTPA data with the US Census and the National Center for Education Statistics (NCES). The following is the set of variables: Y_2 is an indicator for whether the individual is above or below the median of 30-month earnings,[15] D_2 is an indicator for whether the individual participated in the job training program, Z_2 is an indicator for whether the individual was (randomly) assigned to the program, Y_1 is an indicator for whether the individual is above or below the 80th percentile of pre-program earnings, D_1 is an indicator for whether the individual received a HS diploma or GED, and Z_1 is an indicator for the density of high schools (Neal, 1997).[16] The number of individuals in the sample is 9,223. We assume $Z \perp (Y(d), D(z))$.

Following Han (2024), given the dynamic treatment regime $\boldsymbol{\delta}(\cdot)$, the counterfactual earning type in the terminal stage is defined as $Y_2(\boldsymbol{\delta}(y_1)) \equiv Y_2(\delta_1, \delta_2(y_1, \delta_1))$, where y_1 is the counterfactual earning type in the first stage. Even though it is not a direct function of individual characteristics, the treatment regime $\boldsymbol{\delta}(\cdot)$ is a function of Y_1, which is an important summary that reflects individual characteristics relevant to earnings. Recall all possible regimes in $\mathcal{D}$ that are listed in Table 1. Suppose $\boldsymbol{\delta}^*$ is the optimal regime that maximizes the average terminal earning $W(\boldsymbol{\delta}) \equiv \mathbb{E}[Y_2(\boldsymbol{\delta}(\cdot))]$ as welfare. We are interested in constructing CIs for $W(\boldsymbol{\delta})$ evaluated at $\boldsymbol{\delta} = \widehat{\boldsymbol{\delta}}$ that is chosen by the maximin welfare rule (Example 1): $\mathcal{D}(\widehat{p}) = \{\widehat{\boldsymbol{\delta}}\}$ where $\widehat{\boldsymbol{\delta}} \in \operatorname{argmax}_{\boldsymbol{\delta}} \widehat{L}(\boldsymbol{\delta})$ with $\widehat{L}(\boldsymbol{\delta})$ being the estimated version of the lower bound on $W(\boldsymbol{\delta})$. In Section 5.1, we show how to derive the analytical lower bound $L(\boldsymbol{\delta})$ (i.e., (5.3)).

Recall that conditional inference is natural when the policymaker requires accountability for a particular program recommendation chosen from the data. In the context of the present example, suppose a funding body would consider supporting the schooling-and-targeted-training regime only when the empirical analysis recommends it. The body would want the reported CI for the regime's earnings prospects to be reliable specifically among samples in which the recommendation is made; good coverage in samples recommending no training or universal training would not compensate

[15] Note the difference from Section 2, which considers the employment status as Y_2 for illustration.

[16] Specifically, $Z_1 = 1$ if the number of high schools per square mile in each training site (e.g. a city) is above 35.

for systematically misleading assessments of the program it is being asked to fund. On the other hand, unconditional inference is natural when the policymaker evaluates the reliability of an entire selection-and-reporting procedure. In the context of the present example, suppose a planner is willing to implement any of the eight regimes and adopts the rule of choosing the regime according to $\widehat{\boldsymbol{\delta}}$ above and would like a CI for the earnings prospects associated with this rule. The planner may require that this procedure reliably assess whichever regime it recommends across repeated samples, while accepting that coverage need not meet the nominal level separately for every possible recommendation; this is the criterion addressed by the projection and hybrid intervals.

Table 3 reports 95% CIs for the welfare selected by maximizing the estimated lower bound on the welfare. Recall that the welfare is the probability that the 30-month earnings are above the median. It is notable that the hybrid CI is *shorter* than the conventional CI, which is defined in analogy to (7.1) above, even though the latter does not have (uniformly) correct coverage. We can also see that although the hybrid CI is not quite contained in the projection CI, it is somewhat shorter. Finally, the conditional CI has infinite length in this example, demonstrating an extreme example of how the conditional CIs can be uninformatively long (see Kivaranovic and Leeb, 2021).

Table 3: Confidence Intervals

Conv	Cond	Proj	Hyb
(0.32, 0.78)	(0.28, ∞)	(0.29, 0.79)	(0.28, 0.73)

This table reports 95% CIs for the welfare selected by maximizing the estimated lower bound on the welfare: conventional ("Conv"), conditional ("Cond"), projection ("Proj") and hybrid ("Hyb") CIs.

Appendix D contains Monte Carlo simulated coverage frequencies of the CIs with various DGPs in the setting of this application. Overall, the findings are consistent with the ones in Section 7.

A Additional Examples

This appendix details examples in addition to those detailed in the main text, showing how they fall under the general framework of this paper.

A.1 Revisiting Manski Bounds with a Continuous Outcome

We revisit the example with Manski bounds in Section 3, now allowing for a continuous outcome with bounded support. Let $Y \in [y^l, y^u]$ be continuously distributed and assume $\mathbb{E}[Y(d)|Z] = \mathbb{E}[Y(d)]$ for $d \in \{0,1\}$. Note that the sharp bounds on $W(d) \equiv \mathbb{E}[Y(d)]$ are

$$L(d) \equiv \max_{z \in \{0,1\}} \left\{ \mathbb{E}[Y|D=d, Z=z]\mathbb{P}(D=d|Z=z) + (1 - \mathbb{P}(D=d|Z=z))y^l \right\},$$
$$U(d) \equiv \min_{z \in \{0,1\}} \left\{ \mathbb{E}[Y|D=d, Z=z]\mathbb{P}(D=d|Z=z) + (1 - \mathbb{P}(D=d|Z=z))y^u \right\},$$

and the sharp bounds on the ATE are $L(1,0)$ and $U(1,0)$ for $L(\tilde{d}, d) \equiv L(\tilde{d}) - U(d)$ and $U(\tilde{d}, d) \equiv U(\tilde{d}) - L(d)$. We can define the identified set $\mathcal{D}^* \subseteq \mathcal{D}$ of optimal treatments as

$$\mathcal{D}^* \equiv \left\{ d \in \{0,1\} : L(\tilde{d}, d) \leq 0 \ \forall \tilde{d} \in \{0,1\} \right\} = \left\{ d \in \{0,1\} : \max_{\tilde{d} \in \{0,1\}} L(\tilde{d}) \leq U(d) \right\}.$$

Then, Assumption 4.1 holds with

$$p = \begin{pmatrix} \mathbb{E}[Y|D=0, Z=1]\mathbb{P}(D=0|Z=1) \\ \mathbb{E}[Y|D=0, Z=0]\mathbb{P}(D=0|Z=0) \\ \mathbb{E}[Y|D=1, Z=1]\mathbb{P}(D=1|Z=1) \\ \mathbb{E}[Y|D=1, Z=0]\mathbb{P}(D=1|Z=0) \\ \mathbb{P}(D=0|Z=1) \\ \mathbb{P}(D=0|Z=0) \end{pmatrix}$$

and $\tilde{\ell}_{0,j} = y^l$ and $\tilde{\ell}_{1,j} = 0$ for $j \in \{0,1\}$ and

$$\ell_{0,0} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & -y^l \end{pmatrix}, \quad \ell_{0,1} = \begin{pmatrix} 1 & 0 & 0 & 0 & -y^l & 0 \end{pmatrix},$$

$$\ell_{1,0}=(\ 0\quad 0\quad 0\quad 1\quad 0\quad y^l\ ),\quad \ell_{1,1}=(\ 0\quad 0\quad 1\quad 0\quad y^l\quad 0\ )$$

and symmetrically for $\tilde{u}_{d,j}$ and $u_{d,j}$. For $\widehat{L}(d)$ and $\widehat{U}(d)$ being the sample counterparts of $L(d)$ and $U(d)$ upon replacing p with $\widehat{p}$, Assumption 4.2 holds for $\mathcal{D}(\widehat{p})=\widehat{\mathcal{D}}$ by Proposition 4.2.

A.2 Bounds for Dynamic Treatment Effects

Consider binary Y_t and D_t for $t=1,...,T$. Let $Y\equiv(Y_1,...,Y_T)$ and $D\equiv(D_1,...,D_T)$. Suppose that we are equipped with a sequence of binary instruments $Z\equiv(Z_{t_1},...,Z_{t_K})$, which is a subvector of $(Z_1,...,Z_T)$. For $t=1,...,T$, let $Y_t(d_1,...,d_t)$ be the counterfactual outcome at t and $Y(d)\equiv(Y_1(d_1),...,Y_T(d_1,...,d_T))$. We assume that the instruments Z are independent of $(Y(d),D(z))$.

When $T=2$, the setting is similar to that in the motivating example (Section 2). Let $Y\equiv(Y_1,Y_2)$ and $D\equiv(D_1,D_2)$. For given welfare $W(d)$ with $d\equiv(d_1,d_2)$, we are interested in the optimal policy d^* that satisfies

$$d^*\in\operatorname*{argmax}_{d\in\mathcal{D}}W(d),$$

where $\mathcal{D}\equiv\{(1,1),(1,0),(0,1),(0,0)\}$. Note that this is a non-adaptive version of the policy considered in Section 2. The sign of the welfare difference, $W(d)-W(\tilde{d})$ for $d,\tilde{d}\in\mathcal{D}$, is useful for establishing the ordering of $W(d)$ with respect to d and thus to identifying d^*. However, without additional identifying assumptions, we can only establish a partial ordering of $W(d)$ based on the bounds on the welfare difference (Han, 2024). This will produce the identified set $\mathcal{D}^*$ for d^*.

An example of the welfare is $W(d)\equiv\mathbb{E}[Y_2(d)]$, namely, the average counterfactual terminal outcome. The bounds on welfare $W(d)$ are

$$L(d)\equiv\max_z L(d;z),\qquad U(d)\equiv\min_z U(d;z),$$

where

$$L(d;z)\equiv\mathbb{P}(Y_2=1,D=d|Z=z),$$

$$U(d;z) \equiv \mathbb{P}(Y_2=1,D=d|Z=z)+\sum_{d' \neq d}\mathbb{P}(D=d'|Z=z),$$

which have forms analogous to those in the static case. Define the dynamic ATE in the terminal period for a change in treatment from d to $\tilde{d}$ as

$$W(\tilde{d})-W(d)=\mathbb{E}[Y_2(\tilde{d})-Y_2(d)]=\mathbb{P}(Y_2(\tilde{d})=1)-\mathbb{P}(Y_2(d)=1).$$

Then the bounds on the dynamic ATE are as follows:

$$L(\tilde{d},d) \equiv L(\tilde{d})-U(d), \qquad U(\tilde{d},d) \equiv U(\tilde{d})-L(d).$$

Another example of welfare is the joint distribution $W(d) \equiv \mathbb{P}(Y(d)=(1,1))$ where $Y(d) \equiv (Y_1(d_1),Y_2(d))$. The bounds on $W(d)$ in this case are $L(d) \equiv \max_z L(d;z)$ and $U(d) \equiv \min_z U(d;z)$ where

$$\begin{aligned}
L(d;z) &\equiv \mathbb{P}(Y_2=1,D=d|Z=z), \\
U(d;z) &\equiv \mathbb{P}(Y_2=1,D=d|Z=z)+\mathbb{P}(Y_1=1,D_1=d_1,D_2=d_2'|Z=z) \\
&\quad +\mathbb{P}(D_1=d_1',D_2=d_2|Z=z)+\mathbb{P}(D_1=d_1',D_2=d_2'|Z=z),
\end{aligned}$$

with $d_1' \neq d_1$ and $d_2' \neq d_2$. Consider the effect of treatment on the joint distribution, for example,

$$W(1,1)-W(1,0)=\mathbb{P}(Y(1,1)=(1,1))-\mathbb{P}(Y(1,0)=(1,1)).$$

Then, with $\tilde{d}=(1,1)$ and $d=(1,0)$, the bounds on this parameter are

$$\begin{aligned}
L(\tilde{d},d) \equiv &\max_z \mathbb{P}(Y_2=1,D=(1,1)|Z=z) \\
&-\min_z \{\mathbb{P}(Y_2=1,D=(1,0)|Z=z)+\mathbb{P}(Y_1=1,D=(1,1)|Z=z) \\
&+\mathbb{P}(D=(0,1)|Z=z)+\mathbb{P}(D=(0,0)|Z=z)\},
\end{aligned}$$

$$U(\tilde{d},d) \equiv \min_z \{\mathbb{P}(Y_2=1,D=(1,1)|Z=z)+\mathbb{P}(Y_1=1,D=(1,0)|Z=z)$$
$$+\mathbb{P}(D=(0,1)|Z=z)+\mathbb{P}(D=(0,0)|Z=z)\}$$
$$-\max_z \mathbb{P}(Y_2=1,D=(1,0)|Z=z).$$

In these examples, the identified set $\mathcal{D}^*$ can be characterized as a set of maximal elements:

$$\mathcal{D}^* = \{d : \nexists \tilde{d} \neq d \text{ such that } L(\tilde{d},d) > 0\} = \{d : L(\tilde{d},d) \leq 0 \ \forall \tilde{d} \neq d\}.$$

These examples are special cases of the model in Han (2024).[17]

In both cases, it is easy to see that $L(d)$ and $U(d)$ satisfy Assumption 4.1 with p being the vector of probabilities $p(y,d|z) \equiv \mathbb{P}(Y=y,D=d|Z=z)$. To verify Assumption 4.2 for $\mathcal{D}(\widehat{p}) = \widehat{\mathcal{D}}$, let $\widehat{L}(\tilde{d},d)$ be the estimator of $L(\tilde{d},d)$ where the sample frequency replaces the population probability. Then $\widehat{\mathcal{D}} = \{d : \widehat{L}(\tilde{d},d) \leq 0 \ \forall \tilde{d} \neq d\}$.

Continuing with the first example, let $Z = Z_1 \in \{0,1\}$, that is, the researcher is only equipped with a binary instrument in the first period and no instrument in the second period. We focus on inference for $L(d)$ for $d \in \widehat{\mathcal{D}}$. Let $\widehat{z}(d) \in \operatorname{argmax}_{z \in \{0,1\}} \widehat{L}(d;z)$ so that $\widehat{L}(d) = \widehat{L}(d;\widehat{z}(d))$. Then we can write the data-dependent event that d is an element of $\widehat{\mathcal{D}}$, $\widehat{L}(d) = \widehat{L}(d;z^*)$ as a polyhedron

$$\{d \in \widehat{\mathcal{D}}, \widehat{z}(d) = z^*\} = \{\widehat{L}(\tilde{d},d) \leq 0 \ \forall \tilde{d} \neq d, \widehat{z}(d) = z^*\} = \{A^L \widehat{p} \leq 0\}$$

for some matrix A^L, where $\widehat{p}$ is the vector of probabilities $\widehat{p}(y,d|z)$, so that Assumption 4.2 holds. This is due to the forms of $L(d;z)$ and $U(d;z)$ above and $\widehat{L}(\tilde{d},d) \leq 0 \ \forall \tilde{d} \neq d$ if and only if

$$\widehat{L}(\tilde{d};z) \leq \widehat{U}(d;z') \quad \forall \tilde{d} \neq d, \forall z,z'$$

and, for example, $\widehat{z}(d) = 1$ if and only if $\widehat{L}(d;0) - \widehat{L}(d;1) \leq 0$. A similar formulation follows for the second example. In fact, this approach applies to a general parameter $W(d)$ with bounds that

[17] See also Han and Lee (2024) for other examples of dynamic causal parameters that can be used to define optimal treatments.

are minima and maxima of linear combinations of $p(y,d|z)$'s, such as parameters that have the following form:

$$W(d) \equiv f(q_d)$$

for some linear functional f, where $q_d(y) \equiv \mathbb{P}(Y(d)=y)$.

A.3 Bounds Derived from Linear Programming

In more complex settings, calculating analytical bounds on $W(d)$ or $W(d)-W(\tilde{d})$ may be cumbersome. This is especially true when the researcher wants to incorporate additional identifying assumptions. In this situation, the computational approach using linear programming can be useful (Mogstad et al., 2018; Han and Yang, 2024).

To incorporate many complicated settings, suppose that $W(d)=A_d q$ and $p=Bq$ for some known row vector A_d and matrix B, an unknown vector q in a simplex $\mathcal{Q}$, and a vector p that is estimable from data. Typically q is a vector of probabilities of a latent variable that governs the DGP; see Balke and Pearl (1997, 2011), Han (2024) and Han and Yang (2024). The linearity in this assumption is usually implied by the nature of a particular problem (e.g., discreteness). Then we have

$$\begin{aligned} L(d) &= \min_{q\in\mathcal{Q}} A_d q, \\ U(d) &= \max_{q\in\mathcal{Q}} A_d q, \end{aligned} \qquad s.t. \quad Bq=p \tag{A.1}$$

and ATE bounds for a change from treatment d to treatment $\tilde{d}$

$$\begin{aligned} L(\tilde{d},d) &= \min_{q\in\mathcal{Q}} (A_{\tilde{d}} - A_d) q, \\ U(\tilde{d},d) &= \max_{q\in\mathcal{Q}} (A_{\tilde{d}} - A_d) q, \end{aligned} \qquad s.t. \quad Bq=p. \tag{A.2}$$

Note that $L(\tilde{d},d) \neq L(\tilde{d}) - U(d)$ in general because the q that solves (A.1) for $L(\tilde{d})$ and $U(d)$ may be different (and similarly for $U(\tilde{d},d)$). As before, the identified set of optimal treatments here is characterized as $\mathcal{D}^* \equiv \{d : L(\tilde{d},d) \leq 0 \ \forall \tilde{d} \neq d\}$.

An example of this setting can be found in Han and Yang (2024). Let (Y,D,Z) be a vector

of a binary outcome, treatment and instrument and let p be a vector with entries $p(y,d|z) \equiv \mathbb{P}(Y=y,D=d|Z=z)$ across $(y,d,z) \in \{0,1\}^3$.[18] Suppose $W(d) = \mathbb{E}[Y(d)]$ for $d \in \{0,1\}$. Then, we can define the response type $\varepsilon \equiv (Y(1),Y(0),D(1),D(0))$ with a realized value $e \equiv (y(1),y(0),d(1),d(0))$, where $Y(d)$ denotes the counterfactual outcome under treatment d and $D(z)$ denotes the counterfactual treatment under instrument value z. Let $q(e) \equiv \mathbb{P}(\varepsilon = e)$ be the latent distribution. Then

$$W(d) = \mathbb{P}[Y(d)=1] = \sum_{e:y(d)=1} q(e) \equiv A_d q,$$

where q is the vector of $q(e)$'s and A_d is an appropriate selector (a row vector).

Assume that $(Y(d),D(z))$ is independent of Z for $d,z \in \{0,1\}$. The data distribution p is related to the latent distribution by

$$\mathbb{P}[Y=1,D=d|Z=z] = \mathbb{P}[Y(d)=1,D(z)=d] = \sum_{e:y(d)=1,d(z)=d} q(e) \equiv B_{d,z} q,$$

where the first equality follows by the independence assumption, q is a vector of $q(e)$'s and $B_{d,z}$ is an appropriate selector (a row vector). Now define

$$B \equiv \begin{bmatrix} B_{1,1} \\ B_{0,1} \\ B_{1,0} \\ \vdots \end{bmatrix}, \qquad p \equiv \begin{bmatrix} p(1,1|1) \\ p(1,0|1) \\ p(1,1|0) \\ p(1,0|0) \\ \vdots \end{bmatrix}$$

so that all of the constraints relating the data distribution to the latent distribution can be expressed as $Bq = p$.

To verify Assumption 4.1, it is helpful to invoke strong duality for the primal problems (A.1)

[18] See Han and Yang (2024) for the use of linear programming with continuous Y.

(under regularity conditions) and write the following dual problems:

$$L(d) = \max_{\lambda} -\tilde{p}'\lambda, \quad s.t. \quad \tilde{B}'\lambda \geq -A_d',$$
$$U(d) = \min_{\lambda} \tilde{p}'\lambda, \quad s.t. \quad \tilde{B}'\lambda \geq A_d',$$

where $\tilde{B} \equiv \begin{bmatrix} B \\ \mathbf{1}' \end{bmatrix}$ is a $(d_p+1)\times d_q$ matrix with $\mathbf{1}$ being a $d_q \times 1$ vector of ones, and $\tilde{p} \equiv \begin{bmatrix} p \\ 1 \end{bmatrix}$ is a $(d_p+1)\times 1$ vector. By using a vertex enumeration algorithm (e.g., Avis and Fukuda (1991)), one can find all (or a relevant subset) of vertices of the polyhedra $\{\lambda : \tilde{B}'\lambda \geq -A_d'\}$ and $\{\lambda : \tilde{B}'\lambda \geq A_d'\}$. Let $\Lambda_{L,d} \equiv \{\lambda_1,...,\lambda_{J_{L,d}}\}$ and $\Lambda_{U,d} \equiv \{\lambda_1,...,\lambda_{J_{U,d}}\}$ be the sets that collect such vertices, respectively. Then, it is easy to see that $L(d) = \max_{\lambda\in\Lambda_{L,d}} -\tilde{p}'\lambda$ and $U(d) = \min_{\lambda\in\Lambda_{U,d}} \tilde{p}'\lambda$, and thus Assumption 4.1 holds.

To verify Assumption 4.2, we use the dual problems to (A.2):

$$L(\tilde{d},d) = \max_{\lambda} -\tilde{p}'\lambda, \quad s.t. \quad \tilde{B}'\lambda \geq -\Delta_{\tilde{d},d}',$$
$$U(\tilde{d},d) = \min_{\lambda} \tilde{p}'\lambda, \quad s.t. \quad \tilde{B}'\lambda \geq \Delta_{\tilde{d},d}',$$

where $\Delta_{\tilde{d},d} \equiv A_{\tilde{d}} - A_d$. Analogous to the vertex enumeration argument above, let $\Lambda_{L,\tilde{d},d} \equiv \{\lambda_1, ..., \lambda_{J_{L,\tilde{d},d}}\}$ and $\Lambda_{U,\tilde{d},d} \equiv \{\lambda_1, ..., \lambda_{J_{U,\tilde{d},d}}\}$ be the sets that collect all (or a relevant subset) of vertices of the polyhedra $\{\lambda : \tilde{B}'\lambda \geq -\Delta_{\tilde{d},d}'\}$ and $\{\lambda : \tilde{B}'\lambda \geq \Delta_{\tilde{d},d}'\}$, respectively. Then, $L(\tilde{d}, d) = \max_{\lambda\in\Lambda_{L,\tilde{d},d}} -\tilde{p}'\lambda$ and $U(\tilde{d}, d) = \min_{\lambda\in\Lambda_{U,\tilde{d},d}} \tilde{p}'\lambda$. Let $\widehat{\mathcal{D}} = \{d : \widehat{L}(\tilde{d}, d) \leq 0 \ \forall \tilde{d} \neq d\}$, where $\widehat{L}(\tilde{d},d)$ is the sample counterpart of $L(\tilde{d},d)$ with $\widehat{\tilde{p}} \equiv \begin{bmatrix} \widehat{p} \\ 1 \end{bmatrix}$ replacing $\tilde{p} \equiv \begin{bmatrix} p \\ 1 \end{bmatrix}$. Partition λ as $\lambda = (\lambda^{1\prime}, \lambda^0)'$ where λ^0 is the last element of λ. Note that $d \in \widehat{\mathcal{D}}$ if and only if

$$\max_{\lambda\in\tilde{\Lambda}_{L,d}} -(\widehat{p}'\lambda^1 + \lambda^0) \leq 0,$$

where $\tilde{\Lambda}_{L,d}=\bigcup_{\tilde{d}\neq d}\Lambda_{L,\tilde{d},d}$. Also let $\widehat{\lambda}$ be such that $-\widehat{\tilde{p}}'\widehat{\lambda}=\max_{\lambda\in\tilde{\Lambda}_{L,d}}-\widehat{\tilde{p}}'\lambda$. Then, $\widehat{\lambda}=\lambda_{j_L^*}$ if and only if

$$\widehat{p}'\lambda^1_{j_L^*}+\lambda^0_{j_L^*}-(\widehat{p}'\lambda^1+\lambda^0)\leq 0 \quad \forall\lambda\in\tilde{\Lambda}_{L,d}\backslash\{\lambda_{j_L^*}\}$$

so that Assumption 4.2 holds for $\mathcal{D}(\widehat{p})=\widehat{\mathcal{D}}$.

A.4 Optimal Treatment Assignment with Partially-Identified ATE

In recent work, Christensen et al. (2023) note that "plug-in" rules for determining treatment choice can be sub-optimal when ATEs are not point-identified since the bounds on the ATE are not smooth functions of the reduced-form parameter p. Using an optimality criterion that minimizes maximum regret over the identified set for the ATE, conditional on p, they advocate bootstrap and quasi-Bayesian methods for optimal treatment choice. More specifically, they consider settings for which the ATE of a treatment is identified via intersection bounds:

$$b_L(p)\equiv\max_{j\in\{1,...,J_L\}}\{\tilde{\ell}_j+\ell_j p\}\leq ATE\leq\min_{k\in\{1,...,J_U\}}\{\tilde{u}_k+u_k p\}\equiv b_U(p)$$

for some fixed and known J_L, J_U, $\tilde{\ell}_1,...,\tilde{\ell}_{J_L}$, $\tilde{u}_1,...,\tilde{u}_{J_U}$, $\ell_1,...,\ell_{J_L}$ and $u_1,...,u_{J_U}$.[19] Therefore, Assumption 4.1 trivially holds for the ATE.

Now, let $\psi(q)\equiv\max\{b_U(q),0\}+\min\{b_L(q),0\}$. In the notation of Assumptions 6.1 and 6.2, the quasi-Bayesian treatment rule of Christensen et al. (2023) is

$$\widehat{d}^{\infty}_n=\mathbf{1}\{E^*[\psi(\widehat{p}+\varepsilon)]\geq 0\},\qquad \varepsilon\sim\mathcal{N}\left(0,\frac{\widehat{\Sigma}}{n}\right),$$

where $E^*(\cdot)$ denotes expectation with respect to the quasi-posterior draw conditional on the data. In practice, this expectation can be approximated using

$$\widehat{d}\equiv\widehat{d}_{n,m}=\mathbf{1}\left\{\frac{1}{m}\sum_{i=1}^{m}\psi(\widehat{p}+\varepsilon_i)\geq 0\right\},\qquad \varepsilon_1,...,\varepsilon_m\overset{i.i.d.}{\sim}\mathcal{N}\left(0,\frac{\widehat{\Sigma}}{n}\right).$$

[19] Although Christensen et al. (2023) do not write the form of their bounds as they are written here, the representation here is equivalent to the one in that paper upon proper definition of p since the elements in the intersection bounds are smooth functions of a reduced-form parameter.

For each fixed m, this form of $\widehat{d}$ satisfies Assumption 4.2 for $\mathcal{D}(\widehat{p})=\{\widehat{d}\}$ when $\widehat{\gamma}_L(d)=\widehat{\gamma}_U(d)$ are specified properly.

To see this, let

$$\widehat{\gamma}_L(d)=\widehat{\gamma}_U(d)=(\varepsilon_1',...,\varepsilon_m',\underline{k}_1,...,\underline{k}_m,\bar{k}_1,...,\bar{k}_m,s_1^\ell,...,s_m^\ell,s_1^u,...,s_m^u)',$$

where $\underline{k}_i \equiv \text{argmin}_{k\in\{1,...,J_U\}}\{\tilde{u}_k + u_k(\widehat{p}+\varepsilon_i)\}$, $\bar{k}_i \equiv \text{argmax}_{k\in\{1,...,J_L\}}\{\tilde{\ell}_k + \ell_k(\widehat{p}+\varepsilon_i)\}$, $s_i^\ell \equiv \text{sign}(\tilde{\ell}_{\bar{k}_i}+\ell_{\bar{k}_i}(\widehat{p}+\varepsilon_i))$ and $s_i^u \equiv \text{sign}(\tilde{u}_{\underline{k}_i}+u_{\underline{k}_i}(\widehat{p}+\varepsilon_i))$ for $i=1,...,m$. Now note that

1. $\underline{k}_i=\underline{k}_i^*$ and $\bar{k}_i=\bar{k}_i^*$ if and only if $(u_{\underline{k}_i^*}-u_k)\widehat{p}\le\tilde{u}_k-\tilde{u}_{\underline{k}_i^*}+(u_k-u_{\underline{k}_i^*})\varepsilon_i$ for all $k=1,...,J_U$ and $(\ell_k-\ell_{\bar{k}_i^*})\widehat{p}\le\tilde{\ell}_{\bar{k}_i^*}-\tilde{\ell}_k+(\ell_{\bar{k}_i^*}-\ell_k)\varepsilon_i$ for all $k=1,...,J_L$;

2. $s_i^\ell=-$ and $s_i^u=-$ if and only if $\ell_{\bar{k}_i}\widehat{p}\le-\tilde{\ell}_{\bar{k}_i}-\ell_{\bar{k}_i}\varepsilon_i$ and $u_{\underline{k}_i}\widehat{p}\le-\tilde{u}_{\underline{k}_i}-u_{\underline{k}_i}\varepsilon_i$;

3. $\widehat{d}=1$ if and only if $\sum_{i\in\underline{m}}(-u_{\underline{k}_i}\widehat{p})+\sum_{i\in\bar{m}}(-\ell_{\bar{k}_i}\widehat{p})\le\sum_{i\in\underline{m}}(\tilde{u}_{\underline{k}_i}+u_{\underline{k}_i}\varepsilon_i)+\sum_{i\in\bar{m}}(\tilde{\ell}_{\bar{k}_i}+\ell_{\bar{k}_i}\varepsilon_i)$, where $\underline{m}=\{i\in\{1,...,m\}:s_i^u=+\}$ and $\bar{m}=\{i\in\{1,...,m\}:s_i^\ell=+\}$.

Since ψ is Lipschitz,

$$E^*\left[n\left(\frac{1}{m}\sum_{i=1}^m\psi(\widehat{p}+\varepsilon_i)-E^*[\psi(\widehat{p}+\varepsilon)]\right)^2\right]\le\frac{C_\psi^2\text{tr}(\widehat{\Sigma})}{m}$$

for a finite constant C_ψ. Thus, any sequence $m=m_n\to\infty$ makes the Monte Carlo approximation error $o_p(n^{-1/2})$. Under a standard no-tie condition for the limiting treatment-choice statistic, $\Pr(\widehat{d}_{n,m_n}\neq\widehat{d}_n^\infty)\to 0$. Allowing $m=m_n\to\infty$ in the conditional and hybrid uniformity results would require an extension to a growing sequence of polyhedral conditioning events. We leave this extension to future research.

A.5 Empirical Welfare Maximization with Observational Data

Consider allocating a binary treatment based on observed covariates $X\in\mathcal{X}$. A treatment allocation rule can be defined as a function $\delta:\mathcal{X}\to\{0,1\}$ in a class of rules $\mathcal{D}$. Consider the utilitarian welfare

of deploying δ relative to treating no one. The optimal allocation δ^* satisfies

$$\delta^* \in \operatorname*{argmax}_{\delta \in \mathcal{D}} W(\delta).$$

Note that $\mathbb{E}[Y(\delta(X))-Y(0)] = E[\delta(X)\Delta(X)]$, where $\Delta(X) \equiv \mathbb{E}[Y(1)-Y(0)|X]$. This problem is considered in Kitagawa and Tetenov (2018) and Athey and Wager (2021), among others. When only observational data for (Y,D,X) are available with D being endogenous, $W(\delta)$ is only partially identified unless strong treatment effect homogeneity is assumed. This problem has been studied in Kallus and Zhou (2021); Pu and Zhang (2021); D'Adamo (2021); Byambadalai (2022), among others. Using instrumental variables, one can consider bounds on the conditional ATE based on conditional versions of the bounds considered in Sections A.1 and A.3 (i.e., Manski's bounds and bounds produced by linear programming).

In particular, assume that $(Y(d),D(z))$ is independent of Z given X. Let $L(X)$ and $U(X)$ be conditional Manski bounds on $\Delta(X)$. Then, bounds on $W(\delta)$ can be characterized as

$$L(\delta) \equiv \mathbb{E}[\delta(X)L(X)], \qquad U(\delta) \equiv \mathbb{E}[\delta(X)U(X)].$$

Similarly, bounds on $W(\tilde{\delta})-W(\delta) = \mathbb{E}[(\tilde{\delta}(X)-\delta(X))\Delta(X)]$ can be characterized as

$$L(\tilde{\delta},\delta) \equiv \mathbb{E}[(\tilde{\delta}(X)-\delta(X))L(X)], \quad U(\tilde{\delta},\delta) \equiv \mathbb{E}[(\tilde{\delta}(X)-\delta(X))U(X)]. \tag{A.3}$$

Note that $L(\tilde{\delta},\delta) \neq L(\tilde{\delta})-U(\delta)$ in general (and similarly for $U(\tilde{\delta},\delta)$).

Suppose $\mathcal{X}$ is finite and $\mathcal{X} = \{x_1,...,x_K\}$ where x_k can be a vector and K can potentially be large. The discrete covariate space is practically relevant because what may be feasible to the PM are only policies that admit discrete inputs. For simplicity of exposition, suppose $\mathcal{X} = \{0,1,2\}$. Then $\mathcal{D} = \{\delta_1,...,\delta_8\}$ where each δ_j corresponds to a mapping type from $\{0,1,2\}$ to $\{0,1\}$. To verify Assumptions 4.1 and 4.2, we proceed as follows. For given $x \in \mathcal{X}$, by arguments analogous to those in Section A.3 (and Section A.1), bounds L_x and U_x on $\Delta(x)$ satisfy, for some scalars $\tilde{\ell}_j$ and $\tilde{u}_j$

and row vectors ℓ_j and u_j,

$$L_x = \max_{j\in\{1,\ldots,J_L\}} \{\tilde{\ell}_j + \ell_j p_x\}, \qquad U_x = \min_{j\in\{1,\ldots,J_U\}} \{\tilde{u}_j + u_j p_x\},$$

where $p(x)$ is the vector of $p(y,d|z,x)$'s across (y,d,z) fixing x. Then, by Jensen's inequality, for each $\delta \in \mathcal{D}$,

$$L(\delta) \geq \tilde{L}(\delta) \equiv \max_{j\in\{1,\ldots,J_L\}} \{\tilde{\ell}_j \mathbb{E}[\delta(X)] + \ell_j \mathbb{E}[\delta(X)p(X)]\},$$
$$U(\delta) \leq \tilde{U}(\delta) \equiv \min_{j\in\{1,\ldots,J_U\}} \{\tilde{u}_j \mathbb{E}[\delta(X)] + u_j \mathbb{E}[\delta(X)p(X)]\}.$$

Note that $\tilde{L}(\delta)$ and $\tilde{U}(\delta)$ are non-sharp bounds; for calculation of sharp bounds, see Section A.6. We can verify Assumption 4.1 with $\tilde{L}(\delta)$ and $\tilde{U}(\delta)$ by defining

$$p = \begin{pmatrix} \mathbb{E}[\delta_1(X)] \\ \mathbb{E}[\delta_1(X)p(X)] \\ \vdots \\ \mathbb{E}[\delta_8(X)] \\ \mathbb{E}[\delta_8(X)p(X)] \end{pmatrix}$$

and, for $\delta = \delta_1$ as an example, by using $\ell_{\delta_1,j} = (\ \tilde{\ell}_j \quad \ell_j \quad 0 \quad \cdots \quad 0\ )$. Similarly, we can verify Assumption 4.2 by estimating $p(X)$ and $\mathbb{E}[\delta(X)]$ with sample means and $E[\delta(X)p(X)]$ with $\frac{1}{n}\sum_{i=1}^n \delta(X_i)\widehat{p}(X_i)$. Then Assumption 4.2 holds for $\mathcal{D}(\widehat{p}) = \widehat{\mathcal{D}} = \{\delta \in \mathcal{D} : \widehat{L}(\tilde{\delta},\delta) \leq 0\ \forall \tilde{\delta} \neq \delta\}$ for $\widehat{L}(\tilde{\delta},\delta)$ defined the same as $L(\tilde{\delta},\delta)$ in (A.3) after substituting $\widehat{p}$ for p.

This framework can be generalized to settings where $W(\delta)$ is partially identified, not necessarily due to treatment endogeneity but because $W(\delta)$ is a non-utilitarian welfare defined as a functional of the joint distribution of counterfactual outcomes (e.g., Cui and Han, 2026): $W(\delta) = f(F_{Y(1),Y(0)|X})$ where f is some functional and $F_{Y(1),Y(0)|X}$ is the joint distribution of $(Y(1),Y(0))$ conditional on X.

A.6 Empirical Welfare Maximization via Linear Programming

Continuing from Section A.5, we show how sharp bounds on $W(\delta)$ can be computed using linear programming. This can be done by extending the example in Section A.3 with binary Y. Again, let $\mathcal{X}=\{x_1,...,x_K\}$. Let $q(e|x)\equiv\mathbb{P}(\varepsilon=e|X=x)$. In analogy,

$$\mathbb{E}[Y(d)|X=x]=\mathbb{P}[Y(d)=1|X=x]=\sum_{e:y(d)=1} q(e|x)\equiv A_d q(x),$$

where $q(x)$ is a vector with entries $q(e|x)$ across e, and

$$\begin{aligned}\mathbb{P}[Y=1,D=d|Z=z,X=x]&=\mathbb{P}[Y(d)=1,D(z)=d|X=x]\\&=\sum_{e:y(d)=1,d(z)=d}\mathbb{P}[\varepsilon=e|X=x]\equiv B_{d,z}q(x),\end{aligned}$$

where the first equality holds by the independence assumption in the previous section. Then the constraint for each $x\in\mathcal{X}$ becomes

$$Bq(x)=p(x),$$

where $p(x)$ is the vector of $p(y,d|z,x)$'s across (y,d,z) fixing x. Now we can construct a linear program for welfare:

$$\begin{aligned}W(\delta)=\mathbb{E}[\delta(X)\Delta(X)]&=\sum_{x_k\in\mathcal{X}}p(x_k)\delta(x_k)\Delta(x_k)\\&=\sum_{x_k\in\mathcal{X}}p(x_k)\delta(x_k)(A_1-A_0)q(x_k).\end{aligned}$$

Therefore $W(\delta)$ satisfies the structure of Section A.3 and by analogous arguments, Assumptions 4.1 and 4.2 hold for $\mathcal{D}(\widehat{p})=\widehat{\mathcal{D}}$.

B Relationship with Andrews et al. (2024)

The setting of Andrews et al. (2024) (AKM henceforth) has a finite set of options, denoted here by $\mathcal{D}=\{d^0,...,d^K\}$. For a sample of size n, AKM observe $(K+1)$-dimensional estimators $\widetilde{X}_n$ and $\widetilde{Y}_n$, where $\widetilde{X}_n(d)$ is used to select among the options and $\widetilde{Y}_n(d)$ estimates the point-identified target associated with option d, $\widetilde{\mu}_{Y,n}(d)$. Their selected option and inferential target are $\widehat{d}=\text{argmax}_{d\in\mathcal{D}}\widetilde{X}_n(d)$ and $\widetilde{\mu}_{Y,n}(\widehat{d})$. AKM work in Online Appendix F with the scaled estimates $(X_n,Y_n)=\sqrt{n}(\widetilde{X}_n,\widetilde{Y}_n)$ and corresponding centering vectors $\mu_{X,n}=\sqrt{n}\widetilde{\mu}_{X,n}$ and $\mu_{Y,n}=\sqrt{n}\widetilde{\mu}_{Y,n}$, which may depend upon $\mathbb{P}\in\mathcal{P}_n$. Their Assumptions 6–8 deliver the uniform asymptotic validity of the conditional, projection, and hybrid procedures. To embed the AKM framework in the present paper, define

$$\widehat{p}=\begin{pmatrix}\widetilde{X}_n\\ \widetilde{Y}_n\end{pmatrix},\qquad p(\mathbb{P})=\begin{pmatrix}\widetilde{\mu}_{X,n}(\mathbb{P})\\ \widetilde{\mu}_{Y,n}(\mathbb{P})\end{pmatrix}. \tag{B.1}$$

Thus,

$$\sqrt{n}(\widehat{p}-p(\mathbb{P}))=\begin{pmatrix}X_n-\mu_{X,n}(\mathbb{P})\\ Y_n-\mu_{Y,n}(\mathbb{P})\end{pmatrix}. \tag{B.2}$$

For each $d\in\mathcal{D}$, let r_d^X and r_d^Y be selector row vectors satisfying

$$r_d^X\widehat{p}=\widetilde{X}_n(d),\qquad r_d^Y\widehat{p}=\widetilde{Y}_n(d). \tag{B.3}$$

B.1 Step-by-Step Comparison of Assumptions

Step 1: AKM point identification as a special case of Assumption 4.1.

Temporarily suppressing dependence on $\mathbb{P}$ for notational brevity, for option d, define $W(d)=\widetilde{\mu}_{Y,n}(d)=r_d^Y p$. Set

$$J_L=J_U=1,\qquad \widetilde{\ell}_{d,1}=\widetilde{u}_{d,1}=0,\qquad \ell_{d,1}=u_{d,1}=r_d^Y. \tag{B.4}$$

Then Assumption 4.1 becomes $L(d)=U(d)=W(d)$. The estimated lower and upper endpoints both equal $\widetilde{Y}_n(d)$, and the active component indices are nonrandom: $\widehat{j}_L(d)=\widehat{j}_U(d)=1$. Consequently,

the point-identified AKM setting removes the additional max/min nonsmoothness that arises from estimating which component attains an intersection bound.

Step 2: the AKM winner event is a special case of Assumption 4.2.

For each $d \in \mathcal{D}$, let A_d contain the row $r_{d'}^X - r_d^X$ for every $d' \neq d$. Ignoring the treatment of exact ties for the moment, $\{\widehat{d} = d\} = \{\widetilde{X}_n(d) \geq \widetilde{X}_n(d') \text{ for every } d' \neq d\} = \{A_d \widehat{p} \leq 0\}$. Hence, Assumption 4.2 holds with $\widehat{\mathcal{D}} = \{\widehat{d}\}$, $A^L(d,1,\gamma) = A^U(d,1,\gamma) = A_d$ and $c^L(d,1,\gamma) = c^U(d,1,\gamma) = 0$, where $\widehat{\gamma}_L(d)$ and $\widehat{\gamma}_U(d)$ are vacuously set equal to fixed constants. The conditioning events therefore contain only the realized winner; the binding-bound and additional-conditioning labels in the general framework carry no information in this specialization.

Step 3: Assumption 4.3 differs from AKM only on the tie event.

In the AKM specialization, the rows determining the winner event are the pairwise contrasts $r_{d'}^X - r_d^X$. As shown in Step 6 below, AKM Assumption 8 implies that each such contrast has asymptotic variance bounded away from zero. Together with AKM Assumption 6, this implies that the probability of a non-unique winner converges to zero. Assumption 4.3 is stated as exact almost-sure uniqueness in order to define the unconditional target without additional notation. The formulations therefore differ only on an asymptotically negligible tie event, which can alternatively be resolved by a deterministic tie-breaking rule.

Step 4: AKM Assumption 6 is equivalent to Assumption 6.1.

AKM Assumption 6 requires a uniform Gaussian approximation, in bounded-Lipschitz metric, for

$$\begin{pmatrix} X_n - \mu_{X,n}(\mathbb{P}) \\ Y_n - \mu_{Y,n}(\mathbb{P}) \end{pmatrix}.$$

By (B.2), this vector is $\sqrt{n}\{\widehat{p} - p(\mathbb{P})\}$. Therefore, AKM Assumption 6 is identical to Assumption 6.1 in the special case setting of AKM.

Step 5: AKM Assumption 7 is equivalent to Assumption 6.2.

AKM Assumption 7 requires their covariance estimator $\widehat{\Sigma}_n$ to be uniformly consistent for $\Sigma(\mathbb{P})$. Under (B.1), this is exactly Assumption 6.2 of the present paper.

Step 6: AKM Assumption 8 implies Assumption 6.3.

AKM Assumption 8 requires the asymptotic variances $\Sigma_X(d;\mathbb{P})$ and $\Sigma_Y(d;\mathbb{P})$ to be uniformly bounded above and away from zero for every $d\in\mathcal{D}$. It also requires

$$\sqrt{\Sigma_X(d;\mathbb{P})\Sigma_X(d';\mathbb{P})}-\Sigma_X(d,d';\mathbb{P}) \tag{B.5}$$

to be uniformly bounded away from zero for every $d,d'\in\mathcal{D}$ such that $d\neq d'$. Since $\mathcal{D}$ is fixed, the constants appearing in these bounds can be replaced by a common finite constant $\bar{\lambda}$ after increasing it if necessary.

Under the embedding in (B.1), $\Sigma(\mathbb{P})$ is the covariance matrix of the stacked vector $(X_n',Y_n')'$ and is therefore positive semidefinite. Moreover, the upper bounds on the component variances in AKM Assumption 8 imply $\lambda_{\max}\{\Sigma(\mathbb{P})\}\leq \mathrm{tr}\{\Sigma(\mathbb{P})\}=\sum_{d\in\mathcal{D}}\{\Sigma_X(d;\mathbb{P})+\Sigma_Y(d;\mathbb{P})\}\leq\bar{\lambda}$ uniformly over $\mathbb{P}\in\mathcal{P}_n$, after increasing $\bar{\lambda}$ to account for the fixed cardinality of $\mathcal{D}$. Hence, Part 1. of Assumption 6.3 is implied.

Next, under (B.4), $\ell_{d,1}=u_{d,1}=r_d^Y$. Since r_d^Y is a selector row vector, $\|r_d^Y\|^2=1$, and AKM Assumption 8 implies

$$\begin{aligned}
\ell_{d,1}\Sigma(\mathbb{P})\ell_{d,1}'=r_d^Y\Sigma(\mathbb{P})r_d^{Y\prime}=\Sigma_Y(d;\mathbb{P})\geq\frac{1}{\bar{\lambda}}=\frac{\|\ell_{d,1}\|^2}{\bar{\lambda}},\\
u_{d,1}\Sigma(\mathbb{P})u_{d,1}'=r_d^Y\Sigma(\mathbb{P})r_d^{Y\prime}=\Sigma_Y(d;\mathbb{P})\geq\frac{1}{\bar{\lambda}}=\frac{\|u_{d,1}\|^2}{\bar{\lambda}}.
\end{aligned}$$

Thus, Part 2. of Assumption 6.3 is implied.

Finally, by Step 2, every nonzero row of $A^L(d,1,\gamma)=A^U(d,1,\gamma)=A_d$ takes the form $r_{d'}^X-r_d^X$ for some $d'\neq d$. Since r_d^X and $r_{d'}^X$ select distinct coordinates, $\left\|r_{d'}^X-r_d^X\right\|^2=2$. Furthermore,

$$\begin{aligned}
&(r_{d'}^X-r_d^X)\Sigma(\mathbb{P})(r_{d'}^X-r_d^X)'\\
&\quad=\Sigma_X(d';\mathbb{P})+\Sigma_X(d;\mathbb{P})-2\Sigma_X(d,d';\mathbb{P})\\
&\quad=\left\{\sqrt{\Sigma_X(d';\mathbb{P})}-\sqrt{\Sigma_X(d;\mathbb{P})}\right\}^2
\end{aligned}$$

$$+2\left\{\sqrt{\Sigma_X(d;\mathbb{P})\Sigma_X(d';\mathbb{P})}-\Sigma_X(d,d';\mathbb{P})\right\}$$
$$\geq \frac{2}{\bar{\lambda}}=\frac{\left\|r_{d'}^X-r_d^X\right\|^2}{\bar{\lambda}},$$

where the inequality follows from (B.5). Hence, Part 3. of Assumption 6.3 is also implied.

B.2 Specialization of Conditional Inference to AKM Setting

Step C1: the endpoint statistic becomes the selected AKM inferential statistic. Under (B.4), both estimated endpoints for option d equal $\widetilde{Y}_n(d)$. The lower- and upper-endpoint residual statistics of the present paper therefore coincide. Let $\widehat{b}(d)=\widehat{\Sigma}r_d^{Y\prime}\left(r_d^Y\widehat{\Sigma}r_d^{Y\prime}\right)^{-1}$ and define

$$\widehat{Z}_n(d)=\sqrt{n}\widehat{p}-\widehat{b}(d)\sqrt{n}r_d^Y\widehat{p}. \tag{B.6}$$

The X block of (B.6) is

$$Z_n(d)=X_n-\frac{\widehat{\Sigma}_{XY,n}(\cdot,d)}{\widehat{\Sigma}_{Y,n}(d)}Y_n(d), \tag{B.7}$$

which is the feasible residual sufficient statistic used by AKM. The remaining coordinates in (B.6) are residualized inferential statistics for options other than d. They do not enter the winner inequalities and therefore do not alter the scalar truncation interval. Thus, conditioning on the full residual vector in the present paper gives the same truncated distribution for $Y_n(d)$ as conditioning on the AKM residual in (B.7).

Step C2: the polyhedral truncation set is the AKM winner truncation set. Let $\mathcal{X}(d)=\{x\in\mathbb{R}^{K+1}:x(d)\geq x(d')$ for every $d'\neq d\}$. Conditional on $Z_n(d)=z$, the event $\widehat{d}=d$ is equivalent to $Y_n(d)\in\mathcal{Y}_n(d,z)$, where

$$\mathcal{Y}_n(d,z)=\left\{y:z+\frac{\widehat{\Sigma}_{XY,n}(\cdot,d)}{\widehat{\Sigma}_{Y,n}(d)}y\in\mathcal{X}(d)\right\}. \tag{B.8}$$

The polyhedral lemma used in the present paper expresses (B.8) as an interval with endpoints $\widehat{\mathcal{V}}^-(d,z)$ and $\widehat{\mathcal{V}}^+(d,z)$ (dropping the unnecessary dependence on j and γ for the present special case). These are the same lower and upper truncation bounds denoted by AKM's $\mathcal{L}(d,z)$ and $\mathcal{U}(d,z)$,

after translating notation and replacing the covariance matrix Σ by its consistent estimator $\widehat{\Sigma}$.

Step C3: the quantile inversion and equal-tailed CIs coincide.

Let $F_{TN,n}(\cdot;\mu,d,z)$ be the CDF of a normal random variable with mean μ and variance $\widehat{\Sigma}_{Y,n}(d)$ truncated to (B.8). In the AKM point-identified specialization described above, the conditional endpoint estimators in the present paper both solve $F_{TN,n}\big(Y_n(\widehat{d});\mu,\widehat{d},Z_n(\widehat{d})\big)=1-q$ in μ. This is the feasible AKM quantile-unbiased estimator $\widehat{\mu}_{q,n}$. Combining the lower and upper inversions at $q=\alpha/2$ and $q=1-\alpha/2$ gives the CI $\big[\widehat{\mu}_{\alpha/2,n},\widehat{\mu}_{1-\alpha/2,n}\big]$, which is the feasible equal-tailed conditional CI in AKM. Thus, after setting $J_L=J_U=1$ and making the auxiliary conditioning vectors constant, the conditional method of the present paper is exactly the AKM feasible conditional method.

B.3 Specialization of Projection Inference to AKM Setting

Step P1: simultaneous endpoint inference collapses to simultaneous inference on $\mu_{Y,n}$. Under point identification, the lower and upper coefficient matrices in the present paper both reduce to the selector matrix for $\widetilde{Y}_n$: $\ell_{\text{mat}}=u_{\text{mat}}=(r^Y_{d^0},...,r^Y_{d^K})'$. Hence, the lower- and upper-endpoint covariance matrices both reduce to the asymptotic covariance matrix Σ_Y of Y_n. Both projection methods therefore begin from simultaneous inference for the same vector: $\mu_{Y,n}=\big(\mu_{Y,n}(d^0),...,\mu_{Y,n}(d^K)\big)'$.

Step P2: the exact construction of the projection CIs differs.

Treating $\widehat{\Sigma}_{Y,n}$ as fixed, let $\widehat{c}^{+}_{1-q}$ be the $1-q$ quantile of $\max_{d\in\mathcal{D}}\xi_Y(d)/\sqrt{\widehat{\Sigma}_{Y,n}(d)}$ for $\xi_Y\sim\mathcal{N}(0,\widehat{\Sigma}_{Y,n})$. If the separate lower- and upper-endpoint formula in the present paper is applied mechanically to the AKM setting with $\alpha_1=\alpha_2=\alpha/2$, the projection CI for $\mu_{y,n}(\widehat{d})$ is

$$\left[Y_n(\widehat{d})-\widehat{c}^{+}_{1-\alpha/2}\sqrt{\widehat{\Sigma}_{Y,n}(\widehat{d})},Y_n(\widehat{d})+\widehat{c}^{+}_{1-\alpha/2}\sqrt{\widehat{\Sigma}_{Y,n}(\widehat{d})}\right]. \tag{B.9}$$

On the other hand, the feasible projection CI for $\mu_{y,n}(\widehat{d})$ in AKM is given by

$$\left[Y_n(\widehat{d})-\widehat{c}^{\text{abs}}_{1-\alpha}\sqrt{\widehat{\Sigma}_{Y,n}(\widehat{d})},Y_n(\widehat{d})+\widehat{c}^{\text{abs}}_{1-\alpha}\sqrt{\widehat{\Sigma}_{Y,n}(\widehat{d})}\right], \tag{B.10}$$

where $\widehat{c}^{\text{abs}}_{1-\alpha}$ is the $1-\alpha$ quantile of $\max_{d\in\mathcal{D}}|\xi_Y(d)|/\sqrt{\widehat{\Sigma}_{Y,n}(d)}$ when treating $\widehat{\Sigma}_{Y,n}$ as fixed. Therefore, the projection *principle* in the present paper specializes directly to AKM but the projection CIs do

not coincide in the point-identified winners special case primarily because the endpoints of the projection CI of the present paper are designed to separately cover the lower- and upper-bounds of the identified set under partial identification while the CI of AKM is not robust to partial identification.

B.4 Specialization of Hybrid Inference to the AKM Setting

In analogy with their projection counterparts, the hybrid CI of the present paper uses the same hybrid *principle* as that of AKM when specialized to the point-identified winners setting. However, they differ in their details because the hybrid CI of the present paper is designed to separately cover the lower- and upper-bounds of the identified set under partial identification while the CI of AKM is not robust to partial identification. In particular, both hybrid CIs are formed as conditional CIs after intersecting the selection event $\{\widehat{d}=d\}$ with the event that $\mu_{y,n}(\widehat{d})$ lies inside of a projection CI. However, the feasible hybrid CI of AKM is based upon intersecting the selection event with the event that $\mu_{y,n}(\widehat{d})$ lies in (B.10), whereas the hybrid CI of the present paper intersects the selection event with the event that $\mu_{y,n}(\widehat{d})$ lies in one-sided versions of (B.9). Moreover, unlike AKM, the hybrid CI of the present paper treats the lower and upper endpoints separately in order to cover an identified set. It adds a one-sided projection safeguard for the lower-endpoint and the opposite one-sided safeguard for the upper-endpoint experiment, and then combines the two endpoint bounds. In the point-identified setting of AKM, this is unnecessary.

C Reality Check Power Comparison

To our knowledge, the CIs proposed in this paper are the first with proven asymptotic validity for partially-identified parameters selected from the data. Therefore, we have no existing inference method to compare the performance of our CIs to when the partially-identified parameter is data-dependent. However, as discussed in Section 4 above, our inference framework covers cases for which the partially-identified parameter is chosen a priori. For these special cases, there is a large literature on inference on partially-identified parameters or their identified sets that can be applied to form CIs. Although these special cases are not of primary interest for this paper, in this section we compare the performance of our proposed inference methods to one of the

leading inference methods in the partial identification literature as a "reality check" on whether our proposed methods are reasonably informative.

In particular, we compare the power of the test implied by our hybrid CI (i.e., a test that rejects when the value of the parameter under the null hypothesis lies outside of the hybrid CI) to the power of the hybrid test of Andrews et al. (2023), which applies to a general class of moment-inequality models. When d is chosen a priori and the parameter p is equal to a vector of moments of underlying data, Assumption 4.1 implies that $L(d) \leq W(d) \leq U(d)$ can be written as a set of (unconditional) moment inequalities, a special case of the general framework of that paper. Of the many papers on inference for moment inequalities, we choose the test of Andrews et al. (2023) for comparison for two reasons: (i) it has been shown to be quite competitive in terms of power and (ii) it is also based upon a (different) inference method that is a hybrid between conditional and projection-based inference.

When d is chosen a priori, our conditional construction and that of Andrews et al. (2023) share the same general polyhedral-conditioning logic. In both cases, inference is based upon an asymptotically Gaussian scalar statistic by conditioning on a residual sufficient statistic for the remaining mean parameters and on a polyhedral optimizer event but the commonality basically ends there. In our setting, the conditioning event identifies the affine component attaining an intersection bound and the scalar statistic is the corresponding estimate of the lower or upper endpoint. We invert its truncated-normal distribution in the endpoint mean. In Andrews et al. (2023), the statistic is the profiled maximum violation of a system of moment inequalities. Its dual representation is a maximum over Lagrange multipliers and their procedure conditions on the optimal dual multiplier before using the null restriction that its mean is nonpositive. Another difference between the inference methods is that we conduct endpoint-specific inference and combine separate lower- and upper-endpoint confidence bounds whereas their test treats the moment inequalities for a candidate parameter value jointly through a profiled max statistic. Finally, our hybrid procedure combines conditional inference with a simultaneous projection bound while their hybrid procedure combines a conditional test with a least-favorable first-stage test.

We compare the power of tests on the ATE using the Balke–Pearl bounds described in Section 7. The second and third DGPs are the same as in that section. For the first DGP, the power

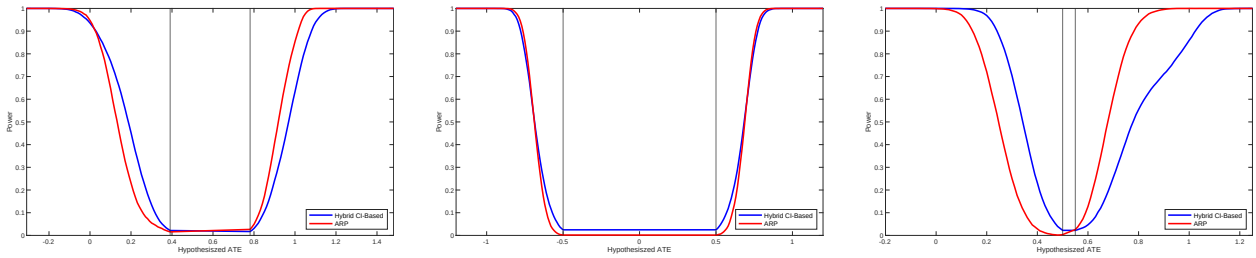


Figure 2: Power curves for hybrid Andrews et al. (2023) test (red) and test implied by hybrid CI (blue) of ATE using bounds from Balke and Pearl (1997, 2011) for $p=(.08,.001,.001,.073,.139,.473)'$ (left), $p=(.25,.25,.25,.25,.25,.25)'$ (middle) and $p=(.01,.44,.01,.01,.01,.54)'$ (right) and $n=100$. The vertical lines illustrate the true identified interval in each of these settings.

comparison uses $p=(.08,.001,.001,.073,.139,.473)'$, a perturbation of the exact Balke–Pearl calibration $p=(.081,0,0,.073,.139,.473)'$ used in Section 7. Figure 2 plots the power curves of the hybrid Andrews et al. (2023) test and the test implied by our hybrid CI, together with the endpoints of the true identified interval for the ATE in each of these three DGPs.

We can see that overall, the power of the test implied by our hybrid CI is quite competitive with that of Andrews et al. (2023). Interestingly, it appears that our test tends to be more powerful than that of Andrews et al. (2023) when the true ATE is larger than the hypothesized one, which can be seen from the hypothesized ATE lying to the left of the lower bound of the identified interval, and vice versa. Although the main innovation of our inference procedures is really their validity in the presence of data-dependent selection, the exercise of this section is reassuring for the informativeness of the procedures we propose as they are quite competitive in the absence of selection.

D Additional Monte Carlo Simulations

In analogy with Section 7, we compute the unconditional coverage frequencies of the conditional, projection and hybrid CIs for DGPs in the dynamic treatment regime setting of the empirical application (Section 8). In particular, we consider two DGPs: DGP 1 generates $\widehat{p}$ from a multinomial distribution based on $p_{d_1,y_1,d_2,y_2|z_1,z_2}=0.25$ for $(d_1,y_1,d_2,y_2)\in\{(1,0,0,1),(1,1,1,1)\}$ and all (z_1,z_2) and $p_{d_1,y_1,d_2,y_2|z_1,z_2}=0.0357$ for all other (d_1,y_1,d_2,y_2) and all (z_1,z_2); DGP 2 generates $\widehat{p}$ based on $p_{d_1,y_1,d_2,y_2|z_1,z_2}=0.375$ for $(d_1,y_1,d_2,y_2)\in\{(1,0,0,1),(1,1,1,1)\}$ and all (z_1,z_2) and $p_{d_1,y_1,d_2,y_2|z_1,z_2}=0.0179$ for all other (d_1,y_1,d_2,y_2) and all (z_1,z_2). The coverage frequencies of the

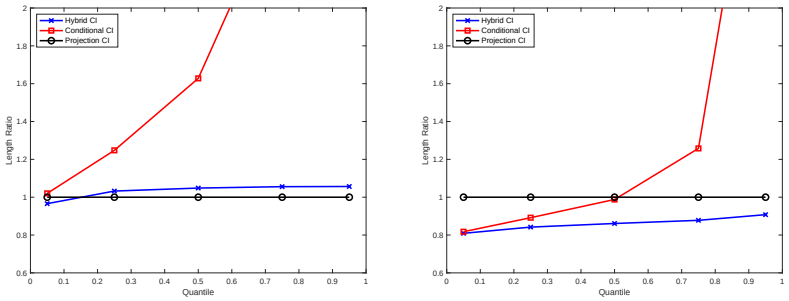


Figure 3: Ratios of CI length quantiles relative to those of the projection CI for the conditional CI (red), projection CI (black) and hybrid CI (blue) of the average counterfactual outcome selected by maximizing the estimated lower bound for DGP 1 (left) and DGP 2 (right) and $n = 1000$.

CIs are reported in Table 4. Again, consistent with the asymptotic results of Theorems 6.1–6.3, the conditional, projection and hybrid CIs all have correct coverage for all DGPs and the sample size of $n = 1000$. Also, the projection CI tends to be conservative with true coverage above the nominal level of 95%, and the conventional CI can substantially under-cover.

Table 4: Unconditional Coverage Frequencies

	Confidence Interval			
Data-Generating Process	Conv	Cond	Proj	Hyb
DGP 1	0.66	0.94	0.99	0.94
DGP 2	0.88	0.94	0.99	0.95

This table reports unconditional coverage frequencies for the counterfactual outcome selected by maximizing the estimated lower bound on the counterfactual outcomes of one of the eight dynamic regimes, all evaluated at the nominal coverage level of 95%. Coverage frequencies are reported for conventional ("Conv"), conditional ("Cond"), projection ("Proj") and hybrid ("Hyb") CIs for a sample size of $n = 1000$.

Figure 3 plots the ratios of the 5th, 25th, 50th, 75th and 95th quantiles of the length of the conditional, projection and hybrid CIs relative to those same length quantiles of the projection CI. The figure shows that the conditional CI has the tendency to become very long, especially at high quantile levels for certain DGPs, whereas the hybrid CI tends to perform the best overall by limiting the worst-case length performance of the conditional CI relative to the projection CI. Relative to projection, the hybrid CI enjoys length reductions of 10-20% for favorable DGPs while showing length increases of 5-10% for unfavorable DGPs.

E Technical Appendix

Proof of Proposition 4.1: In Example 1, $d=\widehat{d}$ if and only if, for some index $j_L^* \in \{1,...,J_L\}$,

$$(\ell_{d',j'} - \ell_{d,j_L^*})\widehat{p} \leq \tilde{\ell}_{d,j_L^*} - \tilde{\ell}_{d',j'} \qquad \text{for all } d' \in \mathcal{D},\ j' \in \{1,...,J_L\}. \tag{E.1}$$

Thus, the selection event is a union of J_L polyhedra. An analogous expression can be derived for Example 2.

In Example 3, $d=\widehat{d}$ if and only if, for some indices $j_{a,L}^* \in \{1,...,J_L\}$, $j_{a,U}^* \in \{1,...,J_U\}$, and $b_a^* \in \mathcal{D}$ for each $a \in \mathcal{D}$,

$$(\ell_{a,j} - \ell_{a,j_{a,L}^*})\widehat{p} \leq \tilde{\ell}_{a,j_{a,L}^*} - \tilde{\ell}_{a,j} \qquad \text{for all } a \in \mathcal{D},\ j = 1,...,J_L, \tag{E.2}$$

$$(u_{a,j_{a,U}^*} - u_{a,j})\widehat{p} \leq \tilde{u}_{a,j} - \tilde{u}_{a,j_{a,U}^*} \qquad \text{for all } a \in \mathcal{D},\ j = 1,...,J_U, \tag{E.3}$$

$$\widehat{U}(b,d) \leq \widehat{U}(b_a^*,a) \qquad \text{for all } a,b \in \mathcal{D}. \tag{E.4}$$

Equations (E.2) and (E.3) imply

$$\widehat{U}(b,a) = \begin{cases} 0, & b = a, \\ \tilde{u}_{b,j_{b,U}^*} - \tilde{\ell}_{a,j_{a,L}^*} + (u_{b,j_{b,U}^*} - \ell_{a,j_{a,L}^*})\widehat{p}, & b \neq a, \end{cases} \qquad a,b \in \mathcal{D}. \tag{E.5}$$

By (E.5), (E.4) also consists of linear inequalities. The existence of indices b_a^* satisfying (E.4) is equivalent to $\widehat{R}(d) \leq \widehat{R}(a)$ for every $a \in \mathcal{D}$, which characterizes selection. Therefore, the selection event is the union of the polyhedra defined by (E.2)–(E.4) over the finitely many index choices.

In Example 4, we have that $d \in \widehat{\mathcal{D}}$ if and only if

$$(\ell_{d',j'} - u_{d,j})\widehat{p} \leq \tilde{u}_{d,j} - \tilde{\ell}_{d',j'} \tag{E.6}$$

for all $d' \in \{d^0,...,d^K\}$, $j \in \{1,...,J_U\}$ and $j' \in \{1,...,J_L\}$. ■

Proof of Proposition 4.2: All event equivalences below hold a.s. We consider consistent

conditioning values; inconsistent values give the empty polyhedron. We suppress the conditioning arguments of A^M, c^M and write $j^*_{a,M}$ for a fixed value of $\widehat{j}_M(a)$, $M \in \{L,U\}$.

For Examples 1–2, $d = \widehat{d}$ if and only if

$$w_L \widehat{L}(d) + w_U \widehat{U}(d) \geq w_L \widehat{L}(a) + w_U \widehat{U}(a), \qquad \text{for all } a \in \mathcal{D}. \tag{E.7}$$

Define the endpoint-activity matrices and vectors by

$$A_L(a,j) = \begin{pmatrix} \ell_{a,1} - \ell_{a,j} \\ \vdots \\ \ell_{a,J_L} - \ell_{a,j} \end{pmatrix}, \qquad c_L(a,j) = \begin{pmatrix} \tilde{\ell}_{a,j} - \tilde{\ell}_{a,1} \\ \vdots \\ \tilde{\ell}_{a,j} - \tilde{\ell}_{a,J_L} \end{pmatrix}, \tag{E.8}$$

$$A_U(a,j) = \begin{pmatrix} u_{a,j} - u_{a,1} \\ \vdots \\ u_{a,j} - u_{a,J_U} \end{pmatrix}, \qquad c_U(a,j) = \begin{pmatrix} \tilde{u}_{a,1} - \tilde{u}_{a,j} \\ \vdots \\ \tilde{u}_{a,J_U} - \tilde{u}_{a,j} \end{pmatrix}. \tag{E.9}$$

Thus, $\widehat{j}_M(a) = j$ is characterized by $A_M(a,j)\widehat{p} \leq c_M(a,j)$.

1. When $w_U = 0 < w_L$, the event $\{\widehat{d} = d, \widehat{j}_L(d) = j^*_{d,L}\}$ is characterized by (E.1), with $j^*_L = j^*_{d,L}$. Consequently,

$$A^L = \begin{pmatrix} \ell_{d^0,1} - \ell_{d,j^*_{d,L}} \\ \vdots \\ \ell_{d^0,J_L} - \ell_{d,j^*_{d,L}} \\ \vdots \\ \ell_{d^K,1} - \ell_{d,j^*_{d,L}} \\ \vdots \\ \ell_{d^K,J_L} - \ell_{d,j^*_{d,L}} \end{pmatrix}, \qquad c^L = \begin{pmatrix} \tilde{\ell}_{d,j^*_{d,L}} - \tilde{\ell}_{d^0,1} \\ \vdots \\ \tilde{\ell}_{d,j^*_{d,L}} - \tilde{\ell}_{d^0,J_L} \\ \vdots \\ \tilde{\ell}_{d,j^*_{d,L}} - \tilde{\ell}_{d^K,1} \\ \vdots \\ \tilde{\ell}_{d,j^*_{d,L}} - \tilde{\ell}_{d^K,J_L} \end{pmatrix}. \tag{E.10}$$

For upper-endpoint inference, additionally impose $\widehat{j}_U(d)=j^*_{d,U}$ using (E.9). Hence,

$$A^U=\begin{pmatrix} A^L \\ A_U(d,j^*_{d,U}) \end{pmatrix}, \qquad c^U=\begin{pmatrix} c^L \\ c_U(d,j^*_{d,U}) \end{pmatrix}. \tag{E.11}$$

2. When $w_L=0<w_U$, fix $\widehat{j}_U(a)=j^*_{a,U}$ for every $a\in\mathcal{D}$. Substituting these indices into (E.7) and adding their activity restrictions gives

$$A^U=\begin{pmatrix} u_{d^0,j^*_{d^0,U}}-u_{d,j^*_{d,U}} \\ \vdots \\ u_{d^K,j^*_{d^K,U}}-u_{d,j^*_{d,U}} \\ A_U(d^0,j^*_{d^0,U}) \\ \vdots \\ A_U(d^K,j^*_{d^K,U}) \end{pmatrix}, \qquad c^U=\begin{pmatrix} \tilde{u}_{d,j^*_{d,U}}-\tilde{u}_{d^0,j^*_{d^0,U}} \\ \vdots \\ \tilde{u}_{d,j^*_{d,U}}-\tilde{u}_{d^K,j^*_{d^K,U}} \\ c_U(d^0,j^*_{d^0,U}) \\ \vdots \\ c_U(d^K,j^*_{d^K,U}) \end{pmatrix}. \tag{E.12}$$

For lower-endpoint inference, additionally impose $\widehat{j}_L(d)=j^*_{d,L}$ using (E.8). Hence,

$$A^L=\begin{pmatrix} A^U \\ A_L(d,j^*_{d,L}) \end{pmatrix}, \qquad c^L=\begin{pmatrix} c^U \\ c_L(d,j^*_{d,L}) \end{pmatrix}. \tag{E.13}$$

3. When $w_L,w_U>0$, fix $\widehat{j}_L(a)=j^*_{a,L}$ and $\widehat{j}_U(a)=j^*_{a,U}$ for every $a\in\mathcal{D}$. Substitution into (E.7),

together with (E.8)–(E.9), gives

$$A^L = A^U = \begin{pmatrix} w_L(\ell_{d^0,j^*_{d^0,L}} - \ell_{d,j^*_{d,L}}) + w_U(u_{d^0,j^*_{d^0,U}} - u_{d,j^*_{d,U}}) \\ \vdots \\ w_L(\ell_{d^K,j^*_{d^K,L}} - \ell_{d,j^*_{d,L}}) + w_U(u_{d^K,j^*_{d^K,U}} - u_{d,j^*_{d,U}}) \\ A_L(d^0,j^*_{d^0,L}) \\ \vdots \\ A_L(d^K,j^*_{d^K,L}) \\ A_U(d^0,j^*_{d^0,U}) \\ \vdots \\ A_U(d^K,j^*_{d^K,U}) \end{pmatrix}, \tag{E.14}$$

$$c^L = c^U = \begin{pmatrix} w_L(\tilde{\ell}_{d,j^*_{d,L}} - \tilde{\ell}_{d^0,j^*_{d^0,L}}) + w_U(\tilde{u}_{d,j^*_{d,U}} - \tilde{u}_{d^0,j^*_{d^0,U}}) \\ \vdots \\ w_L(\tilde{\ell}_{d,j^*_{d,L}} - \tilde{\ell}_{d^K,j^*_{d^K,L}}) + w_U(\tilde{u}_{d,j^*_{d,U}} - \tilde{u}_{d^K,j^*_{d^K,U}}) \\ c_L(d^0,j^*_{d^0,L}) \\ \vdots \\ c_L(d^K,j^*_{d^K,L}) \\ c_U(d^0,j^*_{d^0,U}) \\ \vdots \\ c_U(d^K,j^*_{d^K,U}) \end{pmatrix}. \tag{E.15}$$

4. In Example 3, fix the indices $\left(\widehat{j}_L(a),\widehat{j}_U(a),\widehat{b}(a)\right)_{a\in\mathcal{D}}$ at $(j^*_{a,L},j^*_{a,U},b^*_a)_{a\in\mathcal{D}}$. Their joint realization is characterized by (E.2)–(E.3) and

$$\widehat{U}(b,a) \le \widehat{U}(b^*_a,a), \qquad \text{for all } a,b\in\mathcal{D}. \tag{E.16}$$

Under (E.16), $\widehat{R}(a)=\widehat{U}(b_a^*,a)$, so selection additionally requires

$$\widehat{U}(b_d^*,d)\leq\widehat{U}(b_a^*,a),\qquad\text{for all } a\in\mathcal{D}. \tag{E.17}$$

By (E.5) and $\widehat{U}(a,a)=0$, all these inequalities are affine in $\widehat{p}$. Stacking their coefficients gives $A^L=A^U$ and $c^L=c^U$.

Finally, Examples 4 and 5 both condition on the endpoint realizations and therefore involve the inequalities $A_M(d,j_M^*)\widehat{p}\leq c_M(d,j_M^*)$ for the relevant $M\in\{L,U\}$ characterized by (E.8)–(E.9). These inequalities completely characterize A^M and c^M, $M\in\{L,U\}$, for Example 5 while these inequalities are concatenated with all of the inequalities in (E.6) to produce A^M and c^M for Example 4. ■

Lemma E.1. *Suppose Assumptions 4.1, 4.2 and 6.1–6.3 hold. Then, for any $0<\alpha<1$,*

$$\begin{aligned}\limsup_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\Big|\mathbb{P}\Big(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)^C_\alpha\Big|d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*\Big)-\alpha\Big|\\ \cdot\mathbb{P}\Big(d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*\Big)=0,\end{aligned} \tag{E.18}$$

$$\begin{aligned}\limsup_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\Big|\mathbb{P}\Big(\sqrt{n}(\tilde{u}_{d,\widehat{j}_U(d)}+u_{d,\widehat{j}_U(d)}p)\leq\widehat{U}(d)^C_\alpha\Big|d\in\mathcal{D}(\widehat{p}),\widehat{j}_U(d)=j_U^*,\widehat{\gamma}_U(d)=\gamma_U^*\Big)-\alpha\Big|\\ \cdot\mathbb{P}\Big(d\in\mathcal{D}(\widehat{p}),\widehat{j}_U(d)=j_U^*,\widehat{\gamma}_U(d)=\gamma_U^*\Big)=0,\end{aligned} \tag{E.19}$$

for all $d\in\{d^0,\ldots,d^K\}$, $j_L^*\in\{1,\ldots,J_L\}$, $j_U^*\in\{1,\ldots,J_U\}$, γ_L^* in the support of $\widehat{\gamma}_L(d)$ and γ_U^* in the support of $\widehat{\gamma}_U(d)$.

Proof: The proof of (E.19) is nearly identical to the proof of (E.18) so that we only show the latter. Lemma A.1 of Lee et al. (2016) implies that $F_{TN}(t;\mu,\sigma^2,v^-,v^+)$ is strictly decreasing in μ so that $\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)^C_\alpha$ is equivalent to

$$\begin{aligned}F_{TN}\Big(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}\widehat{p});\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p),\ell_{d,\widehat{j}_L(d)}\widehat{\Sigma}\ell'_{d,\widehat{j}_L(d)}\Big|\widehat{\mathcal{V}}_L^-\Big(\widehat{\mathcal{Z}}_L(d,\widehat{j}_L(d))\Big),\\ \widehat{\mathcal{V}}_L^+\Big(\widehat{\mathcal{Z}}_L(d,\widehat{j}_L(d))\Big)\Big)\geq 1-\alpha,\end{aligned}$$

where we use $\widehat{\mathcal{V}}_L^-\left(\widehat{\mathcal{Z}}_L(d,\widehat{j}_L(d))\right)$ and $\widehat{\mathcal{V}}_L^+\left(\widehat{\mathcal{Z}}_L(d,\widehat{j}_L(d))\right)$ as shorthand for

$$\widehat{\mathcal{V}}_L^-\left(\widehat{\mathcal{Z}}_L(d,\widehat{j}_L(d)),d,\widehat{j}_L(d),\widehat{\gamma}_L(d)\right)$$

and

$$\widehat{\mathcal{V}}_L^+\left(\widehat{\mathcal{Z}}_L(d,\widehat{j}_L(d)),d,\widehat{j}_L(d),\widehat{\gamma}_L(d)\right).$$

In addition, Lemma 2 of McCloskey (2024) implies

$$\begin{aligned}
&F_{TN}\Big(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}\widehat{p});\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p),\ell_{d,\widehat{j}_L(d)}\widehat{\Sigma}\ell'_{d,\widehat{j}_L(d)}\Big|\widehat{\mathcal{V}}_L^-\left(\widehat{\mathcal{Z}}_L(d,\widehat{j}_L(d))\right),\\
&\qquad\qquad\widehat{\mathcal{V}}_L^+\left(\widehat{\mathcal{Z}}_L(d,\widehat{j}_L(d))\right)\Big)\\
&=F_{TN}\Big(\ell_{d,\widehat{j}_L(d)}\sqrt{n}(\widehat{p}-p);0,\ell_{d,\widehat{j}_L(d)}\widehat{\Sigma}\ell'_{d,\widehat{j}_L(d)}\Big|\widehat{\mathcal{V}}_L^-\left(\widehat{\mathcal{Z}}_L^*(d,\widehat{j}_L(d))\right),\widehat{\mathcal{V}}_L^+\left(\widehat{\mathcal{Z}}_L^*(d,\widehat{j}_L(d))\right)\Big),
\end{aligned}$$

where $\widehat{\mathcal{Z}}_L^*(d,j)=\sqrt{n}\widehat{p}-\widehat{b}_L(d,j)\ell_{d,j}\sqrt{n}(\widehat{p}-p)$. Therefore, $\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)_\alpha^C$ is equivalent to

$$F_{TN}\Big(\ell_{d,\widehat{j}_L(d)}\sqrt{n}(\widehat{p}-p);0,\ell_{d,\widehat{j}_L(d)}\widehat{\Sigma}\ell'_{d,\widehat{j}_L(d)}\Big|\widehat{\mathcal{V}}_L^-\left(\widehat{\mathcal{Z}}_L^*(d,\widehat{j}_L(d))\right),\widehat{\mathcal{V}}_L^+\left(\widehat{\mathcal{Z}}_L^*(d,\widehat{j}_L(d))\right)\Big)\geq 1-\alpha. \tag{E.20}$$

Note that the set of matrices satisfying Assumption 6.3 is compact because it is a closed subset of the finite-dimensional set of positive semidefinite matrices whose largest eigenvalue is at most $\bar{\lambda}$. Therefore, under Assumptions 4.1, 6.1 and 6.3, a slight modification of Lemma 5 of Andrews et al. (2024) implies that to prove (E.18), it suffices to show that for all subsequences $\{n_s\}\subset\{n\}$, $\{\mathbb{P}_{n_s}\}\subset\times_{n=1}^{\infty}\mathcal{P}_n$ satisfying

1. $\Sigma(\mathbb{P}_{n_s})\to\Sigma^*$, where Σ^* satisfies Assumption 6.3;
2. $\mathbb{P}_{n_s}\left(d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*\right)\to q^*\in(0,1]$;
3. $\sqrt{n_s}\left\{c^L(d,j_L^*,\gamma_L^*)-A^L(d,j_L^*,\gamma_L^*)p(\mathbb{P}_{n_s})\right\}\to\omega(d,j_L^*,\gamma_L^*)\in(-\infty,\infty]^{\text{rows}(A^L(d,j_L^*,\gamma_L^*))}$ component-wise;

we have

$$\lim_{s\to\infty}\mathbb{P}_{n_s}\Big(\sqrt{n_s}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p(\mathbb{P}_{n_s}))\le\widehat{L}(d)^C_\alpha\Big|d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j^*_L,\widehat{\gamma}_L(d)=\gamma^*_L\Big)=\alpha$$

for all $d\in\{d^0,...,d^K\}$, $j^*_L\in\{1,...,J_L\}$ and γ^*_L in the support of $\widehat{\gamma}_L(d)$. A subsequence satisfying condition 3. always exists by compactness of $[-\infty,\infty]$ applied componentwise. No component of $\omega(d,j^*_L,\gamma^*_L)$ can equal $-\infty$. Indeed, if its kth component were $-\infty$, then Assumption 6.1 would imply

$$\mathbb{P}_{n_s}\big(A^L_k(d,j^*_L,\gamma^*_L)\sqrt{n_s}(\widehat{p}-p(\mathbb{P}_{n_s}))\le\sqrt{n_s}\{c^L_k(d,j^*_L,\gamma^*_L)-A^L_k(d,j^*_L,\gamma^*_L)p(\mathbb{P}_{n_s})\}\big)\to 0,$$

where $A^L_k(d,j^*_L,\gamma^*_L)$ denotes the kth row of $A^L(d,j^*_L,\gamma^*_L)$. Since the conditioning event is a subset of the event in the preceding display, this would contradict condition 2.

Let $\{\mathbb{P}_{n_s}\}$ be a sequence satisfying conditions 1.–3. Under $\{\mathbb{P}_{n_s}\}$, $(\sqrt{n_s}(\widehat{p}-p(\mathbb{P}_{n_s})),\widehat{\Sigma})\xrightarrow{d}(\xi^*,\Sigma^*)$ by Assumptions 6.1 and 6.2, where $\xi^*\sim\mathcal{N}(0,\Sigma^*)$. Thus, under Assumptions 4.1, 4.2 and 6.1–6.3, similar arguments to those used in the proof of Lemma 8 in Andrews et al. (2024) show that for any $d\in\{d^0,...,d^K\}$, $j^*_L\in\{1,...,J_L\}$ and γ^*_L in the support of $\widehat{\gamma}_L(d)$, $(\widehat{\mathcal{V}}^-_L(\widehat{\mathcal{Z}}^*_L(d,j^*_L)),\widehat{\mathcal{V}}^+_L(\widehat{\mathcal{Z}}^*_L(d,j^*_L)))\xrightarrow{d}$ $(\mathcal{V}^{-,*}_L(d,j^*_L,\gamma^*_L),\mathcal{V}^{+,*}_L(d,j^*_L,\gamma^*_L))$ under $\{\mathbb{P}_{n_s}\}$, where

$$\mathcal{V}^{-,*}_L(d,j^*_L,\gamma^*_L)=\max_{k:(A^L(d,j^*_L,\gamma^*_L)b_L(d,j^*_L))_k<0}\frac{(\omega(d,j^*_L,\gamma^*_L))_k-(A^L(d,j^*_L,\gamma^*_L)(I-b_L(d,j^*_L)\ell_{d,j^*_L})\xi^*)_k}{(A^L(d,j^*_L,\gamma^*_L)b_L(d,j^*_L))_k},$$
$$\mathcal{V}^{+,*}_L(d,j^*_L,\gamma^*_L)=\min_{k:(A^L(d,j^*_L,\gamma^*_L)b_L(d,j^*_L))_k>0}\frac{(\omega(d,j^*_L,\gamma^*_L))_k-(A^L(d,j^*_L,\gamma^*_L)(I-b_L(d,j^*_L)\ell_{d,j^*_L})\xi^*)_k}{(A^L(d,j^*_L,\gamma^*_L)b_L(d,j^*_L))_k},$$

with $b_L(d,j)=\Sigma^*\ell'_{d,j}\big(\ell_{d,j}\Sigma^*\ell'_{d,j}\big)^{-1}$. This convergence is joint with that of $(\sqrt{n_s}(\widehat{p}-p(\mathbb{P}_{n_s})),\widehat{\Sigma})$ so that under $\{\mathbb{P}_{n_s}\}$,

$$\begin{aligned}&\Big(\sqrt{n_s}\ell_{d,j^*_L}(\widehat{p}-p(\mathbb{P}_{n_s})),\widehat{\Sigma},\widehat{\mathcal{V}}^-_L\Big(\widehat{\mathcal{Z}}^*_L(d,j^*_L)\Big),\widehat{\mathcal{V}}^+_L\Big(\widehat{\mathcal{Z}}^*_L(d,j^*_L)\Big)\Big)\\&\quad\xrightarrow{d}\big(\ell_{d,j^*_L}\xi^*,\Sigma^*,\mathcal{V}^{-,*}_L(d,j^*_L,\gamma^*_L),\mathcal{V}^{+,*}_L(d,j^*_L,\gamma^*_L)\big)\end{aligned}\tag{E.21}$$

for all $d\in\{d^0,...,d^K\}$, $j^*_L\in\{1,...,J_L\}$ and γ^*_L in the support of $\widehat{\gamma}_L(d)$.

Using (E.21) and the equivalence in (E.20), the remaining arguments to prove (E.18) are nearly identical to those used in the proof of Proposition 1 of McCloskey (2024) and therefore omitted for brevity. ■

Lemma E.2. *Suppose Assumptions 4.1, 4.2 and 6.1–6.3 hold. Then, for any* $0<\alpha<1$,

$$\lim_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\left|\mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)^C_\alpha\middle|d\in\mathcal{D}(\widehat{p})\right)-\alpha\right|\cdot\mathbb{P}(d\in\mathcal{D}(\widehat{p}))=0,$$
$$\lim_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\left|\mathbb{P}\left(\sqrt{n}(\tilde{u}_{d,\widehat{j}_U(d)}+u_{d,\widehat{j}_U(d)}p)\leq\widehat{U}(d)^C_\alpha\middle|d\in\mathcal{D}(\widehat{p})\right)-\alpha\right|\cdot\mathbb{P}(d\in\mathcal{D}(\widehat{p}))=0,$$

for all $d\in\{d^0,\ldots,d^K\}$.

Proof: The results of this lemma follow from Lemma E.1 since, e.g.,

$$\begin{aligned}
&\lim_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\left|\mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)^C_\alpha\middle|d\in\mathcal{D}(\widehat{p})\right)-\alpha\right|\cdot\mathbb{P}(d\in\mathcal{D}(\widehat{p}))\\
&=\lim_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\left|\mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)^C_\alpha,d\in\mathcal{D}(\widehat{p})\right)-\alpha\cdot\mathbb{P}(d\in\mathcal{D}(\widehat{p}))\right|\\
&=\lim_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\left|\sum_{j_L^*=1}^{J_L}\sum_{\gamma_L^*}\left[\mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)^C_\alpha,d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*\right)\right.\right.\\
&\qquad\left.\left.-\alpha\cdot\mathbb{P}\left(d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*\right)\right]\right|\\
&=\lim_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\left|\sum_{j_L^*=1}^{J_L}\sum_{\gamma_L^*}\left[\mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)^C_\alpha\middle|d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*\right)-\alpha\right]\right.\\
&\qquad\left.\cdot\mathbb{P}\left(d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*\right)\right|\\
&\leq\lim_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\sum_{j_L^*=1}^{J_L}\sum_{\gamma_L^*}\left|\mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)^C_\alpha\middle|d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*\right)-\alpha\right|\\
&\qquad\cdot\mathbb{P}\left(d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*\right)\\
&\leq\sum_{j_L^*=1}^{J_L}\sum_{\gamma_L^*}\lim_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\left|\mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)^C_\alpha\middle|d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*\right)-\alpha\right|\\
&\qquad\cdot\mathbb{P}\left(d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*\right)=0,
\end{aligned}$$

where the inner sums $\sum_{\gamma_L^*}$ are over the elements of the support of $\widehat{\gamma}_L(d)$. ■

Proof of Theorem 6.1: The result of this theorem follows from Lemma E.2 since

$$
\begin{aligned}
&\liminf_{n\to\infty}\inf_{\mathbb{P}\in\mathcal{P}_n}\left[\mathbb{P}\left(\sqrt{n}[L(d),U(d)]\subseteq\left(\widehat{L}(d)^C_{\alpha_1},\widehat{U}(d)^C_{1-\alpha_2}\right)\middle|d\in\mathcal{D}(\widehat{p})\right)-(1-\alpha_1-\alpha_2)\right]\cdot\mathbb{P}(d\in\mathcal{D}(\widehat{p}))\\
&\geq\liminf_{n\to\infty}\inf_{\mathbb{P}\in\mathcal{P}_n}\left[1-\mathbb{P}\left(\sqrt{n}L(d)\leq\widehat{L}(d)^C_{\alpha_1}\middle|d\in\mathcal{D}(\widehat{p})\right)-\mathbb{P}\left(\sqrt{n}U(d)\geq\widehat{U}(d)^C_{1-\alpha_2}\middle|d\in\mathcal{D}(\widehat{p})\right)\right.\\
&\qquad\qquad\left.-(1-\alpha_1-\alpha_2)\right]\cdot\mathbb{P}(d\in\mathcal{D}(\widehat{p}))\\
&\geq\liminf_{n\to\infty}\inf_{\mathbb{P}\in\mathcal{P}_n}\left[1-\mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)^C_{\alpha_1}\middle|d\in\mathcal{D}(\widehat{p})\right)\right.\\
&\qquad\qquad\left.-\mathbb{P}\left(\sqrt{n}(\tilde{u}_{d,\widehat{j}_U(d)}+u_{d,\widehat{j}_U(d)}p)\geq\widehat{U}(d)^C_{1-\alpha_2}\middle|d\in\mathcal{D}(\widehat{p})\right)-(1-\alpha_1-\alpha_2)\right]\cdot\mathbb{P}(d\in\mathcal{D}(\widehat{p}))\\
&=\liminf_{n\to\infty}\inf_{\mathbb{P}\in\mathcal{P}_n}\left[\alpha_1-\mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)^C_{\alpha_1}\middle|d\in\mathcal{D}(\widehat{p})\right)\right.\\
&\qquad\qquad\left.+\mathbb{P}\left(\sqrt{n}(\tilde{u}_{d,\widehat{j}_U(d)}+u_{d,\widehat{j}_U(d)}p)\leq\widehat{U}(d)^C_{1-\alpha_2}\middle|d\in\mathcal{D}(\widehat{p})\right)-(1-\alpha_2)\right]\cdot\mathbb{P}(d\in\mathcal{D}(\widehat{p}))\\
&\geq\liminf_{n\to\infty}\inf_{\mathbb{P}\in\mathcal{P}_n}\left[\alpha_1-\mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p)\leq\widehat{L}(d)^C_{\alpha_1}\middle|d\in\mathcal{D}(\widehat{p})\right)\right]\cdot\mathbb{P}(d\in\mathcal{D}(\widehat{p}))\\
&\quad+\liminf_{n\to\infty}\inf_{\mathbb{P}\in\mathcal{P}_n}\left[\mathbb{P}\left(\sqrt{n}(\tilde{u}_{d,\widehat{j}_U(d)}+u_{d,\widehat{j}_U(d)}p)\leq\widehat{U}(d)^C_{1-\alpha_2}\middle|d\in\mathcal{D}(\widehat{p})\right)-(1-\alpha_2)\right]\cdot\mathbb{P}(d\in\mathcal{D}(\widehat{p}))=0,
\end{aligned}
$$

where the second inequality follows from the facts that $L(d)\geq\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p$ and $U(d)\leq\tilde{u}_{d,\widehat{j}_U(d)}+u_{d,\widehat{j}_U(d)}p$ almost surely and the final equality follows from Lemma E.2. ■

Proof of Theorem 6.2: We start by showing

$$
\liminf_{n\to\infty}\inf_{\mathbb{P}\in\mathcal{P}_n}\mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p)\geq\widehat{L}(\widehat{d})^P_{\alpha}\right)\geq 1-\alpha. \tag{E.22}
$$

Consider arbitrary subsequences $\{n_s\}\subset\{n\}$ and $\{\mathbb{P}_{n_s}\}\subset\times_{n=1}^{\infty}\mathcal{P}_n$ along which the left-hand side of (E.22) attains its liminf. The set of matrices satisfying Parts 1. and 2. of Assumption 6.3 is compact because it is a closed subset of the finite-dimensional set of positive semidefinite matrices whose eigenvalue is at most $\bar{\lambda}$ so that Parts 1. and 2. of Assumption 6.3 imply that there always exists a further subsequence $\{n_{s_p}\}\subset\{n_s\}$ such that $\Sigma(\mathbb{P}_{n_{s_p}})\to\Sigma^*$, where Σ^* satisfies Parts 1. and 2. of Assumption 6.3. Therefore, by the same argument as in the proof of Lemma E.1, to prove

(E.22), it suffices to show

$$\lim_{s\to\infty}\mathbb{P}_{n_s}\left(\sqrt{n_s}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p(\mathbb{P}_{n_s}))\geq\widehat{L}(\widehat{d})^P_\alpha\right)\geq 1-\alpha \tag{E.23}$$

for any subsequences $\{n_s\}\subset\{n\}$ and $\{\mathbb{P}_{n_s}\}\subset\times_{n=1}^{\infty}\mathcal{P}_n$ along which the limit exists and $\Sigma(\mathbb{P}_{n_s})\to\Sigma^*$. Consider any such subsequences. Assumptions 6.1 and 6.2 imply

$$\left(\ell^{mat}\sqrt{n_s}(\widehat{p}-p(\mathbb{P}_{n_s})),\widehat{\Sigma}_L\right)\xrightarrow{d}(\xi^*_L,\Sigma^*_L),\qquad \xi^*_L\sim\mathcal{N}(0,\Sigma^*_L),\quad \Sigma^*_L=\ell^{mat}\Sigma^*\ell^{mat\prime}, \tag{E.24}$$

where every diagonal element corresponding to (d,j) satisfies $(\Sigma^*_L)_{dJ_L+j,dJ_L+j}=\ell_{d,j}\Sigma^*\ell'_{d,j}\geq\|\ell_{d,j}\|^2/\bar{\lambda}>0$ by Part 2. of Assumption 6.3.

We next verify that

$$\widehat{c}_{1-\alpha,L}\xrightarrow{p}c^*_{1-\alpha,L} \tag{E.25}$$

along any $\{\mathbb{P}_{n_s}\}$ described above for $c^*_{1-\alpha,L}$ the $1-\alpha$ quantile of $\max_{i\in\{1,\ldots,(K+1)J_L\}}\frac{\xi^*_{L,i}}{\sqrt{(\Sigma^*_L)_{ii}}}$. Let $\Sigma_{L,s}=\ell^{mat}\Sigma(\mathbb{P}_{n_s})\ell^{mat\prime}$ and $Z\sim\mathcal{N}(0,I_{(K+1)J_L})$. Continuity of the positive semidefinite square root provides $\Sigma^{1/2}_{L,s}\to(\Sigma^*_L)^{1/2}$, and hence

$$\max_i\frac{(\Sigma^{1/2}_{L,s}Z)_i}{\sqrt{(\Sigma_{L,s})_{ii}}}\longrightarrow\max_i\frac{((\Sigma^*_L)^{1/2}Z)_i}{\sqrt{(\Sigma^*_L)_{ii}}}\qquad\text{almost surely.}$$

The limiting random variable has a strictly increasing cdf that is continuous in Σ^*_L so that $\widehat{\Sigma}_L\xrightarrow{p}\Sigma^*_L$, established in (E.24) implies (E.25).

Since $\sqrt{n_s}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p)\geq\widehat{L}(\widehat{d})^P_\alpha$ is equivalent to $\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}\sqrt{n_s}(\widehat{p}-p)\leq\widehat{c}_{1-\alpha,L}\sqrt{\widehat{\Sigma}_{L,\widehat{d}J_L+\widehat{j}_L(\widehat{d})}}$, the left-hand side of (E.23) is equal to

$$\begin{aligned}
&\lim_{s\to\infty}\mathbb{P}_{n_s}\left(\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}\sqrt{n_s}(\widehat{p}-p(\mathbb{P}_{n_s}))\leq\widehat{c}_{1-\alpha,L}\sqrt{\widehat{\Sigma}_{L,\widehat{d}J_L+\widehat{j}_L(\widehat{d})}}\right)\\
&\geq\lim_{s\to\infty}\mathbb{P}_{n_s}\left(\ell^{mat}\sqrt{n_s}(\widehat{p}-p(\mathbb{P}_{n_s}))\leq\widehat{c}_{1-\alpha,L}\sqrt{\mathrm{Diag}\left(\widehat{\Sigma}_L\right)}\right)\\
=\mathbb{P}\left(\xi^*_L\leq c^*_{1-\alpha,L}\sqrt{\mathrm{Diag}(\Sigma^*_L)}\right)&=\mathbb{P}\left(\max_{i\in\{1,\ldots,(K+1)J_L\}}\frac{\xi^*_{L,i}}{\sqrt{\Sigma^*_{L,i}}}\leq c^*_{1-\alpha,L}\right)=1-\alpha,
\end{aligned}$$

where all inequalities are taken element-wise across vectors, the inequality follows from the fact that $\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}\sqrt{n_s}(\widehat{p}-p(\mathbb{P}_{n_s}))$ is a (random) element of $\ell^{mat}\sqrt{n_s}(\widehat{p}-p(\mathbb{P}_{n_s}))$ and the first equality follows from (E.24) and (E.25). By way of (E.23), we have thus proved (E.22). In addition,

$$\liminf_{n\to\infty}\inf_{\mathbb{P}\in\mathcal{P}_n}\mathbb{P}\Big(\sqrt{n}(\tilde{u}_{\widehat{d},\widehat{j}_U(\widehat{d})}+u_{\widehat{d},\widehat{j}_U(\widehat{d})}p)\leq\widehat{U}(\widehat{d})^P_{1-\alpha}\Big)\geq 1-\alpha.$$

follows by nearly identical arguments. The statement of the theorem then follows by nearly identical arguments to those used in the proof of Theorem 6.1. ■

Lemma E.3. *Suppose Assumptions 4.1–6.3 hold. Then, for any* $0<\beta<\alpha<1$,

$$\limsup_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\left|\mathbb{P}\Big(\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p)\leq\widehat{L}(\widehat{d})^H_\alpha\Big|\widehat{d}=d^*,\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p)\geq\widehat{L}(\widehat{d})^P_\beta\Big)-\frac{\alpha-\beta}{1-\beta}\right|$$
$$\cdot\mathbb{P}\Big(\widehat{d}=d^*,\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p)\geq\widehat{L}(\widehat{d})^P_\beta\Big)=0, \tag{E.26}$$

$$\limsup_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\left|\mathbb{P}\Big(\sqrt{n}(\tilde{u}_{\widehat{d},\widehat{j}_U(\widehat{d})}+u_{\widehat{d},\widehat{j}_U(\widehat{d})}p)\leq\widehat{U}(\widehat{d})^H_\alpha\Big|\widehat{d}=d^*,\sqrt{n}(\tilde{u}_{\widehat{d},\widehat{j}_U(\widehat{d})}+u_{\widehat{d},\widehat{j}_U(\widehat{d})}p)\leq\widehat{U}(\widehat{d})^P_{1-\beta}\Big)-\frac{\alpha-\beta}{1-\beta}\right|$$
$$\cdot\mathbb{P}(\widehat{d}=d^*,\sqrt{n}(\tilde{u}_{\widehat{d},\widehat{j}_U(\widehat{d})}+u_{\widehat{d},\widehat{j}_U(\widehat{d})}p)\leq\widehat{U}(\widehat{d})^P_{1-\beta})=0, \tag{E.27}$$

for all $d^*\in\{d^0,...,d^K\}$.

Proof: The proof of (E.27) is nearly identical to the proof of (E.26) so that we only show the latter. Upon noting that $F_{TN}(t;\mu,\sigma^2,\widehat{\mathcal{V}}^-_L(z,d,j,\gamma),\widehat{\mathcal{V}}^{+,H}_L(z,d,j,\gamma,\mu))$ is strictly decreasing in μ by the same argument used in the proof of Proposition 5 of Andrews et al. (2024) and replacing condition 2. in the proof of Lemma E.1 with

2'. $\mathbb{P}_{n_s}\Big(\widehat{d}=d^*,\widehat{j}_L(\widehat{d})=j^*_L,\widehat{\gamma}_L(\widehat{d})=\gamma^*_L,\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p)\geq\widehat{L}(\widehat{d})^P_\beta\Big)\to q^*\in(0,1]$,

completely analogous arguments to those used to prove (E.18) in Lemma E.1, along with the proof of (E.25) in the proof of Theorem 6.2 above imply

$$\limsup_{n\to\infty}\sup_{\mathbb{P}\in\mathcal{P}_n}\left|\mathbb{P}\Big(\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p)\leq\widehat{L}(\widehat{d})^H_\alpha\Big|\widehat{d}=d^*,\widehat{j}_L(\widehat{d})=j^*_L,\widehat{\gamma}_L(\widehat{d})=\gamma^*_L,\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p)\geq\widehat{L}(\widehat{d})^P_\beta\Big)\right.$$
$$\left.-\frac{\alpha-\beta}{1-\beta}\right|\cdot\mathbb{P}\Big(\widehat{d}=d^*,\widehat{j}_L(\widehat{d})=j^*_L,\widehat{\gamma}_L(\widehat{d})=\gamma^*_L,\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})}+\ell_{\widehat{d},\widehat{j}_L(\widehat{d})}p)\geq\widehat{L}(\widehat{d})^P_\beta\Big)=0$$

for all $d^* \in \{d^0,...,d^K\}$, $j_L^* \in \{1,...,J_L\}$ and γ_L^* in the support of $\widehat{\gamma}_L(d^*)$. Then, the same argument as in the proof of Lemma E.2 implies (E.26). ■

Lemma E.4. *Suppose Assumptions 4.1–6.3 hold. Then, for any* $0<\alpha<1$,

$$\liminf_{n\to\infty} \inf_{\mathbb{P}\in\mathcal{P}_n} \mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})} + \ell_{\widehat{d},\widehat{j}_L(\widehat{d})} p) > \widehat{L}(\widehat{d})^H_\alpha\right) \geq 1-\alpha, \tag{E.28}$$

$$\liminf_{n\to\infty} \inf_{\mathbb{P}\in\mathcal{P}_n} \mathbb{P}\left(\sqrt{n}(\tilde{u}_{\widehat{d},\widehat{j}_U(\widehat{d})} + u_{\widehat{d},\widehat{j}_U(\widehat{d})} p) < \widehat{U}(\widehat{d})^H_{1-\alpha}\right) \geq 1-\alpha. \tag{E.29}$$

Proof: The proof of (E.29) is nearly identical to the proof of (E.28) so that we only show the latter. Lemma 6 of Andrews et al. (2024) and Lemma E.3 imply

$$\begin{aligned}
&\liminf_{n\to\infty} \inf_{\mathbb{P}\in\mathcal{P}_n} \mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})} + \ell_{\widehat{d},\widehat{j}_L(\widehat{d})} p) > \widehat{L}(\widehat{d})^H_\alpha\right) \\
\geq &\frac{1-\alpha}{1-\beta} \liminf_{n\to\infty} \inf_{\mathbb{P}\in\mathcal{P}_n} \sum_{d^*\in\{d^0,...,d^K\}} \mathbb{P}\left(\widehat{d}=d^*, \sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})} + \ell_{\widehat{d},\widehat{j}_L(\widehat{d})} p) \geq \widehat{L}(\widehat{d})^P_\beta\right) \\
= &\frac{1-\alpha}{1-\beta} \liminf_{n\to\infty} \inf_{\mathbb{P}\in\mathcal{P}_n} \mathbb{P}\left(\sqrt{n}(\tilde{\ell}_{\widehat{d},\widehat{j}_L(\widehat{d})} + \ell_{\widehat{d},\widehat{j}_L(\widehat{d})} p) \geq \widehat{L}(\widehat{d})^P_\beta\right) \geq 1-\alpha,
\end{aligned}$$

where the final inequality follows from (E.22) in the proof of Theorem 6.2. ■

Proof of Theorem 6.3: Using Lemma E.4 in the place of Lemma E.2, the proof is nearly identical to the proof of Theorem 6.1. ■

F Degenerate Affine Endpoint Components

This appendix permits an affine component of an identified-set endpoint to have zero asymptotic variance when that component is uniformly separated from the corresponding population endpoint. Parts 1. and 3. of Assumption 6.3 are maintained. Part 2. may be replaced by the following condition: for every $d \in \{d^0,...,d^K\}$ and every $\mathbb{P}\in\mathcal{P}_n$,

$$\ell_{d,j}\Sigma(\mathbb{P})\ell_{d,j}' \geq \|\ell_{d,j}\|^2/\bar{\lambda} \quad \text{or} \quad \begin{cases} \ell_{d,j}\Sigma(\mathbb{P})\ell_{d,j}' = 0, \\ L(d) - \{\tilde{\ell}_{d,j} + \ell_{d,j} p(\mathbb{P})\} \geq 1/\bar{\lambda}, \end{cases} \tag{F.1}$$

$$u_{d,j}\Sigma(\mathbb{P})u_{d,j}' \geq \|u_{d,j}\|^2/\bar{\lambda} \quad \text{or} \quad \begin{cases} u_{d,j}\Sigma(\mathbb{P})u_{d,j}' = 0, \\ \{\tilde{u}_{d,j} + u_{d,j}p(\mathbb{P})\} - U(d) \geq 1/\bar{\lambda}. \end{cases} \tag{F.2}$$

Call a lower (upper) component "degenerate" when the second alternative in (F.1) ((F.2)) holds and call it "regular" otherwise. Because Assumption 6.2 holds uniformly, there exists a deterministic sequence $\tau_n \downarrow 0$ such that

$$\sup_{\mathbb{P}\in\mathcal{P}_n} \mathbb{P}\Big(\|\widehat{\Sigma} - \Sigma(\mathbb{P})\| > \tau_n\Big) \longrightarrow 0. \tag{F.3}$$

In the definitions of the conditional, projection, and hybrid bounds, classify a lower component as degenerate when $\ell_{d,j}\widehat{\Sigma}\ell_{d,j}' \leq \tau_n\|\ell_{d,j}\|^2$ and regular otherwise and classify an upper component analogously. Omit components classified as degenerate when computing the studentized maxima defining the projection critical values. If the selected lower component is classified as degenerate, set

$$\widehat{L}(d)_\alpha^C = \widehat{L}(d)_\alpha^P = \widehat{L}(d)_\alpha^H = \sqrt{n}\{\tilde{\ell}_{d,\widehat{j}_L(d)} + \ell_{d,\widehat{j}_L(d)}\widehat{p}\}. \tag{F.4}$$

If the selected upper component is classified as degenerate, set

$$\widehat{U}(d)_\alpha^C = \widehat{U}(d)_\alpha^P = \widehat{U}(d)_\alpha^H = \sqrt{n}\{\tilde{u}_{d,\widehat{j}_U(d)} + u_{d,\widehat{j}_U(d)}\widehat{p}\}. \tag{F.5}$$

Otherwise, retain the definitions in Section 6.

Theorem F.1. *Suppose Assumptions 4.1–6.2, Parts 1 and 3 of Assumption 6.3, and (F.1)–(F.2) hold. Then Theorems 6.1, 6.2, and 6.3 continue to hold for the bounds defined above.*

Proof: We first show that a degenerate component is selected with probability converging to zero uniformly over $\mathbb{P}\in\mathcal{P}_n$. Suppose that the second alternative in (F.1) holds for (d,j), and choose $j_L^0(d) \in \operatorname{argmax}_{k\in\{1,\ldots,J_L\}}\{\tilde{\ell}_{d,k} + \ell_{d,k}p(\mathbb{P})\}$. If $\widehat{j}_L(d) = j$, then

$$0 \leq \{\tilde{\ell}_{d,j} + \ell_{d,j}\widehat{p}\} - \{\tilde{\ell}_{d,j_L^0(d)} + \ell_{d,j_L^0(d)}\widehat{p}\} \leq -1/\bar{\lambda} + \{\ell_{d,j} - \ell_{d,j_L^0(d)}\}\{\widehat{p} - p(\mathbb{P})\},$$

so that

$$\mathbb{P}(\widehat{j}_L(d)=j)\leq\mathbb{P}\left(\left\|\sqrt{n}\{\widehat{p}-p(\mathbb{P})\}\right\|\geq\frac{\sqrt{n}}{\bar{\lambda}\|\ell_{d,j}-\ell_{d,j^0_L(d)}\|}\right). \tag{F.6}$$

Assumption 6.1 and Part 1 of Assumption 6.3 imply uniform tightness of $\sqrt{n}(\widehat{p}-p(\mathbb{P}))$. Since the vectors $\ell_{d,j}$ are fixed and finite in number, the right-hand side of (F.6) converges to zero uniformly over the distributions for which the component (d,j) is degenerate. Consequently, for every $d\in\mathcal{D}$ and $j\in\{1,...,J_L\}$,

$$\sup_{\mathbb{P}\in\mathcal{P}_n}\mathbb{P}\left(\widehat{j}_L(d)=j,\ell_{d,j}\Sigma(\mathbb{P})\ell'_{d,j}=0\right)\longrightarrow 0.$$

The analogous argument for the upper endpoint gives, for every $d\in\mathcal{D}$ and $j\in\{1,...,J_U\}$,

$$\sup_{\mathbb{P}\in\mathcal{P}_n}\mathbb{P}\left(\widehat{j}_U(d)=j,u_{d,j}\Sigma(\mathbb{P})u'_{d,j}=0\right)\longrightarrow 0. \tag{F.7}$$

Next, using the operator norm for matrices, on the event $\{\|\widehat{\Sigma}-\Sigma(\mathbb{P})\|\leq\tau_n\}$,

$$\left|\ell_{d,j}\{\widehat{\Sigma}-\Sigma(\mathbb{P})\}\ell'_{d,j}\right|\leq\tau_n\|\ell_{d,j}\|^2$$

for every $d\in\mathcal{D}$ and $j\in\{1,...,J_L\}$. If the component (d,j) is degenerate, this implies

$$\ell_{d,j}\widehat{\Sigma}\ell'_{d,j}\leq\tau_n\|\ell_{d,j}\|^2.$$

If it is regular, then, for all sufficiently large n,

$$\ell_{d,j}\widehat{\Sigma}\ell'_{d,j}\geq\|\ell_{d,j}\|^2(1/\bar{\lambda}-\tau_n)>\tau_n\|\ell_{d,j}\|^2,$$

where the strict inequality follows from $\tau_n<1/(2\bar{\lambda})$. The same argument applies to the upper components after replacing $\ell_{d,j}$ by $u_{d,j}$. Thus, for all sufficiently large n,

$$\begin{aligned}&\inf_{\mathbb{P}\in\mathcal{P}_n}\mathbb{P}(\text{all endpoint components are classified correctly})\\&\geq 1-\sup_{\mathbb{P}\in\mathcal{P}_n}\mathbb{P}\left(\|\widehat{\Sigma}-\Sigma(\mathbb{P})\|>\tau_n\right)\longrightarrow 1\end{aligned}$$

by (F.3).

Consider Lemma E.1. If the component indexed by j_L^* is degenerate, then the absolute conditional-coverage error in (E.18) is at most one and hence

$$\left|\mathbb{P}\Big(\sqrt{n}\{\tilde{\ell}_{d,\widehat{j}_L(d)}+\ell_{d,\widehat{j}_L(d)}p(\mathbb{P})\}\le\widehat{L}(d)_\alpha^C\Big|d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*\Big)-\alpha\right|\mathbb{P}(d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*)$$
$$\le\mathbb{P}(d\in\mathcal{D}(\widehat{p}),\widehat{j}_L(d)=j_L^*,\widehat{\gamma}_L(d)=\gamma_L^*)\le\mathbb{P}(\widehat{j}_L(d)=j_L^*).$$

The last probability converges to zero uniformly by (F.6). On the other hand, if the component indexed by j_L^* is regular, then by the arguments above, it will be classified as regular with probability converging to one uniformly so that the proof of the lemma goes through essentially unchanged. The proof of (E.19) follows identically from (F.7). Thus Lemmas E.1 and E.2, and therefore Theorem 6.1, remain valid.

Let $\mathcal{E}_n$ denote the event that all endpoint components are classified correctly and that the components indexed by $\widehat{j}_L(\widehat{d})$ and $\widehat{j}_U(\widehat{d})$ are regular. For all sufficiently large n, the preceding classification argument and the union bound imply

$$\begin{aligned}\sup_{\mathbb{P}\in\mathcal{P}_n}\mathbb{P}(\mathcal{E}_n^c)\le&\sup_{\mathbb{P}\in\mathcal{P}_n}\mathbb{P}\Big(\|\widehat{\Sigma}-\Sigma(\mathbb{P})\|>\tau_n\Big)\\&+\sum_{d\in\mathcal{D}}\sum_{j=1}^{J_L}\sup_{\mathbb{P}\in\mathcal{P}_n}\mathbb{P}\Big(\widehat{j}_L(d)=j,\ell_{d,j}\Sigma(\mathbb{P})\ell_{d,j}'=0\Big)\\&+\sum_{d\in\mathcal{D}}\sum_{j=1}^{J_U}\sup_{\mathbb{P}\in\mathcal{P}_n}\mathbb{P}\Big(\widehat{j}_U(d)=j,u_{d,j}\Sigma(\mathbb{P})u_{d,j}'=0\Big)\longrightarrow0,\end{aligned}\tag{F.8}$$

where convergence follows from (F.3), the preceding bounds on the probabilities of selecting degenerate components, and finiteness of K, J_L, and J_U. On $\mathcal{E}_n$, lower projection noncoverage implies

$$\max_{d\in\{d^0,\dots,d^K\}}\max_{j:\ell_{d,j}\Sigma(\mathbb{P})\ell_{d,j}'>0}\frac{\ell_{d,j}\sqrt{n}\{\widehat{p}-p(\mathbb{P})\}}{\sqrt{\ell_{d,j}\widehat{\Sigma}\ell_{d,j}'}}>\widehat{c}_{1-\alpha,L},$$

and upper projection noncoverage implies the analogous event with $u_{d,j}$ and $\widehat{c}_{1-\alpha,U}$. The simultaneous Gaussian-approximation argument in the proof of Theorem 6.2, applied to the finite subvectors

appearing in these maxima, therefore gives

$$\limsup_{n\to\infty} \sup_{\mathbb{P}\in\mathcal{P}_n} \mathbb{P}\Big(\sqrt{n}L(\widehat{d}) < \widehat{L}(\widehat{d})^P_\alpha, \mathcal{E}_n\Big) \le \alpha,$$
$$\limsup_{n\to\infty} \sup_{\mathbb{P}\in\mathcal{P}_n} \mathbb{P}\Big(\sqrt{n}U(\widehat{d}) > \widehat{U}(\widehat{d})^P_{1-\alpha}, \mathcal{E}_n\Big) \le \alpha.$$

Combining these inequalities with (F.8) and the Bonferroni inequality proves Theorem 6.2.

On $\mathcal{E}_n$, the modified hybrid bounds coincide with the original hybrid bounds for regular selected components. Hence, for each endpoint, the probability of hybrid noncoverage is bounded above by the corresponding noncoverage probability in Lemmas E.3–E.4 plus $\mathbb{P}(\mathcal{E}_n^c)$. Equation (F.8) and those lemmas therefore imply Theorem 6.3. ■

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